# R&D-Adjusted Profitability Persistence and Valuation Feasibility in Enterprise SaaS

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## Abstract

We examine how assumptions affect inverse valuation for Salesforce, ServiceNow, Box, and HubSpot, using R&D-adjusted accounting and 27 configurations of risk, growth, and payment timing per firm. Gradual revenue convergence increases Salesforce's compatible cases from four to nine; explicit investment modeling and restrictions on future margins reduce them to four. Subsequent matched comparisons attribute the main compatibility reversals to imposed upper profitability restrictions under consistent guidance. A Box path matches the valuation target but violates a later margin ceiling. Setting incremental non-R&D book investment to zero adds one Salesforce match under the original ceilings; other firms' classifications remain unchanged within the tested configurations. Baseline portability was demonstrated, but later studies have separate replication records without a combined replay. Constructed cash flows, mixed dates, elapsed-year timing, unvalidated ceilings, and unresolved methodological overlap limit interpretation. Compatibility neither identifies investor beliefs nor establishes mispricing.

## AI Use Statement

The original manuscript was author-written, with AI providing feedback. Subsequent work used OpenAI ChatGPT/Codex for selected derivations, coding, source and calculation checks, controlled comparisons, figure preparation and editorial revisions. Google Gemini and Anthropic Claude provided critical review feedback; Claude feedback was evaluated and incorporated in this editorial pass. The authors remain responsible for the submitted text, references, data, calculations and interpretations; these tools are not authors and their feedback is not independent human peer review.

## 1. Research question and contribution

A valuation combines revenue scale, profit, investment and risk. Solving it for a persistence parameter therefore requires a completion of the other operating quantities. A precise solution remains conditional on those imposed quantities; this study does not compare their influence with measured price variation. This paper asks **what fixed market-value proxies can reveal about intangible-adjusted profitability persistence when the future operating path is not independently known**.

Enterprise SaaS provides concrete cases because research is largely expensed, growth can precede mature profitability, and acquisitions complicate productive-capital measurement. We examine Salesforce (CRM), ServiceNow (NOW), Box (BOX) and HubSpot (HUBS). The object is the set of annual decay rates for which at least one completely admitted operating path reconciles a fixed enterprise-value proxy. Empty support excludes that model-domain combination; it does not estimate zero persistence. Nonempty support gives conditional feasibility, not the operating path investors actually expected.

The contribution is the firm-specific audit: quantified compatibility reversals, changing persistence bounds, and a rejected Box equality witness that would survive a root-only test. Accounting valuation, monotone inversion and assumption-dependent support have established precedents. Ohlson (1995), Feltham and Ohlson (1995), Penman and Sougiannis (1998), and Lundholm and O'Keefe (2001) provide the valuation foundation. Nissim and Penman (2001), Lev and Sougiannis (1996), and Peters and Taylor (2017) motivate the profit/capital decomposition and intangible adjustment. Gebhardt, Lee and Swaminathan (2001), Mills (2005), and Vlčková and Buus (2025)

establish relevant inverse-valuation and persistence precedents; Tamer (2010) supplies the broader conditional-identification context.

The closest identified study, Garcia-Estevez and Puente Vicente (2026), describes persistence inversion and predictive content in its accessible abstract. Its full methods and appendices were unavailable in the retained review. We cannot establish equivalence, methodological priority, or whether it already treats failed roots and capital-policy effects. The present claim is a documented case audit, not a new general identification theory. The technical supplement retains the detailed literature assessment and complete bibliography.

The article follows the financial mechanism before the catalogue: accounting and model in Section 2, growth/capital/admission findings in Section 3, matched attribution and selected robustness in Section 4, and economic interpretation in Section 5. Supplement sections S1-S20 retain the full technical account, case tables, original figures, source crosswalks and replication history. A content map identifies every original equation, table, figure and substantive text block.

## 2. Accounting construction and model

### 2.1 Measurement perimeter and fixed valuation cases

The four valuation firms are a purposive subset of a larger accounting panel, not a representative SaaS sample. The initial collection yielded 11 firms and 56 usable core firm-years; expansion yielded 15 firms and 72 usable core firm-years. Construction eligibility requires reconciled profit/capital inputs, compatible adjacent balances and positive average adjusted capital. It is separate from whether a valuation inversion succeeds. Exact membership of the 11-firm and expanded 15-firm panel vintages is listed in supplement Section S2; target-specific eligible peer donors and the separate chronological-study donor sets are listed in Sections S7.1 and S18.2.

**Table 1. Opening accounting scale and fixed valuation controls**

| Firm | Fiscal origin | Revenue ($bn) | Adjusted ending capital ($bn) | EV proxy ($bn) | Base WACC (%) |
|---|---|---|---|---|---|
| CRM | 2026-01-31 | 41.525 | 72.998 | 199.868 | 10.33 |
| NOW | 2025-12-31 | 13.278 | 9.826 | 141.264 | 10.46 |
| BOX | 2026-01-31 | 1.177 | 0.980 | 5.529 | 10.43 |
| HUBS | 2025-12-31 | 3.131 | 2.125 | 9.308 | 10.52 |

*Notes:* USD billions; rounded display. Ending capital includes synthetic research capital, while ROIC uses average capital. Prices are October 6 closes; share and financial-balance observations are earlier, so these are mixed-date proxies. Firm-specific WACC, claims, guidance scales and source dates are retained in supplement Tables S1-S2. Required returns are externally specified, not estimated from these prices.

Each firm has 27 maintained cases: base WACC shifted by minus 100, zero or plus 100 basis points; terminal revenue growth of 2%, 3% or 4%; and YEAR_END, UNIFORM_FIRST or UNIFORM_ALL payment timing. The **base** case is BASE WACC, 3% growth and UNIFORM_FIRST. The previously defined **favorable** corner is LOW WACC, 4% growth and UNIFORM_ALL. These labels identify model configurations, not probabilities. Uniform timing is a stylized payment convention, not a reconstruction of elapsed October operations.

### 2.2 Research-adjusted profit and capital

Historical annual observations use index $t$. Forward model years use $j$, with $j=0$ the approved fiscal accounting origin and $j=1$ the guidance fiscal year; these coordinates do not start on the October price date. The same accounting quantities carry forward subscripts. Let $w$ denote the annual weighted average cost of capital (WACC). Let reported-input net operating assets be $N_t$, synthetic research capital $K_t$, reported operating income $O_t$, research spending $I_t^K$, and revenue $R_t$. Under the retained accounting convention,

$$
\begin{aligned}
N_t &= B_t + NCI_t + FO_t - FA_t, \\
K_t &= (1-d)K_{t-1} + I_t^K, d = 0.308, \\
O_t^* &= O_t + I_t^K - dK_{t-1}, \\
Y_t &= O_t^* - \tau\, max(O_t^*, 0), \tau = 0.21.
\end{aligned} \tag{1}
$$

Here B is common book equity, NCI applicable noncontrolling interests, FO disjoint financing obligations and FA classified financial assets. Goodwill, acquired intangibles, recognized software and contract-cost assets remain in N. Research initialization, eligibility corrections, accounting classifications and residual overlap limitations are retained rather than silently replaced. The depreciation scalar comes from the retained Li and Hall (2016) working-paper version; it is not issuer-specific measured obsolescence. SG&A is not capitalized.

Total adjusted capital is $C_t = N_t + K_t$, average capital is $\acute{C}_t = (C_{t-1} + C_t)/2$, adjusted NOPAT margin is $m_t = Y_t / R_t$, and average-capital ROIC is $r_t = Y_t / \acute{C}_t$. Forward net book investment is $\Delta C_j = C_j - C_{j-1}$; adjusted turnover is $k_j = R_j / \acute{C}_j$, and its reciprocal is the average-capital-to-revenue ratio $\Gamma_j = \acute{C}_j / R_j$. Constructed free cash flow and beginning-capital economic profit are

$$
F_j = Y_j - \Delta C_j, EP_j = Y_j - wC_{j-1}, EP_j = (r_j - w)\acute{C}_j + \frac{w}{2}\Delta C_j. \tag{2}
$$

The last relation matters: average-capital ROIC equal to WACC can coexist with positive economic profit during capital growth. It is a disclosed convention, not an established algebra error. For example, beginning capital 100, ending capital 120 and WACC 10% give average capital 110. A 10% average-capital return produces 11 of NOPAT but a 10 beginning-capital charge, leaving 1 of economic profit.

For year-end flows, the finite identity is

$$
\sum_{j=1}^{H} \frac{F_j}{(1+w)^j} = C_0 + \sum_{j=1}^{H} \frac{EP_j}{(1+w)^j} - \frac{C_H}{(1+w)^H}. \tag{3}
$$

When discounted capital vanishes, the infinite constructed-FCFF and EP representations agree. Timing adjustments must be applied consistently. This algebra does not establish equivalence to distributable cash; Section 5.1 distinguishes the historical cash comparator.

**2.3 Fade process, equality and complete admission**

Transition margins follow a declared linear or normalized-exponential path over $T = 2, \ldots, 6$, anchored in first-year guidance and constrained by annual margins and changes. After T the maintained process fades the average-capital ROIC spread relative to WACC:

$$
\begin{aligned}
r_{T+n} - w &= s_T e^{-\lambda n}, s_T = r_T - w, \\
EP_{T+n} &= s_T \acute{C}_{T+n} e^{-\lambda n} + \frac{w}{2}\Delta C_{T+n}, n \geq 1, \lambda \geq 0.
\end{aligned} \tag{4}
$$

Lambda is an annual rate starting after the profitability-transition endpoint, not at the price date or revenue-convergence endpoint. Positive and negative spreads have opposite fixed-path valuation responses; a zero spread removes the amplitude. Zero decay preserves the spread and has no finite upper half-life bound. Infinite decay is a limiting value, not a parameter member. Dollar EP has two terms: the fading spread multiplied by average capital, and a capital-growth term. With a positive leading capital-growth mode and a nonzero spread amplitude, the first term scales asymptotically with $(1+g)e^{-\lambda}$; its magnitude can grow even while the rate declines. For positive growth and positive WACC, the second term can grow with capital independently of the fade. Signed spreads, zero amplitudes and cancellation at zero fade prevent a universal conclusion about total dollar EP (supplement Section S19.9).

Let $P$ be the fixed enterprise-value proxy, $c$ identify a valuation case and $\theta$ the operating path/discrete transition choices. Its conditional support is

$$\Lambda_c(P)=\{\lambda\in[0,\infty):\exists\,\theta\in\Theta_c(\lambda),V_c(\lambda,\theta)=P\}. \quad (5)$$

**Valuation equality and complete admission are distinct tests.** A root establishes equality for its completed path. Admission requires that same path to satisfy guidance, transition restrictions, capital/investment conditions and every future margin inequality. The admissible domain can depend on lambda. A maximum above the target alone does not prove equality; an admitted matching witness and support coverage are required.

The kernels, signed/zero-spread cases, finite-prefix tails and component topology are developed in supplement Sections S3 and S9-S12. Saved numerical inner/outer enclosures describe endpoint uncertainty at the declared computational resolution; they are not economic confidence intervals or certified sharp bounds at every printed digit.

### 2.4 Growth and capital completion

The historical comparison switches rapidly to terminal growth and infers ending capital from an imposed average-capital turnover path: $C_j=2R_j/k_j-C_{j-1}$. Its homogeneous root is minus 1, retaining an alternating capital component. This recurrence is a historical diagnostic, not the preferred or economically validated investment technology.

Growth comparisons retain the first-year guided scale and converge log growth to g after a finite prefix. For $H_R\geq 3$,

$$\begin{aligned} u_j&=\log(1+g)+[\log(1+a_0)-\log(1+g)]\,max\{0,1-(j-2)/(H_R-2)\},\\ R_j&=R_{j-1}e^{u_j},\ j\geq 2. \end{aligned} \quad (6)$$

Chan, Karceski and Lakonishok (2003) provide evidence on growth persistence and the limits of long-horizon extrapolation; the finite convergence paths here are scenarios, not fitted forecasts. The guidance anchor is $a_0=R_1/R_0-1$; the history anchor is the median of the latest three approved annual growth observations. H3/H5 label the first model year at terminal growth. No price or solved lambda selects the anchor.

The explicit investment completion preserves opening N0/K0 but uses

$$\begin{aligned} N_j&=N_{j-1}+\eta(R_j-R_{j-1}),\\ K_j&=0.692\,K_{j-1}+\rho R_j,\ C_j=N_j+K_j,\\ \Delta C_j&=\eta\,\Delta R_j+\rho R_j-d\,K_{j-1}. \end{aligned} \quad (7)$$

Rho is the latest historical research-spending share extended into the future. Baseline eta is the exact saved calibration of the opening NOA/revenue stock ratio, not measured marginal cash-investment efficiency. Turnover becomes derived from completed average capital; imposing the old independent turnover target as well would overdetermine the model. The completion removes the alternating mode, but does not validate gross investment, asset consumption, working-capital cash or acquisition economics.

Let $M$ denote the declared all-future adjusted NOPAT-margin ceiling in the selected support family. Writing $a=1+g$ and $v=0.692$, terminal growth requires continuing book investment and full future admission:

$$\begin{aligned} \lim C_j/R_j&=\eta+\frac{\rho\,a}{a-v},\\ \Gamma_\infty&=\frac{1+a}{2a}\left(\eta+\frac{\rho\,a}{a-v}\right),\\ \Gamma_{T+n}\{w+s_T e^{-\lambda n}\}&\leq M \text{ for all integers } n\geq 1. \end{aligned} \quad (8)$$

A finite simulation cannot establish the last condition. The computations cover annual inequalities and every omitted tail year. “Complete” refers to the declared model rules, not a complete empirical characterization of the firm’s technology.

## 3. Core growth, capital and admission findings

### 3.1 What changes across the operating completions?

**Table 2. Compatible cases under growth and capital completions**

| Revenue / capital completion | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|
| Immediate growth / historical recurrence | 4 | 0 | 0 | 27 |
| Guidance H3 / historical recurrence | 6 | 0 | 0 | 27 |
| Guidance H5 / historical recurrence | 9 | 0 | 0 | 27 |
| History H3 / historical recurrence | 6 | 0 | 0 | 27 |
| History H5 / historical recurrence | 7 | 0 | 0 | 27 |
| Guidance H3 / investment, full ceiling | 3 | 0 | 0 | 27 |
| Guidance H5 / investment, full ceiling | 4 | 0 | 0 | 27 |
| History H3 / investment, full ceiling | 3 | 0 | 0 | 27 |
| History H5 / investment, full ceiling | 4 | 0 | 0 | 27 |

*Notes:* Each count is out of 27 maintained cases. Growth-only comparisons recompute capital under the original recurrence. Investment comparisons preserve the revenue path but change capital completion, turnover interpretation and full future-margin admission together. They retain the historical guidance bridge; Section 4 separately matches coherent guidance. These are conditional results, not empirical success rates.

Salesforce is specification-sensitive: slower convergence raises its count from four to nine, and explicit investment/full admission reduces the H5 count to four. ServiceNow and Box remain excluded; HubSpot remains compatible throughout these families. Different history/guidance counts prevent interpreting one profile as the uniquely correct forecast.

At the base case, guidance H5 raises the historical-recurrence maxima by about $15.187bn for CRM, $19.831bn for NOW, $0.334bn for BOX and $3.205bn for HUBS. The matched investment maxima instead are approximately $167.42bn, $70.30bn, $3.71bn and $14.23bn. Investment maxima are lower in all 432 saved growth-profile comparisons. This is catalogue-specific evidence, not a theorem that every coherent investment model reduces value. Complete maximum/gap brackets and all earlier comparison families are retained in supplement Sections S5-S7.

The mechanism is joint. More revenue increases transition profit and inherited terminal scale, but also changes required capital. Constant research spending as growth slows can increase accumulated research capital relative to revenue. A fixed endpoint margin then produces a different endpoint return and different subsequent margin path. Under full ceilings a small positive fade can admit a higher endpoint margin than zero fade, so the maximum constrained envelope need not occur at zero.

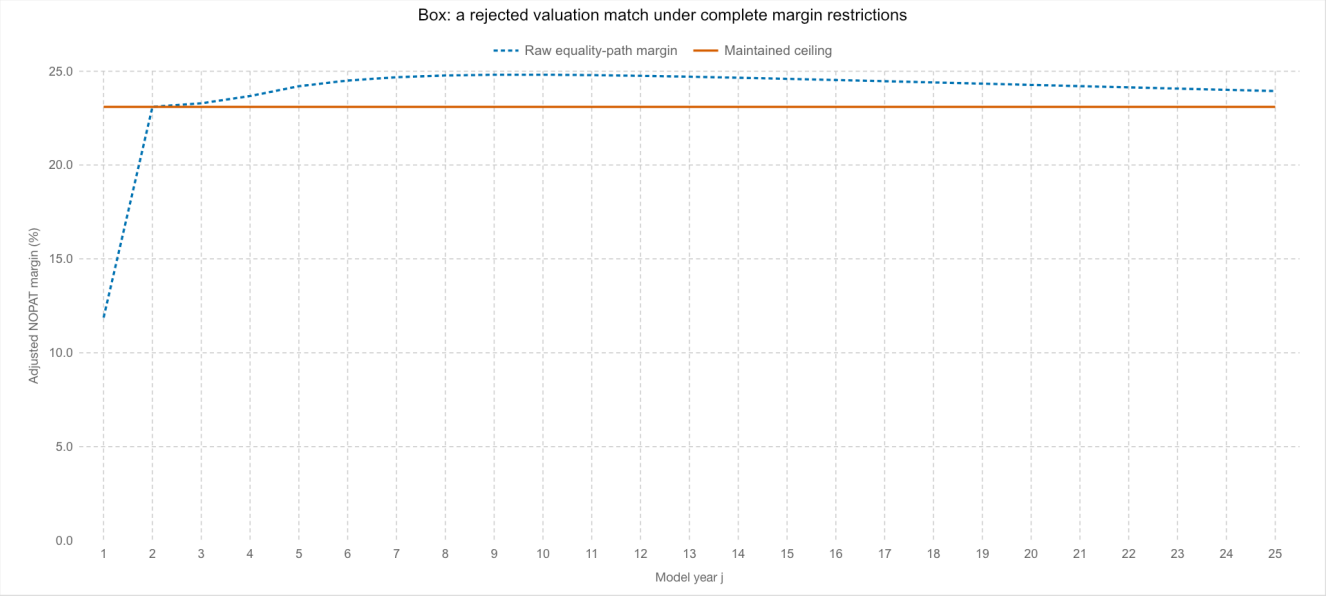
For HUBS’s base case, the historical benchmark implies a minimum spread half-life of about 11.9 years after T; investment with historical-guidance H5 gives about 5.7 years, and coherent peer H5 about 30.4 years. All three sets include zero decay, so none has a finite upper duration bound. These are conditional lower bounds, not point estimates. The investment-to-peer comparison bundles guidance and cap changes; Section 4.1 separates them.

The NOW/BOX exclusions concern only the tested configurations: immediate terminal growth at model year 2 and the H3/H5 convergence profiles, with terminal growth first reached in model year 3 or 5 from the fiscal origin. They do not test longer revenue-convergence horizons or imply exclusion under every operating completion.

### 3.2 Box: a root rejected by the operating path

A relaxed growth/investment calculation gives a raw Box maximum near $5.884bn, above its $5.529bn proxy. A saved equality witness has lambda about 0.004174 per year. Its endpoint satisfies the cap, but the later adjusted NOPAT margin reaches about 24.816% in model year 10, exceeding the maintained 23.102% ceiling.

**Figure 1. Box equality does not establish full admission**




*Rejected Box equality witness*

*Notes:* Saved guidance H5 witness, LOW WACC, 4% growth and UNIFORM_ALL. The path matches the target in a relaxed domain and is rejected by the complete future-margin rule. The first 25 years shown are saved constructed-path records, not new forecasts. The ceiling itself remains an unvalidated future restriction.

Reoptimization over the complete domain leaves a favorable maximum of approximately $5.22184-$5.22190bn, about $307m short. No legitimate equality remains in these tested investment profiles. This is a rejection under the maintained ceiling, not evidence that Box could never earn higher margins or that its market price is wrong.

## 4. Matched attribution and selected robustness evidence

### 4.1 Guidance consistency versus upper profitability restrictions

The original support-family comparison changes the first-year R&D bridge as well as upper restrictions. A separately prespecified matched H5 study separates those changes. The coherent bridge translates the public GAAP anchor q using the unchanged research policy and synthetic tax:

$$b = \Pi_\tau\left(q + \rho - d\,K_0 / R_1\right), \Pi_\tau(x) = x - \tau\, max(x, 0). \quad (9)$$

This is an algebraically coherent guidance translation, not validation of the research-spending forecast. A saved operating-dollar range is translated at the same fixed revenue scale where available.

On first use, **G** means coherent guidance with the old nominal terminal, transition and all-future upper caps. **O** uses the same guidance with saved sustained own-history upper caps; **P** uses saved multi-year peer upper caps. All share the common transition floor, min(old annual floor, coherent guide lower), and the unchanged terminal

lower bound. These upper-support bundles include transition ceilings, not only terminal margins. The optional paired turnover/hull screen is off. **H** retains historical guidance/old caps; **RP** retains historical guidance with fixed peer caps as a reverse-order diagnostic.

For CRM and BOX, both own-history ceilings equal the coherent guided margin; for HUBS they equal the upper endpoint of its coherent guidance range. These rules impose future performance restrictions, rather than independently establishing economic ceilings. NOW differs: its terminal ceiling is 17.776% and its transition/future ceiling 17.850%, compared with coherent guidance of 15.080% (supplement Table S30).

**Table 3. Matched H5 compatibility under guidance and cap families**

| Firm | Historical H | Coherent G | Own O | Peer P | Historical bridge / peer RP |
|---|---|---|---|---|---|
| CRM | 4 | 4 | 0 | 0 | 0 |
| NOW | 0 | 0 | 0 | 0 | 0 |
| BOX | 0 | 0 | 0 | 0 | 0 |
| HUBS | 27 | 27 | 0 | 19 | 21 |

*Notes:* Counts out of 27; fixed revenue, investment coefficients, EV targets, WACC/g/timing, tax and fade convention. Coherent O/P records were reused after exact control checks and focused reproduction. Guidance correction alone leaves classifications unchanged under old caps; upper restrictions govern the main reversal at coherent guidance. Reverse-order contrasts are diagnostics, not alternative economically validated forecasts.

The pilot used the fixed base/favorable cases for all four firms. Prespecified compatibility changes, maximum-value differences exceeding 1% of fixed EV, or operating-domain inconsistency triggered full 27-case expansion of all core cells. The 1% rule is research triage, not significance. G-to-O removes four CRM and 27 HUBS matches; G-to-P removes four CRM and eight HUBS matches. RP-to-P adds two HUBS exclusions, demonstrating interaction with guidance at tighter caps. Neither order identifies a causal economic decomposition.

Failed combinations remain informative. Coherent NOW guidance under the old annual floor has no operating domain because about 15.080% is below 15.340%; the common-floor change restores operating admission without changing the terminal lower bound. Historical HUBS guidance with fixed own caps is also empty because its guide lower, about 14.594%, exceeds the 13.808% upper restriction. Missing domains are not zero values or zero persistence estimates. Supplement Section S15 retains all cells, numerical differences and interaction brackets.

### 4.2 Non-R&D accumulation sensitivity

The completed eta experiment holds opening N0, K0, C0, fixed valuation targets/timing, H5 revenue, research spending/depreciation/tax, coherent guidance and all G/O/P restrictions fixed. It compares the exact serialized baseline coefficient A with acquisition-excluded B and zero-incremental C:

$$\eta_A = \text{saved opening stock-ratio calibration}, \eta_B = \frac{N_0 - GW_0 - AI_0}{R_0}, \eta_C = 0. \quad (10)$$

GW and AI are disjoint opening goodwill and the verified net acquired-intangible balance. Historical acquisitions remain in opening N0/C0; only future accumulation changes. B is an acquisition-excluded historical stock-ratio scenario, not corrected organic marginal investment. C holds N constant while research continues: an extreme diagnostic, not a realistic cash-flow forecast. All capital paths, average returns, valuation modes and complete future certificates were recomputed consistently.

**Table 4. Eta scenarios and conditional compatibility**

| Firm | Baseline eta A | Acquisition-excluded eta B | Zero-eta C: G / O / P |
|---|---|---|---|
| CRM | 1.371704 | -0.187742; N-stock domain fails | 5 /0 /0 |
| NOW | 0.215243 | -0.138650; N-stock domain fails | 0 /0 /0 |
| BOX | 0.144444 | Unavailable category net balance | 0 /0 /0 |
| HUBS | 0.032806 | -0.071522; N-stock domain fails | 27 /0 /19 |

*Notes:* Compare C counts with A's G/O/P counts in Table 3; each resolved cell covers 27 cases. A uses exact saved calibration, not newly rounded displayed values. B's negative values are not floored. HUBS's externally acquired scope includes separately purchased rights. Box's net intangible caption mixes internal software and acquired assets without category amortization. Source dates, balances and exact coefficients are in supplement Section S19 and separate reproduction records.

The prespecified pilot has four firms, two fixed corners and three cells:24 combinations per scenario. All 24 A baselines reproduced saved records. A classification change, a maximum-value difference wholly beyond 1% of EV, or changed topology/structural admission triggered expansion per firm/cell comparison; a threshold inside numerical uncertainty would also trigger conservative expansion. All 12 A/C comparisons expanded, as did nine computable B comparisons through analytic domain failures. The three Box B comparisons remained unavailable. Records comprise 324 A cases (24 reproduced, 300 reused),324 new C solves, 243 analytic B failures and six unavailable Box B pilot rows. No B valuation grid or repeated full baseline replay was needed. Related older median-three eta evidence was inspected rather than duplicated.

Zero eta adds exactly one CRM G match: LOW WACC, 3% growth, YEAR_END. The maximum rises from about $197.086bn to $200.568bn against the fixed $199.868bn target. Its supported rate set is [0, L] with L approximately 0.000561394 per year: a minimum spread half-life of about 1,235 years after T, with no finite upper bound because zero decay is admitted. This extreme conditional implication is neither a measured duration nor an investor belief. This maximum surplus is about $0.700bn ($699.853m before display rounding). The saved timing offset is 248/365 years from January 31 to October 6, 2026, about 68% of the first model year. YEAR_END retains the full year-1 flow and adjusts discount time; it does not remove elapsed operations. How that elapsed flow aligns with the July 31 claims snapshot is unresolved. No elapsed-flow correction or synchronized surplus is established. CRM O/P remain excluded; all other firms retain their individual classifications within these H5 controls. Values and some persistence bounds nevertheless move, and CRM G maximum changes can be negative or positive across cases.

**Table 5. Casewise maximum-value change under zero eta (% of fixed EV)**

| Firm | G | O | P |
|---|---|---|---|
| CRM | -6.04 to 8.96 | 9.38 to 14.26 | 3.02 to 10.89 |
| NOW | 1.72 to 2.76 | 1.64 to 2.73 | 1.62 to 2.64 |
| BOX | 1.16 to 1.94 | 1.25 to 2.03 | 1.40 to 2.02 |
| HUBS | 0.74 to 1.81 | 0.72 to 1.64 | 0.72 to 1.66 |

*Notes:* For each matched case, form the outer numerical bracket for C maximum minus A maximum and divide by that case's unchanged EV. Entries are the minimum lower and maximum upper endpoints across 27 cases, rounded outward to two decimals. They are neither differences between pooled extrema nor economic confidence intervals. Supplement Table S32 retains the corresponding case-family summaries and the separate reproduction records retain full precision.

For example, HUBS's base P-case minimum spread half-life changes from about 30.4 to 28.3 years after T under zero eta, while compatibility stays unchanged. Exact reciprocal brackets remain in supplement Section S19.

Sensitivity operates through valuation and admission. Opening N0 and C0 remain fixed. At a fixed endpoint margin, changing eta changes endpoint average capital and thus endpoint ROIC; the subsequent ROIC fade then implies different profit margins through the changed capital-to-revenue ratios. Investment deductions and both valuation-tail terms change as well. Removing incremental non-R&D book investment therefore need not increase value. All 324 saved A maximum witnesses remain admitted under C when T, transition shape, endpoint/guide margins and lambda are held fixed, but their values can rise or fall. These are fixed operating-coordinate comparisons, not identical ROIC or cash-flow paths. Reoptimizing those coordinates is a separate comparison (supplement Section S19.9). Nine HIGH-WACC CRM O cases that previously admitted only zero fade now admit positive-fade operating paths; none reaches valuation equality. These are joint controlled effects, not additive causal shares.

For negative eta and positive terminal growth, $N_j = N_0 + \eta(R_j - R_0)$ eventually becomes negative. The maintained nonnegative N-stock domain therefore fails for CRM, NOW and HUBS, even though the saved certificates show positive total C at the first N violation. Rejection is the component constraint $N_j \geq 0$, not a demonstrated total-capital failure at that year. A positive asymptotic C/R ratio does not establish C or average capital at every intermediate year; no complete-path positivity certificate for B is available. Negative eta is not attributed to a single accounting liability, and zero eta does not bracket the inadmissible B alternatives. Box B has no verified category net balance and no imputed coefficient. The findings show sensitivity within tested scenarios; they do not identify the correct investment policy or establish that acquisition accounting caused the exclusions.

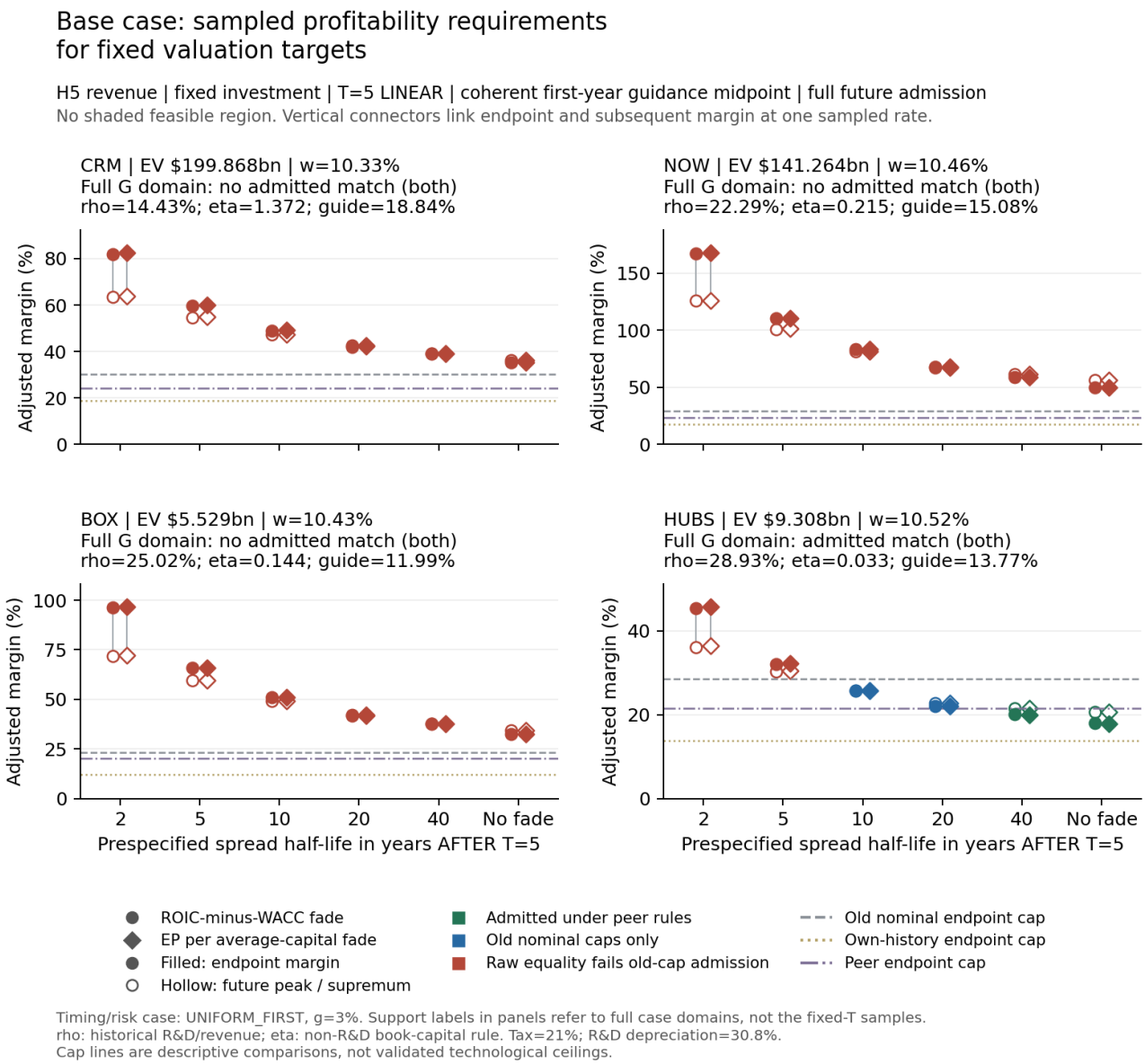
**4.3 Profitability benchmark and an assumption-requirement diagnostic**

A derived alternative fades economic profit per unit of average capital, $r_j - w\,C_{j-1}/\acute{C}_j$, rather than $r_j - w$. The break-even return differs during capital growth. Its corresponding valuation tail and full future-margin conditions were derived separately; retaining the original formulas after replacing only the benchmark would be inconsistent (supplement Section S16).

In the prespecified 24-pair H5 G/O/P base/favorable pilot, every compatibility classification is unchanged. The largest absolute maximum-value difference bound is about 0.7674% of EV. Neither that, classification changes nor the declared duration criterion triggered expansion, so no alternative full 27-case menu was run. This is a bounded null result, not general benchmark robustness.

The sampled diagnostic fixes revenue/risk/investment, coherent guidance, T=5 and LINEAR transition. It solves for endpoint profitability at half-lives of 2,5,10,20 and 40 years after T, plus zero decay, then distinguishes endpoint margins from subsequent peaks/suprema and applies full admission.

**Figure 2. Sampled base-case profitability requirements**




*Sampled profitability requirements*

*Notes:* Filled markers are endpoint requirements; hollow markers are subsequent peaks/suprema. Red raw equalities fail admission under the original nominal ceilings; blue admits those ceilings only; green admits peer rules. ROIC-spread and EP-rate benchmarks have distinct markers. These are sampled points, with numerical peak-boundary uncertainty retained in the records; no continuous feasible region is shaded. Historical/peer caps are descriptive assumptions, not validated technological limits.

For the original HUBS base sample at a 20-year spread half-life, the required endpoint margin is about 22.06% and the subsequent peak about 22.79%. It passes G but fails P. A compatible full case can still have rejected fixed-T samples, so the figure does not replace complete support calculations. The favorable figure, exact sample settings and all failures remain in supplement Section S17.

### 4.4 What the limited chronological evidence tests

**38 days of Workday's and 45 days of Docusign's FY2026 follow-up years had already elapsed at their FY2025 filing origins.** These are two descriptive annual endpoint comparisons, not wholly future annual-flow forecasts. Origins were March 11 and March 18, 2025; both episodes passed margin dependency verification under the unchanged maturity definition and target-excluded minimum-three-distinct-donor gates.

**Table 6. FY2026 endpoints relative to guidance-omitted components**

| Episode | Outcome margin (%) | History ceiling (%) | Outcome minus history (pp) | Peer component (%) | Outcome minus peer (pp) |
|---|---|---|---|---|---|
| Workday | 10.415 | 10.457 | -0.042 | 17.133 | -6.718 |
| Docusign | 12.697 | 10.765 | +1.932 | 17.015 | -4.317 |

*Notes:* Adjusted NOPAT/revenue percentages; signed percentage-point differences calculated before rounding. The history-only ceiling includes the origin and largest available own adjacent-year mean; the peer component additionally uses verified target-excluded level/headroom evidence. Both omit guidance. The episodes share donors and cannot estimate a coverage probability. Turnover and paired-hull restrictions were not tested.

Docusign exceeds the history component; Workday's non-exceedance is small. Both remain below the peer component. This does not validate full guidance-inclusive historical/peer support rule, two-year sustained levels or infinite-horizon admission. Broad validation stopped because the verified panel lacks origins combining the required earlier donor support with two subsequent annual observations. The separate earlier 12/16 exceedance result concerns overlapping non-mature history-only cases, not mature-peer or full-rule evidence; it remains unpooled in supplement Section S18.

## 5. Economic interpretation and limitations

### 5.1 Accounting equivalence does not establish distributable cash

The constructed FCFF/EP identity is internally consistent. Its correspondence to cash available to claimholders is unresolved. Book capital changes include consumption, acquisitions, working capital and noncash adjustments. The historical cash reconciliation makes this distinction visible.

**Table 7. Historical cash construction and a partial comparator**

| Firm / FY | Constructed FCFF ($m) | Partial reported-cash comparator ($m) | Constructed minus comparator ($m) |
|---|---|---|---|
| CRM /2026 | 701.844 | 14,402.000 | -13,700.156 |
| NOW /2025 | -911.047 | 4,576.000 | -5,487.047 |
| BOX /2026 | 9.012 | 315.202 | -306.190 |
| HUBS /2025 | -133.638 | 576.928 | -710.566 |

*Notes:* Annual flows in USD millions, not October forecasts. The partial comparator is consolidated CFO less PPE cash purchases and, for BOX/HUBS, separately disclosed capitalized-software cash. Other investment categories and CRM/NOW software deductions remain unknown. It is not distributable unlevered FCFF. Supplement Table S20 and Section S9 retain publication dates, components, source scope and exact arithmetic.

CFO adds back SBC and D&A, incorporates nonoperating and consolidated tax items, and reconciles working capital on its reported scope. Constructed profitability retains reported compensation cost, uses normalized adjusted-profit tax, and deducts changes in book operating capital. Acquisition, lease and remeasurement movements can affect that capital without matching selected cash purchases. The decomposition closes arithmetically; it is not an unexplained error or a set of independent favorable corrections. Annual differences cannot be added to EV.

The normalized-tax wedge and outstanding compensation claims require coordinated treatment. A research tax expression is not automatically an omitted usable tax shield; adding back SBC while separately adjusting dilution can double-count. Residual R&D overlap, initialization, ordinary gross investment and depreciation remain unresolved. Eta sensitivity makes the investment limitation more concrete without supplying the missing cash bridge.

### 5.2 Dates, risk and observed versus assumed restrictions

October prices, earlier shares/financial balances and annual operating origins are not one synchronized state. Public availability does not align those stocks and flows. Full-year guidance does not identify the genuinely future October remainder, and uniform timing does not repair missing elapsed operations. ServiceNow's reviewed interim NOA rises from $2.858bn to $11.253bn while the frozen annual origin remains earlier; that stock increase is not an additive correction to its valuation gap. A stronger current-market interpretation requires a separately coordinated operating/claim/risk/remaining-flow design.

WACC relies on common industry inputs, financing classifications and constant-risk assumptions. Discrete stresses do not identify an issuer's required return or a probability distribution. Unpriced awards, hybrid rights and private strategic assets also prevent an aggregate synchronized EV-error bound. Shared unresolved inputs mean repeated exclusions are not independent economic confirmations.

Historical margins and peer means are observations under selected vintages. Their extension into future ceilings is a maintained scenario restriction with unestablished empirical reliability. The chronological endpoint check and eta experiment do not validate hard future supports. Nor does conditional exclusion establish technological impossibility: earlier saved margin relaxations restore some Box matches. Choosing a cap or eta because it fits the price would destroy the independence of this audit.

### 5.3 How an analyst can apply the audit

First define the capital/income perimeter and target claims together. Then make the timing, growth and investment completion explicit; identify which profitability quantity fades. Locate equality for a completed path and admit that same path under every future restriction. Report a supported set, numerical enclosure, structural failure or unavailable input, rather than manufacturing a point estimate.

For support [0, L], the positive-rate spread half-life is at least ln(2)/L after T. Outer/inner rate endpoints give reciprocal numerical brackets in reverse order. Zero membership supplies no finite upper duration bound; empty support supplies no duration at all. Neither a duration bound nor a sample root identifies investors' beliefs, calendar time from October 6, or the lifetime of dollar EP.

This sequence is useful because it exposes what a valuation requires and why a result changes. It does not certify distributable cash, forecasting success, acquisition-caused mispricing or an economically correct investment policy. The case sample and model menu cannot establish SaaS-wide prevalence. Unresolved closest-method overlap prevents a methodological priority claim.

### 5.4 Conclusion

The fixed proxies supply conditional feasibility restrictions whose strength and existence depend on growth, capital completion and assumed profitability support. The Box witness shows why numerical equality is insufficient. Matched attribution places the main reversals in upper restrictions at coherent guidance, with a smaller guidance interaction. Eta sensitivity adds one Salesforce G match and changes values, bounds and operating admission, while several classifications survive the extreme zero-eta diagnostic. The tested alternatives do not select the correct investment policy. Mathematical consistency and reproducibility make these restrictions challengeable; they do not turn them into investor expectations or validated future operating limits.

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## Declaration of interests

The authors declare no financial conflicts of interest.

## Data and code availability

The source records, constructed data, code and case-level results are retained in replication materials that can be provided to the editor and reviewers. The verified baseline portable package predates later studies; those studies have separate implementation and verification records. No public repository or unified replay of the combined later materials is claimed. The technical supplement explains their scope and limitations. Software provenance is recorded in supplement Section S20.3: historical portable-replay receipts identify Python 3.9.6, while current document-preparation versions are reported separately and are not assigned retrospectively to later scientific runs.

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# Technical and Replication Supplement

*Companion to the revised article, October 15, 2026. All calculations below derive from completed records; this consolidation adds no valuation experiment or source collection.*

## Reading guide

Sections S1-S8 preserve the detailed literature, accounting, model, design and result catalogues from the prior manuscript. Sections S9-S18 retain the derivations, source/cash crosswalks, exact brackets and later evidence. Section S19 reports the completed eta study and identifies its separate full reproduction records. Section S20 retains the complete bibliography. The main article is the concise narrative. Main-article numbering is independent: equations, tables and figures here have an S prefix. CONTENT_MAPPING.csv gives original identifiers and locations, including consolidated summaries. Archival study descriptions retain their original scope; later studies do not silently enter the older portable ZIP. The preserved starting document is in baseline/.

### Archive label key

| Saved label | Meaning and record locator |
|---|---|
| D7 | Revenue-convergence comparison with recurrence and initial investment diagnostics; phase_d_step7/RESULTS.csv |
| D8 | Explicit investment completion with complete future-margin admission; phase_d_step8/RESULTS.csv, selection INVESTMENT_FULL_CEILING |
| D9 | Historical/peer operating-support calibration and scenario comparisons; phase_d_step9/CALIBRATION_EVIDENCE.csv and RESULTS.csv |
| B6 | Earlier investment-completion sensitivities; phase_b_step6/specification.json; C_INV_NOA_MEDIAN3 uses latest rho and median-three eta under older controls |
| C3 / C4 | Economic-measurement audit and its historical cash-gap decomposition; phase_c_step3/annual_cash_bridge.csv and phase_c_step4/historical_cash_gap_components.csv |
| step2 | Matched H5 cap/guidance attribution study; paper_usefulness_2026-10-13/step2_matched_attribution/ |

These are archive identifiers, not additional economic variables or independent samples. G/O/P and H/RP denote the operating cells defined in Sections S7.5 and S15; H3/H5 denote the first model year at terminal revenue growth. Paths are replication-record locators, not public download claims.

## S1. Related literature and contribution

### S1.1 Accounting valuation and profitability dynamics

The valuation foundation is established. Ohlson (1995) links equity value to accounting earnings and book value, while Feltham and Ohlson (1995) distinguish operating and financial activities within a clean-surplus framework. Penman and Sougiannis (1998) and Lundholm and O'Keefe (2001) examine the relation among dividend, cash-flow and residual-income valuation. These studies motivate consistent accounting and timing across representations. Our reconciliation of constructed free cash flow and economic profit is an application of that foundation, not a new valuation identity.

An accounting representation and a future profitability process are distinct objects. Dechow, Hutton and Sloan (1999) assess residual-income valuation and its information dynamics. Nissim and Penman (2001) organize profitability and forecasting through financial-statement ratios. Their work provides a basis for distinguishing operating margin from capital turnover and for examining how those components combine into returns. We use that decomposition to make the operating completion visible rather than to infer a mature endpoint from a contemporaneous accounting relation.

The study's persistence process remains a maintained hypothesis. It is not estimated as a common historical SaaS law. Earlier chronological component forecasts provide mixed evidence: short-horizon margin gains do not establish reliable joint multi-year ROIC predictions. Such evidence supports retaining uncertainty over operating states. It does not establish a general theorem that profitability cannot be forecast or that every alternative forecasting process would fail.

### S1.2 Market inversion and competitive-advantage duration

Inverse valuation is also established. Gebhardt, Lee and Swaminathan (2001) construct an implied cost of capital using valuation and earnings information. Hughes, Liu and Liu (2009) explain why an implied capital-cost quantity need not equal expected return under their stochastic-return setting. Their estimand differs from ours, but the broader interpretive distinction is relevant: solving an accounting valuation equation is not sufficient to establish the economic meaning of the resulting parameter.

Market-implied competitive-advantage periods and profitability adjustment have direct precedents. Mills (2005) examines growth assumptions in an IPO valuation case through a competitive-advantage-period approach. Vlčková and Buus (2025) compare models of market-implied profitability mean reversion. These references prevent the use of persistence inversion, a new parameter symbol or an industry application alone as an originality claim.

The closest identified study is Garcia-Estevez and Puente Vicente (2026), *When Does Value Creation Fade?* Its accessible author abstract describes a finite-duration persistence parameter, market inversion, boundary outcomes and external predictive content. Full methods and appendices were not accessible in the retained literature review and subsequent bounded access check. We therefore cannot establish whether its parameter is equivalent to our exponential rate, whether it treats unsuccessful inversions differently, or whether it already characterizes nuisance-domain and capital-policy effects.

That access limitation restricts the contribution statement. We make no claim that the competing study omits failed cases, future restrictions or investment consistency. Nor do we infer methodological differences from abstract-level terminology. The strongest current description is a documented conditional feasibility audit with firm-specific reversals. A priority claim or a claimed methodological extension would require an actual equation- and design-level comparison.

### S1.3 Intangible capital, growth and investment

Lev and Sougiannis (1996) examine R&D capitalization, amortization and value relevance. Hall and Mairesse (1995) address research-capital measurement in productivity analysis. Peters and Taylor (2017) bring broader intangible capital into investment-q analysis. These studies motivate attention to both the profit adjustment and the corresponding capital stock. Expensing research in one representation and capitalizing it in another changes the return denominator and the future investment completion as well as current earnings.

Our research stock uses the retained software depreciation estimate from Li and Hall's (2016) BEA working-paper version. The parameter is an industry-based estimate under its source assumptions, not an issuer-specific measured rate of obsolescence. Chinloy, Jiang and John (2020) provide additional context for depreciation and industry variation. A single transferred scalar does not validate every element of the company's research technology, and the study does not silently substitute a later publication version or capitalize selling and administrative expenses.

Growth also requires a joint earnings and investment interpretation. Chan, Karceski and Lakonishok (2003) provide relevant evidence on growth persistence and the difficulty of long-horizon extrapolation. We retain finite revenue convergence as a transparent scenario rather than a fitted permanent premium. The investment comparison asks whether revenue growth remains accompanied by coherent capital accumulation, not whether a faster revenue path can mechanically rescue price equality.

### S1.4 Conditional support and the incremental application

Tamer (2010) organizes partial identification around the information supplied by maintained assumptions. Masten and Poirier (2021) study falsified instrumental-variable models and minimum relaxations in a statistical setting. Those precedents establish that assumptions determine support, and that empty support or relaxation thresholds are not new concepts. Our deterministic equality to a financial proxy is not their statistical model and supplies no sampling confidence region.

The incremental evidence lies in the application: the changing case counts and gap magnitudes, the rejection of an inadmissible Box match, the interaction of growth with capital accumulation, and the inability of historical operating evidence to validate future hard supports. The numerical apparatus allows those claims to be challenged without forcing an implied estimate for every firm. Its value is auditability and a concrete demonstration of inferential dependence, rather than a new general identification theory.

The bibliography and Section S14 record literature access scope; no empirical coefficient or detailed methodological claim is attributed to an uninspected study.

## S2. Data and accounting construction

### S2.1 Selection, sources and information timing

The historical collection is purposive. An initial roster of 14 enterprise SaaS and related software candidates yields 11 firms and 56 usable core firm-years; an expanded roster of 23 yields 15 firms and 72 usable core firm-years. The corresponding consecutive-year transition counts are 45 and 57. These stages are distinct from the final four-firm valuation exercise. Eligibility is governed by profit/capital inputs, adjacent balance compatibility, positive average adjusted capital, source reconciliation and reviewed accounting scope, rather than by whether a market inversion succeeds.

The final cases are Salesforce, ServiceNow, Box and HubSpot, denoted CRM, NOW, BOX and HUBS. Their approved histories provide annual accounting origins and a forward GAAP operating anchor or a saved reconciliation. A firm lacking those prerequisites is construction-blocked, not assigned zero persistence. The selected firms cannot identify SaaS-wide prevalence or causal differences in competitive advantage.

The initial usable panel (56 firm-years) comprises CRM, DDOG, DOCU, HUBS, NET, NOW, OKTA, SNOW, VEEV, WDAY and ZS, as recorded in the frozen saas_research_panel_usable.csv. The expanded calibration vintage comprises BOX, CRM, DDOG, DOCU, HUBS, NET, NOW, OKTA, PATH, SMAR, SNOW, VEEV, WDAY, ZM and ZS. Its 72 observations are the OBSERVATION records in phase_d_step9/CALIBRATION_EVIDENCE.csv, using the study's saas_transition_pit_observations.csv; they are not a union of unrelated folders. The later chronological feasibility inventory records the same expanded membership and 72 rows, with its own availability checks (Section S18). PATH contributes only one usable year, so panel membership alone does not establish a consecutive transition or donor eligibility.

Ticker key: DDOG = Datadog; DOCU = Docusign; NET = Cloudflare; OKTA = Okta; PATH = UiPath; SMAR = Smartsheet; SNOW = Snowflake; VEEV = Veeva Systems; WDAY = Workday; ZM = Zoom; ZS = Zscaler. CRM/NOW/BOX/HUBS denote the four valuation firms defined above. These rosters distinguish retained usable observations from the larger candidate rosters.

The annual/interim source bibliography cites Salesforce (2026), ServiceNow (2026), Box (2026) and HubSpot (2026). The required-return inputs use U.S. Department of the Treasury (2026) and Damodaran (2026); Internal Revenue Service (Publication 542) supplies statutory tax context, not issuer cash-tax measurement. Source details are in Section S20.2.

Primary records retain fiscal dates, source URLs, accessions, filing vintages, units, locators and checksums where available. Missing observations remain missing. Structured SEC facts support retrieval and corroboration but do not alone validate the disclosure scope. Later comparative accounts can enter the approved history under their recorded availability dates; the archive is not a complete reconstruction of every earlier information set.

The financial cutoff is October 6, 2026. Annual operating origins precede that date, while the market proxies use the October 6 closes and older disclosed financial balances and shares. The information was public before the price date, but its public availability does not synchronize the underlying stocks and flows. This is a fixed, mixed-date proxy experiment. The controlled comparisons preserve those origins and proxies so that operating-completion changes are not confounded with an unimplemented valuation-date reconstruction.

**Table S1. Accounting characteristics at the approved annual origin**

| Firm | Origin year | Revenue | Reported NOA | R&D capital | Adjusted capital | NOPAT margin (%) | Turnover | ROIC (%) |
|---|---|---|---|---|---|---|---|---|
| CRM | 2026 | 41.525 | 56.960 | 16.038 | 72.998 | 18.75 | 0.598 | 11.21 |
| NOW | 2025 | 13.278 | 2.858 | 6.968 | 9.826 | 17.85 | 1.622 | 28.96 |
| BOX | 2026 | 1.177 | 0.170 | 0.810 | 0.980 | 9.95 | 1.271 | 12.65 |
| HUBS | 2025 | 3.131 | 0.103 | 2.022 | 2.125 | 10.51 | 1.654 | 17.37 |

*Notes:* Revenue and capital amounts are USD billions. Adjusted capital includes the research R&D stock. Margin and ROIC are constructed research measures under the common tax and depreciation convention. Turnover uses average adjusted capital, while the capital columns report ending stocks. Fiscal years ending in January are the issuers' fiscal-year labels. Sources are the approved annual accounting records and reconciled operating-origin archive.

**S2.2 Operating capital and research profit**

Historical annual observations use $t$; forward years use $j$, with $j=0$ the approved fiscal accounting origin and $j=1$ the guidance fiscal year. WACC $w$ is the annual weighted average cost of capital; $\Delta C_j = C_j - C_{j-1}$ is constructed net book investment. Define forward adjusted turnover $k_j = R_j / \acute{C}_j$, its reciprocal $\Gamma_j = \acute{C}_j / R_j$, and $M$ as the declared all-future adjusted NOPAT-margin ceiling for a support family. These definitions apply throughout; historical counterparts carry t subscripts. Let $N_t$ denote reported-input net operating assets. Common book equity $B_t$, applicable noncontrolling interests $NCI_t$, disjoint financing obligations $FO_t$, and classified financial assets $FA_t$ give

$$N_t = B_t + NCI_t + FO_t - FA_t. \text{(S1)}$$

Historical financing classifications use carrying amounts and distinguish identified debt, finance leases and applicable redeemable claims. Operating leases, ordinary payables, contract liabilities and operating accruals remain operating. Goodwill, acquired intangibles, recognized software and contract-cost assets remain within operating capital. Agreement of the balance-side and financing-side construction supports extraction/classification consistency, not a uniquely measured economic asset base.

Let $I_t^K$ be eligible expensed R&D and $K_t$ the research knowledge-capital stock. The maintained annual law and its amortization are

$$K_t = (1-d)K_{t-1} + I_t^K,\ A_t^K = d\,K_{t-1},\ d = 0.308. \text{(S2)}$$

The stock starts from assumed zero before the longest verified expenditure backfill. A missing expenditure breaks the chain rather than restarting it at zero. Specified opening-stock alternatives and fiscal-stub treatment assess initialization sensitivity. Passing those checks does not measure all economically omitted research or prove that the beginning stock has no error.

Reported operating income $O_t$ becomes adjusted pretax profit $O_t^* = O_t + I_t^K - A_t^K$. With normalized tax parameter $\tau = 0.21$,

$$Y_t = O_t^* - \tau\, max(O_t^*, 0). \text{(S3)}$$

Positive adjusted profit receives normalized tax and losses receive no immediate tax benefit. The statutory-rate context does not make this consolidated operating cash tax. It also does not determine eligibility, jurisdiction, deferred tax basis or future loss utilization. SBC remains an expense except for the timing change of its eligible R&D component; no general SBC addback is imposed.

Adjusted invested capital, average capital, NOPAT margin, turnover and ROIC are

$$C_t = N_t + K_t, \acute{C}_t = \frac{C_{t-1} + C_t}{2}, m_t = \frac{Y_t}{R_t}, k_t = \frac{R_t}{\acute{C}_t}, r_t = \frac{Y_t}{\acute{C}_t} = m_t k_t. \text{(S4)}$$

These definitions align the profit adjustment with the capital denominator. They remain a restricted accounting construction: SG&A is not capitalized, unquantified research/acquired-technology overlap is not assumed absent, and two-balance averaging does not reconstruct acquisition timing. The paper consequently uses "R&D-adjusted" rather than comprehensive intangible-adjusted capital.

**S2.3 Market-value proxy and required return**

The maintained operating-value restriction is

$$P = p_{Oct6} n_{disclosed} + Q^{proxy} + D^{proxy} - F^{proxy}. \text{(S5)}$$

Here $Q^{proxy}$ denotes a separately recognized preferred claim, $D^{proxy}$ other financing claims, and $F^{proxy}$ classified financial assets. Shares are disclosed outstanding counts, not diluted-EPS period averages. Some claim values are disclosed fair-value observations; others are carrying or conversion-parity proxies. Their dates and scopes are preserved.

**Table S2. Market inputs and retained guidance revenue**

| Firm | Close (USD) | Shares (millions) | Share date | Balance date | EV proxy (USD bn) | Base WACC (%) | Guided revenue scale (USD bn) |
|---|---|---|---|---|---|---|---|
| CRM | 224.99 | 823 | 2026-08-20 | 2026-07-31 | 199.868 | 10.33 | 46.250 |
| NOW | 137.97 | 1033.862 | 2026-06-30 | 2026-06-30 | 141.264 | 10.46 | 16.254 |
| BOX | 35.27 | 137.200773 | 2026-07-31 | 2026-07-31 | 5.529 | 10.43 | 1.290 |
| HUBS | 217.09 | 49.866201 | 2026-07-31 | 2026-06-30 | 9.308 | 10.52 | 3.682 |

*Notes:* The price date is October 6, 2026 for every firm; share and balance dates shown are earlier. CRM's share count is approximate. Guided revenue is the unchanged full-year midpoint or documented proxy. NOW's subscription guidance is translated to total revenue using the retained historical subscription share. It is not issuer total-revenue guidance. Sources are the dated quote/claim records, quarterly disclosures, issuer guidance and saved risk construction.

The required-return construction uses common industry-risk inputs, including a cash-corrected unlevered software beta, a nominal Treasury rate, an equity-risk premium and a debt-spread proxy. Financing weights produce the firm-specific base WACCs in Table S2. The maintained construction and subsequent case menu do not independently measure issuer required returns or a probability distribution for risk.

Financial rights that are not measured include current awards and, where relevant, ASR or capped-call rights and private strategic asset values. A convertible-note fair value already contains option value; adding fully converted shares while retaining the entire note value would double-count that claim. Disclosed award counts, grant-date expense, contractual capacity and repurchase authorizations cannot be substituted for current economic claim values.

The proxy is held fixed across comparisons to isolate changes in operating completion. Section S8.3 discusses the unresolved economic consequences of its mixed-date measurement.

## S3. Valuation and conditional identification

### S3.1 Capital completion and the valuation object

Forward annual coordinates are indexed by $j$, with $j=0$ the approved historical origin and $j=1$ the guidance fiscal year. The benchmark uses $R_1$ followed immediately by terminal revenue growth $g$. For an imposed turnover path, it completes capital through

$$R_j = R_1(1+g)^{j-1}, \acute{C}_j = \frac{R_j}{k_j}, C_j = 2\acute{C}_j - C_{j-1}. \text{(S6)}$$

The equation exactly enforces the declared average-capital relation. It is not an asset-level investment budget. Once turnover stabilizes, its homogeneous root −1 retains an alternating capital component. Removing that component while claiming to retain equation (S6) would change the model.

Constructed FCFF and beginning-capital economic profit are

$$F_j = Y_j - (C_j - C_{j-1}), EP_j = Y_j - wC_{j-1}. \text{(S7)}$$

For year-end valuation at the annual origin and finite horizon $H$,

$$\sum_{j=1}^{H} \frac{F_j}{(1+w)^j} = C_0 + \sum_{j=1}^{H} \frac{EP_j}{(1+w)^j} - \frac{C_H}{(1+w)^H}. \text{(S8)}$$

Transversality removes the last term in the infinite representation. Its omission before taking that limit would be an accounting error. The identity concerns the same constructed income and capital sequence in both representations. It does not prove that $F_j$ is distributable cash or that every book-capital reduction supplies a cash receipt.

Because ROIC uses average rather than beginning capital,

$$EP_j = (r_j - w)\acute{C}_j + \frac{w}{2}(C_j - C_{j-1}). \text{(S9)}$$

Thus average-capital ROIC equal to WACC does not generally imply zero dollar EP during capital growth. The selected spread benchmark, charge timing and capital law jointly define the economic-profit completion.

### S3.2 Persistence and exact conditional support

Let $T$ be the profitability-transition endpoint and $s_T = r_T - w$. The tail law is

$$r_{T+n} - w = s_T e^{-\lambda n}, n \geq 1, \lambda \geq 0. \text{(S10)}$$

For positive $s_T$, greater lambda reduces the future spread; for negative $s_T$, it accelerates upward convergence to WACC. At lambda zero the endpoint spread is constant. For positive lambda the spread's half-life is $\log 2 / \lambda$, measured after $T$, not from the market date or as a lifetime of dollar EP.

Let $c$ index a maintained WACC/growth/timing case, and $\theta$ the admitted operating quantities and discrete transition choices. Capital is independent of lambda once its completion and operating parameters are fixed. Positive future average capital and positive valuation weights then yield

$$V_c(\lambda, \theta) = D_c(\theta) + s_T(\theta) K_c(\lambda, \theta), K_c > 0, \frac{\partial K_c}{\partial \lambda} < 0. \text{(S11)}$$

$D_c$ includes transition profits, capital movements and the zero-spread tail. The kernel discounts actual post-transition average capital with the appropriate fade offset. Its form changes with revenue and investment completion; a baseline geometric inverse must not be imposed on a longer finite prefix.

Operating admission may itself depend on lambda, especially under full future-margin ceilings. Write $\Theta_c(\lambda)$ for the complete admissible domain and define

$$\Lambda_c(P)=\{\lambda\in[0,\infty):\exists\,\theta\in\Theta_c(\lambda),V_c(\lambda,\theta)=P\}.\text{(S12)}$$

Infinity is a limit, not a parameter member. An empty set means that the model, domain and proxy cannot be reconciled under that case. A nonempty nonsingleton set gives conditional partial identification. It is not a statistical confidence region: the exercise supplies no population distribution, likelihood or calibrated sampling uncertainty.

**S3.3 Fixed-path inversion and the margin frontier**

Under the benchmark geometric completion, valuation can be reduced further to

$$V_c(\lambda,\theta)=D_c(\theta)+E_c(\theta)h_c(\lambda),h_c(\lambda)=\frac{1}{\exp(\lambda+\delta_c)-1},\delta_c=\log\frac{1+w_c}{1+g_c}>0.\text{(S13)}$$

The coefficient $E_c$ carries the signed spread premium and timing scale. At zero lambda, $h_c(0)=(1+g_c)/(w_c-g_c)$; at infinite decay it tends to zero. For a fixed nondegenerate path, equality is attainable precisely when $z=(P-D_c)/E_c$ belongs to $(0,h_c(0)]$, and then

$$\lambda=\log(1+1/z)-\delta_c.\text{(S14)}$$

There is at most one finite nonnegative root when $E_c\neq0$, irrespective of its sign. If $P=D_c$ on a nonconstant path, only the limiting value is equal and no finite root exists. If $E_c=0$, equality admits every finite lambda when $P=D_c$ and none otherwise. Numerical degeneracy is not defined by an arbitrarily small dollar variation.

At fixed capital/revenue completion, the transition is affine in terminal NOPAT margin $m$ and first-year margin $b$. A useful general representation is

$$V=A_0+A_m m+A_b b+(mR_T-w\acute{C}_T)H(\lambda),H(\lambda)=\frac{K(\lambda)}{\acute{C}_T}.\text{(S15)}$$

Where the denominator is nonzero, equality gives the conditional margin frontier

$$m(\lambda,b)=\frac{P-A_0-A_b b+w\acute{C}_T H(\lambda)}{A_m+R_T H(\lambda)}.\text{(S16)}$$

The resulting margin must satisfy all transition and future restrictions. The frontier displays observational substitution within a maintained completion; it does not estimate a production technology or prove that its margin combinations are economically credible. Equation (S16), rather than a newly sampled frontier surface, provides the analytical relationship used in this paper.

**S3.4 Topology, extrema and convergence**

At fixed lambda and capital parameters, affinity permits valuation extrema over a convex margin-guidance polygon to be evaluated at its vertices. Benchmark turnover enters completed capital through reciprocal linear paths; continuous optimization therefore requires turnover-domain coverage, including endpoints and stationary points where relevant. A denser sampled grid is not a proof of a global envelope.

For a compact connected lambda-independent component with continuous valuation, a strict below-price maximum infinite-limit value and a common below-price zero witness, any matching path has a positive spread. If some zero-fade path reaches or exceeds price, the component's projected support is $[0, L]$ for a finite attained $L$. Continuity supplies equality between an above-price path and the below-price witness at each rate through the largest root. Section S10 states the proof and the premises needed for this result.

Those premises cannot be transferred mechanically to a lambda-dependent domain. A constrained maximum can change through both fixed-path fade and relaxation of admission. Some operating components are empty at zero but become nonempty later. A ceiling below the positive-fade asymptotic margin can instead admit only zero fade. If every admitted endpoint spread is negative, the largest value can be an unattained infinite-decay supremum. Each regime needs explicit treatment.

We retain componentwise support and numerical inner/outer enclosures where an analytical sharp endpoint is not obtained. No supported interval is inferred from sampled roots alone, and unresolved boundary cells remain uncertainty rather than being silently declared feasible or excluded. All nonempty reported benchmark sets satisfy the baseline theorem's premises; later results carry their own coverage qualifications.

Long-run growth below WACC, positive capital and a convergent capital completion ensure valuation transversality in the maintained positive-growth domain. Finite periods of revenue growth above WACC do not cause an infinite-horizon divergence once the terminal tail converges. Capital positivity is an admission condition, not evidence of realistic gross investment or asset productivity.

### S3.5 A matched normalized-economic-profit benchmark

Equation (S9) records a convention distinction, not an established algebra error. The maintained spread subtracts WACC from average-capital ROIC, while beginning-capital EP charges the capital present at the start of each year. A positive capital increment can therefore produce positive EP even at zero maintained ROIC spread.

The alternative defines $e_j = EP_j / \acute{C}_j = r_j - wC_{j-1} / \acute{C}_j$. At the same transition endpoint T and margin m, $e_T = (mR_T - wC_{T-1}) / \acute{C}_T$. It imposes $e_{T+n} = e_T e^{-\lambda n}$, so subsequent NOPAT is $wC_{T+n-1} + e_T \acute{C}_{T+n} e^{-\lambda n}$. The capital path and operating-margin path through T are unchanged for the same operating witness, but post-T profit and future-margin admission change necessarily. Reoptimized maxima/equality paths may select different witnesses from that matched transition menu.

This parameter fades beginning-capital EP per unit of average capital. Its half-life is again ln(2)/lambda for a nonzero spread and positive lambda, after T; it concerns a different estimand from equation (S10). Zero rate has no finite upper half-life, zero spread has no amplitude to halve, and an infinite-decay limit is not a parameter member. Declining EP per capital does not imply declining dollar EP when capital grows.

The alternative tail and complete future-margin inequalities are derived in Section S16. Its zero-EP FCFF tail has a capital-value term $C_J$, rather than the original average-ROIC zero-spread modal base. Substituting only the faded quantity while retaining the old valuation or admission formulas would compare incompatible constructions. Section S7.6 reports a bounded matched pilot rather than silently replacing the original full-menu findings.

## S4. Empirical design

### S4.1 Maintained valuation cases and comparison logic

Each firm is evaluated in 27 cases: base WACC shifted by −100, 0 or +100 basis points; terminal revenue growth of 2%, 3% or 4%; and YEAR_END, UNIFORM_FIRST or UNIFORM_ALL payment timing. A LOW label refers to a firm-specific shifted WACC, not the same rate for all firms. These alternatives are neither probabilities nor estimated confidence distributions.

Year-end timing values full modeled annual flows at their endpoints. The uniform conventions retain the benchmark's stylized allocation of first-year and subsequent flows. Elapsed time is the calendar-day fraction from the annual origin to the market date. Full-year guidance is not decomposed into realized and remaining October flows in these experiments. That unresolved clock is preserved in every matched comparison rather than silently repaired while changing operating supports.

Transition horizons $T = 2, \ldots, 6$ and linear/normalized-exponential margin shapes remain maintained menus. For $1 \le j \le T$,

$$m_j = b + (m - b) f\left(\frac{j-1}{T-1}\right), \quad \text{(S17)}$$

with $f(x) = x$ or $f(x) = (1 - e^{-2x}) / (1 - e^{-2})$. The origin is separately fixed. The normalized curvature is not another estimated persistence parameter. In the benchmark, turnover interpolates from the observed average-capital ratio to its declared endpoint. Under investment completion it is derived from capital instead.

The comparisons change specified operating completions while preserving the annual accounting origins, market proxies, tax parameter, required-return cases and timing. A growth-only contrast recomputes capital under the old recurrence. An investment contrast changes the accumulation law and the mathematically required turnover interpretation. The original historical-support contrast changes profitability restrictions and the guidance bridge. A separately prespecified H5 comparison in §S7.5 now holds guidance and admission conventions matched while changing upper restrictions, and documents the necessary floor repair. Its contrasts are controlled model effects, not a causal or additive decomposition in the economy.

The benchmark is retained as a historical reference, not as a preferred capital technology. Its recurrence exactly enforces the imposed average-capital turnover relation; evaluating it reveals how that relation and its alternating component enter earlier valuations. Discarding the benchmark would conceal the source of changes after repair. The investment completion is a mathematical and accounting improvement in the stock-flow relationship, subject to unvalidated investment coefficients. The own-history rule asks whether sustained performance within a firm's history suffices; the peer rule admits descriptive multi-year expansion and mature levels from other firms. Neither rules out better future performance. These comparisons examine specification dependence without ranking alternatives by price compatibility.

### S4.2 Original operating restrictions

The first-year interval combines public GAAP operating-margin guidance with recent historical net R&D-adjustment intensities. The terminal lower margin is the recent minimum. The upper margin is the larger of current margin and the guidance upper margin, plus a firm-balanced peer allowance based on other firms' largest observed positive single-year improvements. The percentile and extension rule are assumptions about future support, not confidence bounds.

**Table S3. Benchmark operating restrictions**

| Firm | Terminal margin range (%) | First-year margin range (%) | Terminal turnover range |
|---|---|---|---|
| CRM | 14.27-30.31 | 18.77-18.96 | 0.543-0.638 |
| NOW | 15.34-29.20 | 14.90-16.53 | 1.622-1.622 |
| BOX | 7.72-23.10 | 11.21-11.87 | 1.271-1.271 |
| HUBS | 5.38-28.58 | 14.59-17.23 | 1.654-1.654 |

*Notes:* Margins are adjusted NOPAT margins, not GAAP or non-GAAP operating margins. These intervals intersect annual transition-margin and annual-change constraints; in particular, a guidance interval can extend below a transition floor and still be restricted by that floor. Turnover is revenue divided by average adjusted capital. The upper margin is also the maintained future ceiling in the benchmark. Values are rounded; full inequality records are retained.

The absolute annual transition-margin change cap is approximately 13.527 percentage points. It applies to the move from the origin and between transition periods. The historical lower margin and change restrictions define the transition; they are not extended into an additional infinite-tail floor. Post-transition positive ROIC and the declared all-future upper ceiling are maintained. Imposing a historical minimum forever would be a different model and could conflict with the WACC-based spread limit.

Recent turnover rules produce singleton domains for ServiceNow, Box and HubSpot. The collapse reflects their observed recent changes and the specified extrapolation rule. It does not demonstrate known future efficiency. Historical ratios vary with acquisitions, deferred revenue, research-stock accumulation and the capital classification itself.

**S4.3 Revenue reconstruction**

The first-year scale $R_1$ remains the original full-year guidance midpoint or proxy. Let $a_0$ be an independently specified initial post-guide growth anchor, $H_R$ the first model year at terminal growth, and $u_j$ log gross growth. For $j \geq 2$ and $H_R \geq 3$,

$$u_j = \log(1+g) + [\log(1+a_0) - \log(1+g)]\, max\left\{0, 1 - \frac{j-2}{H_R - 2}\right\}, R_j = R_{j-1} e^{u_j}. \text{(S18)}$$

The guidance anchor sets $a_0 = R_1 / R_0 - 1$; the history anchor uses the median of the three latest approved annual revenue-growth observations. We examine $H_R = 3$ and $5$ separately. The immediate-growth benchmark is an explicit $H_R = 2$ branch. No price or solved lambda enters anchor construction.

These choices ensure positive revenue and an exact terminal tail after a finite prefix. They do not estimate the convergence horizon. For the four firms, guide-implied anchors are approximately 11.379%, 22.409%, 9.577% and 17.588%; recent-three medians are 9.579%, 22.439%, 5.048% and 21.072%. The near equality of ServiceNow's two anchors is arithmetic, not evidence of precise medium-term expectations. Acquisition and FX scope remain important to interpreting all anchors.

The safe tail boundary is $J = max(T, H_R)$. Spread fade starts at $T$, even if revenue is still converging, and the actual states at $J$ are carried into the terminal tail. Resetting the fade clock or terminal revenue level at $J$ would change the comparison.

**S4.4 Investment completion and terminal consistency**

The alternative capital law starts from the same reported-input $N_0$ and R&D $K_0$. With $v = 1 - d = 0.692$, fixed historical spending share $\rho$ and non-R&D intensity $\eta$,

$$N_j = N_{j-1} + \eta(R_j - R_{j-1}), K_j = v\, K_{j-1} + \rho\, R_j, C_j = N_j + K_j. \text{(S19)}$$

Net book investment is

$$\Delta C_j = \eta\, \Delta R_j + \rho\, R_j - d\, K_{j-1}. \text{(S20)}$$

This is an explicit accumulation relationship. Greater revenue entails larger non-R&D additions and larger R&D spending under the fixed policy. The law does not introduce a common depreciation rate for ordinary capital or independently measured non-R&D gross investment.

The retained eta is the latest reported NOA/revenue stock ratio, approximately $N_0 / R_0$, rather than a measured incremental cash-investment efficiency. The near proportionality $N_j \approx \eta\, R_j$ therefore reflects the chosen completion. Rho is an observed R&D share extended into a future spending assumption. Acquisitions, asset consumption, working-capital cash, noncash stock changes and productive returns are not independently validated by those anchors.

The historical capitalization and tax convention is identical across comparisons. The original recurrence completes total capital only; explicitly completing future R&D spending and its stock is an inseparable part of the investment alternative, not a cash-investment forecast already established by the benchmark.

For terminal gross revenue growth $a = 1 + g$, the completion gives

$$c_\infty \equiv \frac{\lim C_j}{R_j} = \eta + \frac{\rho a}{a - \nu}, \frac{\lim \Delta C_j}{R_j} = \frac{g}{a} c_\infty, \Gamma_\infty \equiv \frac{\lim \acute{C}_j}{R_j} = \frac{1+a}{2a} c_\infty. \text{(S21)}$$

The limit explicitly requires continuing investment for positive terminal growth. Turnover tends to $1 / \Gamma_\infty$. As revenue growth decelerates while the R&D spending share is held constant, the research stock can rise relative to annual revenue; a lower turnover than recent history is then a consequence of the accumulation assumption, not a free efficiency gain.

**S4.5 Complete admission and historically motivated alternatives**

For every future period after $T$, let $\Gamma_{T+n} = \acute{C}_{T+n} / R_{T+n}$. Complete future-margin admission requires

$$\Gamma_{T+n}\{w + s_T e^{-\lambda n}\} \le M \text{ for every integer } n \ge 1. \text{(S22)}$$

Capital positivity, the accumulation identities, investment feasibility, guidance/transition restrictions and terminal convergence also apply. A numerical valuation root cannot waive any of these conditions. The original fixed-turnover fade automatically bounds future margins between $m$ and $w / k_T$ when both lie below the ceiling; the investment law changes the capital/revenue ratio and requires a new full-tail check.

“Complete” refers to the declared model restrictions, not an exhaustive empirical characterization of the firm’s operating technology, financing capacity or cash-investment requirements.

Two historical profitability families are specified without price calibration. Sustained own-history support uses adjacent-year mean adjusted margins, with a hull containing coherently translated current guidance. The peer family uses a declared maturity diagnostic, donor-firm medians of eligible two-/three-year expansions and mature two-year margin means, then intersects level and headroom restrictions. The optional paired screen restricts terminal margin at the actual derived turnover through an empirical endpoint hull. Historical observations motivate these maintained future restrictions; their empirical reliability is unestablished. Section S7 distinguishes the valuation scenarios from the limited operating-component evidence.

The coherent first-year bridge under the fixed R&D policy uses

$$b = \Pi_\tau\left(q + \rho - \frac{d K_0}{R_1}\right), \Pi_\tau(x) = x - \tau\, max(x, 0), \text{(S23)}$$

where $q$ is the public GAAP operating-margin anchor at the fixed revenue scale. A saved GAAP operating-dollar range is translated at that same scale where available. This changes the operating restriction while preserving spending and taxes. It does not turn the historical spending assumption into issuer R&D guidance. In the matched H5 design, the old nominal terminal/future caps are retained in a coherent-guidance comparator. Cap alternatives then share that guidance and the common transition floor, while the original terminal lower bound remains fixed. Failed combinations are recorded instead of expanding restrictions to manufacture a factorial design (Section S15).

**S4.6 Numerical evidence and reporting**

The benchmark supports are derived from continuous margin-guidance and turnover domains, with discrete transition components retained. Revised completions use positive-capital kernels, exact finite prefixes and

convergent tail sums. Full future ceilings are covered by annual inequalities and a bound on every omitted tail year, not by a finite simulation alone.

The saved numerical work separates operating admission, exact-equality feasibility, numerical convergence and limiting values. Independent constructed-FCFF/EP streams and boundary witnesses support the computations. Later maximum valuations are global inner/outer brackets, not local-optimizer outputs. Endpoint and value precision do not quantify the missing economic measurement error.

The theory concerns $V = P$. Numerical equality witnesses use a residual criterion below \$0.001; full-domain envelope coverage uses a \$0.02 numerical guard in the revised calculations. These are computational controls, not an economic band around price. Long-form results, component coverage and exact serialized brackets are retained in the replication records. Selected tables display fewer digits for economic readability.

## S5. Historical benchmark results

### S5.1 Compatibility and conditional persistence

The immediate-growth/average-capital benchmark gives the four-firm pattern in Table S4. Salesforce is compatible only in four favorable cases; ServiceNow and Box cannot attain their proxies anywhere in their declared operating domains; HubSpot is compatible throughout. This classification concerns equality somewhere in the complete domain, not the percentage of sampled paths with roots.

**Table S4. Benchmark compatibility and persistence support**

| Firm | Compatible / 27 | Base-case set | Cross-case upper endpoint | All-case common set |
|---|---|---|---|---|
| CRM | 4/27 | Empty | ≈0.011984 | Empty |
| NOW | 0/27 | Empty | Empty | Empty |
| BOX | 0/27 | Empty | Empty | Empty |
| HUBS | 27/27 | [0, ≈0.058189] | ≈0.120578 | [0, ≈0.019341] |

*Notes:* The base case uses BASE WACC, g=3% and UNIFORM_FIRST timing. An upper endpoint summarizes a set starting at zero, not a point estimate. Cross-case projection and all-case intersection are distinct diagnostics. Empty is not lambda zero. Endpoints are rounded numerical approximations to conditional endpoints; they are not economic confidence limits.

Salesforce’s compatible cases all use low WACC. Its base-WACC case remains excluded. The original cross-case upper rate is approximately 0.011984. HubSpot’s base-case upper rate is approximately 0.058189 and its cross-case upper rate 0.120578. Its common-rate intersection across all 27 alternatives ends around 0.019341.

The cross-case union answers whether some maintained valuation case can reconcile a rate. The common intersection asks whether a rate can be reconciled in every case, potentially through different operating witnesses. It does not identify one common operating forecast fitting all cases, and it is not required if the cases are alternatives rather than simultaneous truths.

Zero membership means that an admitted constant-spread path can match the proxy. It does not establish perpetual actual rents. The finite upper endpoint is obtained only after checking the strict limit exclusion, an above-price path and a common below-price witness under the benchmark theorem. Merely beginning the parameter domain at zero would not produce that result.

### S5.2 Gap magnitudes and governing restrictions

At the base case, benchmark maximum values are approximately \$166.247bn for Salesforce, \$60.567bn for ServiceNow, \$3.645bn for Box and \$13.362bn for HubSpot. Against their fixed proxies, Salesforce, ServiceNow and Box have shortfalls of approximately \$33.621bn, \$80.697bn and \$1.884bn. The smallest benchmark shortfalls across the excluded ServiceNow and Box cases are \$55.830bn and \$0.458bn.

These distances distinguish a large conditional exclusion from one close to a boundary. They do not supply a price target or establish robustness to an unbounded aggregate measurement error. A book-capital change or annual cash-flow reconciliation difference cannot be subtracted from a valuation gap as though it were a coordinated present-value correction.

Benchmark upper-rate configurations select the admitted terminal and guidance upper margins; Salesforce also selects its governing turnover upper endpoint. Much of that activity follows from positive affine profit coefficients and admission of the relevant corner. Binding is therefore explanatory background. The substantive information is the cap's construction, its credibility and the gap or endpoint it permits.

The benchmark findings already depend on profitability beyond observed history. Earlier own-history terminal caps exclude every firm, and a screened mature-peer cap leaves only HubSpot, in 22 cases. Illustrative margin expansions instead restore some Box cases. These saved controls establish that benchmark exclusion is not a universal impossibility claim. Extended comparisons are retained in Section S13 rather than used to choose a favorable primary specification.

## S6. Revenue growth and investment consistency

### S6.1 Gradual revenue convergence

Growth reconstruction asks whether the common benchmark's rapid revenue convergence is responsible for its feasibility conclusions. All other declared accounting, risk, timing and operating restrictions are retained in this first comparison, and capital is recomputed rather than held fixed. Table S5 reports each anchor/horizon family separately.

**Table S5. Growth completion and compatibility**

| Revenue completion | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|
| Immediate terminal growth | 4/27 | 0/27 | 0/27 | 27/27 |
| Guidance H3 | 6/27 | 0/27 | 0/27 | 27/27 |
| Guidance H5 | 9/27 | 0/27 | 0/27 | 27/27 |
| Historical H3 | 6/27 | 0/27 | 0/27 | 27/27 |
| Historical H5 | 7/27 | 0/27 | 0/27 | 27/27 |

*Notes:* H3 and H5 refer to the model coordinates of terminal revenue-growth convergence. Historical means the recent-three annual growth median, not an estimated future distribution. Capital follows the original average-capital recurrence. Counts are out of 27 maintained cases.

Guidance H3 raises Salesforce compatibility from four to six cases; guidance H5 raises it to nine. Historical H3 also gives six, while historical H5 gives seven. ServiceNow and Box remain excluded and HubSpot remains compatible throughout. The differing Salesforce counts under alternative anchors prevent the guidance profile from being treated as the unique economic growth forecast.

The valuation effects are substantial even where classification is unchanged. At the base case, guidance H5 raises Salesforce's maximum by about $15.187bn, ServiceNow's by $19.831bn, Box's by $0.334bn and HubSpot's by $3.205bn relative to immediate terminal growth. Table S6 shows the remaining profile differences.

**Table S6. Base-case maximum values under revenue completions**

| Firm | Immediate growth | Guidance H3 | Guidance H5 | Historical H3 | Historical H5 |
|---|---|---|---|---|---|
| CRM | 166.247 | 173.891 | 181.433 | 172.249 | 178.076 |
| NOW | 60.567 | 70.093 | 80.398 | 70.108 | 80.431 |
| BOX | 3.645 | 3.814 | 3.978 | 3.698 | 3.747 |
| HUBS | 13.362 | 14.934 | 16.567 | 15.309 | 17.392 |

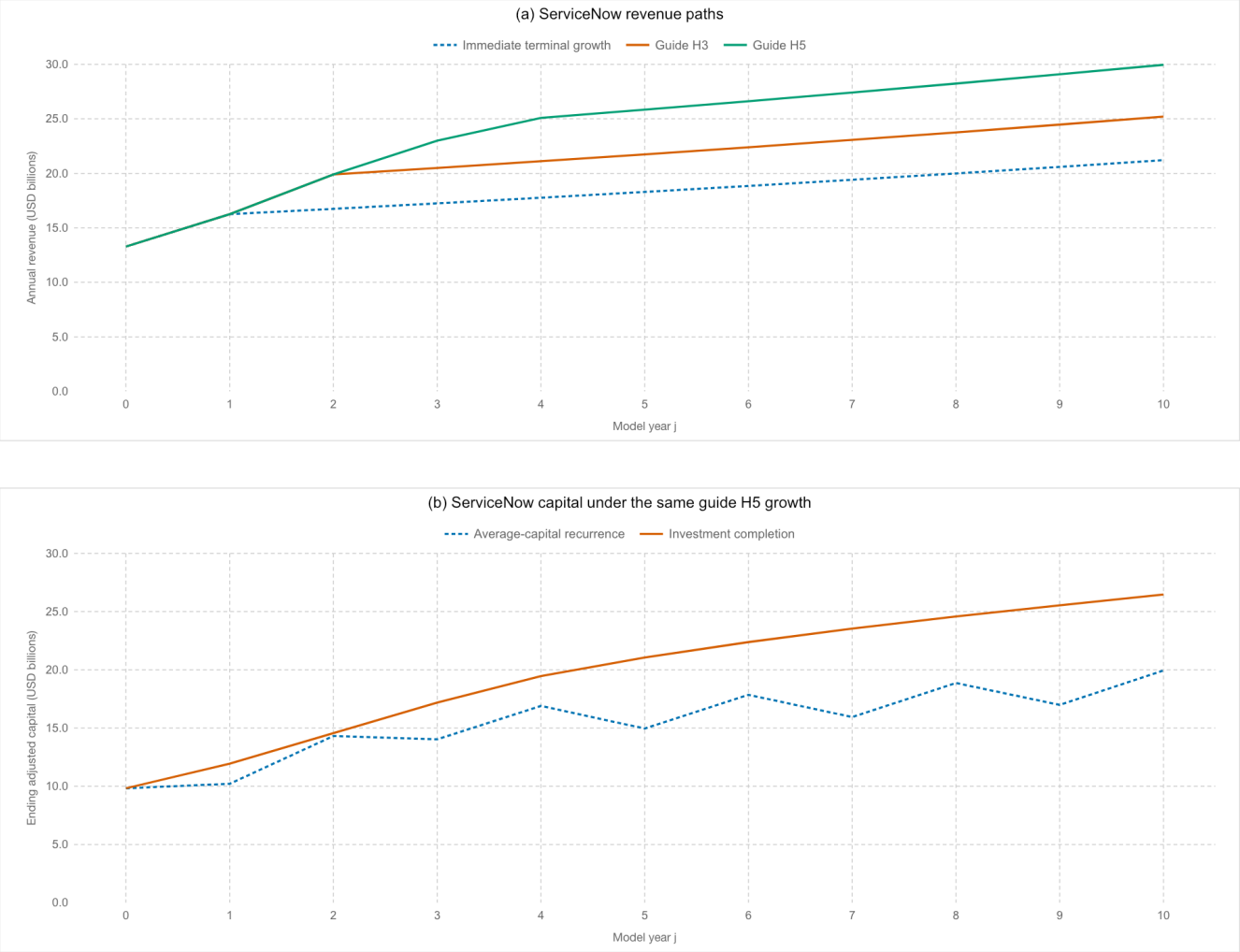
*Notes:* Values are USD billions. BASE WACC, g=3% and UNIFORM_FIRST timing apply throughout. These are maxima over the maintained operating family and lambda, not point-path forecasts. Figures are rounded for display.

The mechanism involves more than compounding an earnings stream. Higher revenue increases transition NOPAT at a fixed margin, changes the scale of the post-transition capital premium, and changes capital additions under the maintained completion. Once growth reaches g, the larger inherited revenue level persists in the terminal tail. Both the prefix and tail therefore differ from a small perturbation of terminal g.

Growth and persistence can substitute locally when the value derivative with respect to growth is positive and the matching spread is positive. The direction is not a general theorem: investment requirements and operating admission can offset the earnings effect. Nor can the originally empty ServiceNow and Box sets be described as positive persistence estimates biased by omitted growth. There was no admitted benchmark root to reinterpret.

The boundaries change as well. Salesforce's cross-case upper rate rises to approximately 0.017393 under guidance H3 and 0.023533 under guidance H5. HubSpot's corresponding upper rates are 0.165870 and 0.221134. Historical H5 produces different bounds, particularly for HubSpot. These changes show why a persistence bound needs its revenue completion stated alongside it.

**Figure S1. Revenue scale and capital completion in the ServiceNow base case**




*Preserved Figure S1*

*Notes:* Panel (a) plots saved immediate-growth, guidance H3 and guidance H5 revenue states. Panel (b) uses the same saved guidance H5 revenue path for both capital completions. Model year 0 is the December 2025 annual origin; year 1 is the full 2026 guidance coordinate. These are conditional annual-origin paths, not reconstructed

October remaining-period forecasts or observed future outcomes. Amounts are USD billions. Only years 0-10 are displayed; lines connect annual state points and do not measure intra-year flows.

**S6.2 From imposed average capital to explicit investment**

The capital comparison addresses an economic weakness that growth reconstruction alone cannot repair. Equation (S6) is mathematically correct for an imposed average-capital path, but the inferred ending stocks need not correspond to a plausible sequence of productive investment. Its alternating homogeneous component remains even after revenue and turnover settle into their terminal patterns.

For the saved ServiceNow guidance H5 base-case path, the original completion has a tail alternating coefficient near −$1.204bn, implying an investment oscillation of approximately $2.408bn. The recorded minimum net book investment is about −7.500% of revenue, with 29 capital-release years in the first 100 model years. Figure S1 illustrates the ending-stock pattern. These releases are algebraic stock reductions, not verified cash disposals.

The investment law removes that mode. It accumulates the research stock through spending and depreciation and links non-R&D additions to revenue changes. Its modes are the terminal revenue-growth factor, positive research retention and a constant offset; none has root −1. In the executed profiles, finite stocks remain positive and the continuing tail carries positive net book investment. The positivity and terminal identities are established for the modeled paths, while actual cash requirements remain unmeasured.

Turnover necessarily changes interpretation. Imposing both the investment law and an independent old turnover target would overdetermine capital. Revenue divided by completed average capital becomes a derived ratio. Consequently, the comparison changes capital completion, the turnover parameterization and the operating restrictions activated by the new capital/revenue path together. It is not a pure experiment deleting one algebraic term while all capital states remain unchanged.

**Table S7. Matched growth cases before and after investment completion**

| Revenue completion | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|
| Guidance H3 | 6 → 3 | 0 → 0 | 0 → 0 | 27 → 27 |
| Guidance H5 | 9 → 4 | 0 → 0 | 0 → 0 | 27 → 27 |
| Historical H3 | 6 → 3 | 0 → 0 | 0 → 0 | 27 → 27 |
| Historical H5 | 7 → 4 | 0 → 0 | 0 → 0 | 27 → 27 |

*Notes:* Each cell is the compatible count under the original capital recurrence followed by the compatible count under explicit investment and complete future ceilings. Both counts are out of 27. The accounting origin, proxy, revenue path, WACC/g/timing and nominal profitability restrictions are held fixed. Turnover becomes derived under investment completion.

Salesforce retains three H3 and four H5 matches under either anchor. ServiceNow and Box remain excluded; HubSpot retains all 27 cases. This result does not mean the investment specification is an economically validated replacement primary model. It establishes that the original classifications are sensitive to a more explicit accumulation relationship, and that the previously increased Salesforce compatibility does not survive intact.

The fixed spending and stock-ratio anchors imply the terminal requirements in Table S8. A model with positive growth continues to reinvest; it does not receive higher revenue while keeping an arbitrary capital budget unchanged.

**Table S8. Investment anchors and terminal requirements**

| Firm | R&D spending share (%) | Non-R&D stock ratio eta | Terminal C/R | Terminal net investment/R (%) | Recorded turnover range |
|---|---|---|---|---|---|
| CRM | 14.43 | 1.372 | 1.812 | 5.28 | 0.560-0.600 |
| NOW | 22.29 | 0.215 | 0.895 | 2.61 | 1.134-1.501 |
| BOX | 25.02 | 0.144 | 0.907 | 2.64 | 1.119-1.263 |
| HUBS | 28.93 | 0.033 | 0.914 | 2.66 | 1.110-1.563 |

*Notes:* Terminal limits use g=3%. Rho is the latest historical R&D/revenue share, eta the latest non-R&D NOA/revenue stock ratio. Turnover ranges summarize the first 100 recorded years of the base guidance H5 maximizing witness. Non-R&D gross investment and depreciation are unmeasured, not zero.

Under constant rho, a deceleration from recent high growth to low terminal growth raises the research stock relative to revenue. The same spending share supports a larger accumulated stock when prior revenue grows more slowly. This increases average capital per revenue dollar and can lower turnover. Comparing the resulting ratio directly with a narrow recent-history singleton would ignore the mechanism generating it.

### S6.3 Valuation effects and complete future profitability

The repaired accumulation law changes both investment outflows and the capital base on which post-transition returns are applied. Holding a terminal margin fixed therefore changes endpoint ROIC. As the capital/revenue ratio evolves, keeping a slowly fading ROIC spread can also imply a future margin exceeding the endpoint margin.

Complete future admission limits those paths. At zero fade, the return is constant, so future margins rise or fall with $\Gamma_j$. If $\Gamma_j$ eventually exceeds $\Gamma_T$, a terminal margin already at M can become inadmissible. A small positive fade may then allow a larger endpoint margin, even though it reduces value on any fixed positive-spread path. The maximizing constrained envelope need not occur at zero.

All selected investment-completion maximum-value witnesses use a positive lambda. Their maxima are lower than matched original-capital growth maxima in all 432 saved growth-profile cases. These are catalogue-specific numerical results. They do not prove that every coherent investment model reduces value, nor that removal of the alternating term alone has a positive or negative value effect.

**Table S9. Matched base guidance H5 valuations after full investment admission**

| Firm | Growth with recurrence | Investment maximum bracket | Investment signed gap bracket |
|---|---|---|---|
| CRM | 181.433 | [167.418, 167.421] | [-32.450, -32.447] |
| NOW | 80.398 | [70.304, 70.306] | [-70.960, -70.958] |
| BOX | 3.978 | [3.714, 3.715] | [-1.815, -1.814] |
| HUBS | 16.567 | [14.226, 14.227] | [4.919, 4.920] |

*Notes:* All values are USD billions. The investment columns are global numerical brackets; a negative signed gap is an irreducible conditional shortfall. Their presentation is rounded outward to the displayed scale. Capital, turnover interpretation and future-margin admission change inseparably. For HUBS, the saved maximum bracket is [14.226711739056750, 14.226801196523108] billion and the fixed target is 9.307694575090 billion; subtraction before rounding gives [4.919017163966750, 4.919106621433108] billion. Outward rounding therefore gives [4.919, 4.920]; subtracting independently display-rounded inputs (14.226 minus 9.308) instead gives 4.918 and is not the saved calculation.

The comparison reverses much of the growth-only valuation uplift. Salesforce’s maximum falls from about $181.433bn to $167.418-167.420bn, and ServiceNow’s from $80.398bn to $70.304-70.306bn. Box’s base maximum falls from $3.978bn to about $3.714bn. HubSpot remains compatible, but its maximum and persistence bounds contract.

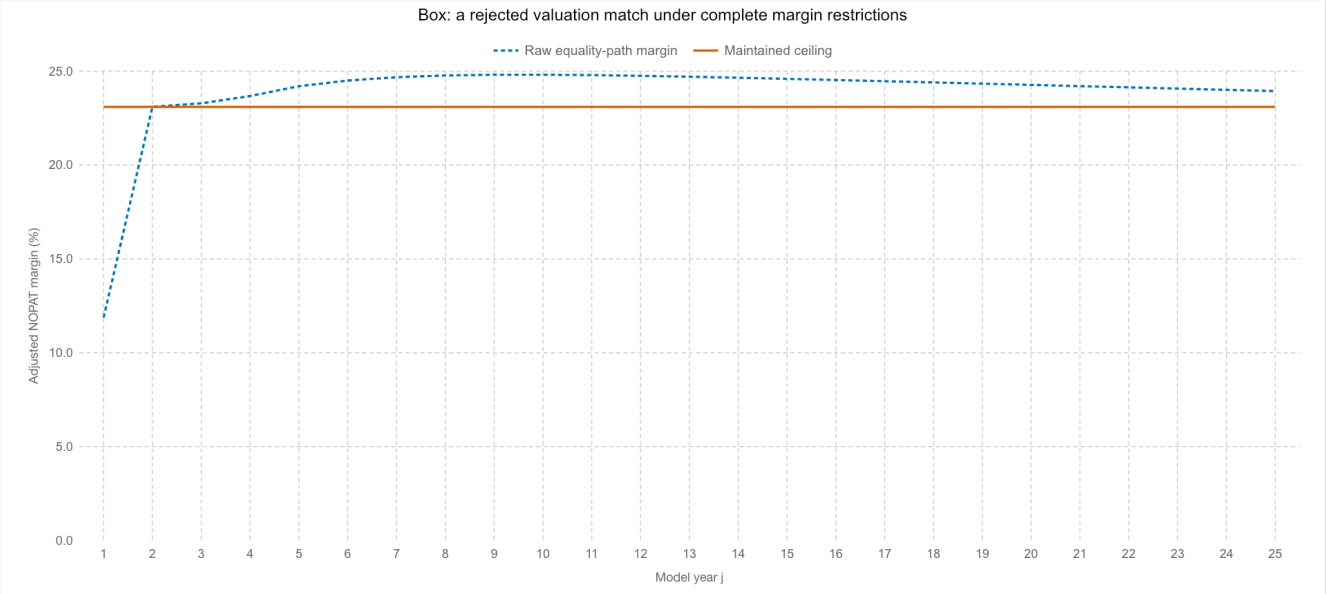
For the favorable guidance H5 corner, growth-only Box is approximately $29.422m short of the proxy. Explicit investment with complete future admission instead leaves about $306.9m. The direction is economically consequential for the tested completion, but still does not identify true required investment or a coordinated correction to the mixed-date proxy.

### S6.4 Why the apparent Box match is rejected

A relaxed combined growth/investment calculation produces a raw maximum around $5.884bn, above Box's $5.529bn proxy. Its valuation equality boundary has lambda about 0.004174. That would appear to supply a persistence estimate if numerical equality were the only criterion.

The saved operating path is not admitted under the full ceiling. Its recorded adjusted NOPAT margin reaches approximately 24.816%, whereas the maintained upper margin is 23.102%. The violation occurs in model year 10, after a terminal-margin endpoint that itself satisfies the cap. The capital/revenue evolution is therefore material to admission, not merely the value formula.

**Figure S2. An inadmissible Box price match**




*Preserved Figure S2*

*Notes:* The dashed series is the saved relaxed-domain equality witness for guidance H5, LOW WACC, g=4% and UNIFORM_ALL. The horizontal series is the maintained 23.102% ceiling. The witness matches the proxy numerically but violates future admission; it is not included in compatibility counts. The displayed first 25 years are existing saved observations of that constructed path, not a new valuation experiment.

After reoptimization over the complete domain, the favorable maximum is approximately $5.22184-5.22190bn. Thus no legitimate Box equality remains in the tested 27-case investment profiles. A small numerical root residual cannot override a future operating inequality.

This example clarifies the distinction between mathematical equality and admissibility. The raw root is a solution of a relaxed valuation equation. It is not a solution of equation (S12) with the complete domain. Because the margin cap itself is unvalidated as an economic hard bound, the rejection is still conditional on that cap; it is not evidence that Box could never reach higher margins in reality.

## S7. Historically motivated operating-support sensitivity

### S7.1 Evidence and rule construction

Operating support is the remaining source of apparently precise persistence information. The benchmark allowance extrapolates a percentile of peers' maximum one-year gains into terminal headroom. A one-year improvement can reflect a temporary peak, loss reduction, acquisition scope or operating restructuring. It does not by itself establish sustainable multi-year expansion.

The evidence archive contains 72 comparable constructed firm-years across 15 firms. A declared profitable-maturity diagnostic identifies 12 two-year episodes, while stricter two-/three-year trajectory eligibility leaves 14 expansion records. The observations use the common R&D, tax and capital definitions; GAAP or non-GAAP operating margins are never inserted directly as adjusted NOPAT.

The sustained own-history family takes the maximum adjacent-year mean own adjusted margin and includes coherently bridged guidance. It is a tighter historical-range hypothesis, not a technological ceiling. The future upper margin also contains the observed origin margin, so historical observations are not rejected merely by the new ceiling.

The peer family uses an existing stage diagnostic: at least billion-dollar revenue, two consecutive positive GAAP operating-profit years, two nonnegative growth rates no greater than 20%, and growth-change magnitude no greater than five percentage points. Those thresholds are explicit comparability assumptions, not estimated maturity boundaries. Business mix and scale differences remain, including the transfer of vertical-software and communications histories to other SaaS cases.

For each target firm, other firms supply donor-firm medians of positive two-/three-year margin changes, requiring positive GAAP operating profit throughout each path. The 90th percentile of those donor medians supplies expansion headroom. A separate 90th percentile of donor-firm medians of mature two-year mean margins supplies a level restriction. This avoids weighting overlapping observations as independent firms or transferring each donor's single largest rebound.

Let $h_f$ be peer headroom, $L_f$ the peer sustained level and $a_f = max\left(m_{0,f}, b_{U,f}\right)$ the origin/guidance anchor. The terminal rule is

$$M_{T,f}^{peer} = max\left\{b_{U,f}, min\left(L_f, a_f + h_f\right)\right\}. \quad \text{(S24)}$$

The own-history terminal rule uses the corresponding maximum own adjacent-year mean with a hull containing $b_{U,f}$. The future ceiling contains that terminal upper, the origin margin and the coherent guidance upper. Using a percentile, a two-year mean and this intersection is a prespecified descriptive choice. None provides a sampling confidence bound or proof of indefinite future coverage.

**Table S10. Profitability support rules and coherent guidance**

| Firm | Benchmark cap (%) | Own-history terminal cap (%) | Peer terminal cap (%) | Peer headroom (pp) | Coherent first-year margin (%) |
|---|---|---|---|---|---|
| CRM | 30.31 | 18.84 | 24.08 | 5.23 | 18.84-18.84 |
| NOW | 29.20 | 17.78 | 23.45 | 7.74 | 15.08-15.08 |
| BOX | 23.10 | 11.99 | 20.22 | 8.23 | 11.99-11.99 |
| HUBS | 28.58 | 13.81 | 21.55 | 7.74 | 13.72-13.81 |

*Notes:* All margin levels are adjusted NOPAT percentages. Peer headroom is in percentage points. Benchmark, own-history and peer caps are competing conditional restrictions, not a ranking of economic truth. The peer statistics leave the target out and balance donor firms. The coherent guide uses public GAAP anchors under the unchanged investment policy.

For CRM and BOX, both own-history ceilings equal the coherent guided margin; for HUBS they equal the upper endpoint of its coherent guidance range. These rules impose future performance restrictions, rather than independently establishing economic ceilings. NOW differs: its terminal ceiling is 17.776% and its transition/future ceiling 17.850%, compared with coherent guidance of 15.080% (supplement Table S30).

The exact saved target-excluded gain and mature-level donor sets coincide in this calibration: CRM uses BOX, DOCU, VEEV, WDAY and ZM; NOW uses BOX, CRM, DOCU, VEEV, WDAY and ZM; BOX uses CRM, DOCU, VEEV, WDAY and ZM; HUBS uses BOX, CRM, DOCU, VEEV, WDAY and ZM. The gain pool uses eligible two-/three-year trajectories with positive GAAP profit throughout and a mature endpoint; the level pool uses mature adjacent-year means. Maturity requires revenue of at least $1bn, positive GAAP operating income in both adjacent years, both growth rates between 0% and 20%, and their difference at most five percentage points. Statistics take donor-firm medians before the interpolated 90th percentile. Membership is verified separately from MULTIYEAR_TRAJECTORY and MATURE20_EPISODE records, and agrees with the RESTRICTION donor fields. The common DOCU/VEEV/WDAY/ZM donors and overlapping CRM/BOX donors preclude treating targets as independent replications. These are the valuation-calibration donor sets, not the origin-date donor sets in Section S18.

The relevant evidence is sparse. Each target has five or six donor firms in the expansion summaries. The original retrospective date-gated check has 14 candidate events and none passes its minimum-donor gate; it also selects events on realized endpoint maturity. The later feasibility audit and fixed-origin annual component study (§S7.8; Section S18) remain separate from that unavailable mature multi-year validation. No zero coverage rate or general forecasting failure is inferred.

Long-term management targets are not converted into a third terminal-profitability family where forward GAAP/R&D reconciliation is missing. Revenue targets, growth-plus-non-GAAP-margin frameworks and qualitative cost objectives do not disclose a complete future adjusted NOPAT bound. Unavailable bridge components remain unmeasured.

### S7.2 The guidance bridge is part of operating consistency

The historical bridge interval and a fixed future spending share can imply different first-year research adjustments. Using public GAAP guidance with the accumulation law requires the addition/depreciation relation in equation (S23). It is not coherent to choose a historical adjustment independently while retaining a spending policy that implies a different $\Delta K_1$.

**Table S11. First-year profitability before and after spending-consistent translation**

| Firm | Historical-bridge interval (%) | Spending-consistent interval (%) |
|---|---|---|
| CRM | 18.775-18.958 | 18.843-18.843 |
| NOW | 14.898-16.530 | 15.080-15.080 |
| BOX | 11.210-11.872 | 11.990-11.990 |
| HUBS | 14.594-17.231 | 13.723-13.808 |

*Notes:* Percentages are adjusted NOPAT margins. CRM, NOW and BOX use published rounded point anchors as conditioned scenarios; no confidence interval is fabricated. HUBS retains its saved GAAP operating-dollar range at fixed midpoint revenue. Rho remains the historical spending assumption, not company forward R&D guidance.

Box’s translated margin, approximately 11.990%, lies above the old bridge upper bound of 11.872%. HubSpot’s translated range, approximately 13.723-13.808%, lies below the old 14.594-17.231% interval. These differences are accounting consequences of the unchanged spending/depreciation policy and the same GAAP anchor. They show why adjusted profitability and investment cannot be completed entirely independently.

ServiceNow’s coherent guide is approximately 15.080%, below the old annual transition lower floor of 15.340%. The annual floor is lowered only enough to admit the source-consistent first-year value; the terminal lower

restriction is retained. This is an explicit operating-support change. It is not a tax, growth or dated-state recalibration.

The original comparison with investment completion therefore combines cap rules and guidance repair. Section S7.5 separately identifies their controlled H5 model effects with matched comparisons. It changes terminal-plus-transition/all-future upper restrictions as one support bundle; it does not isolate a terminal-only cap or causal contributions in the economy. The H3 comparison remains bundled.

### S7.3 Compatibility reversals

**Table S12. Compatibility under historically motivated operating families**

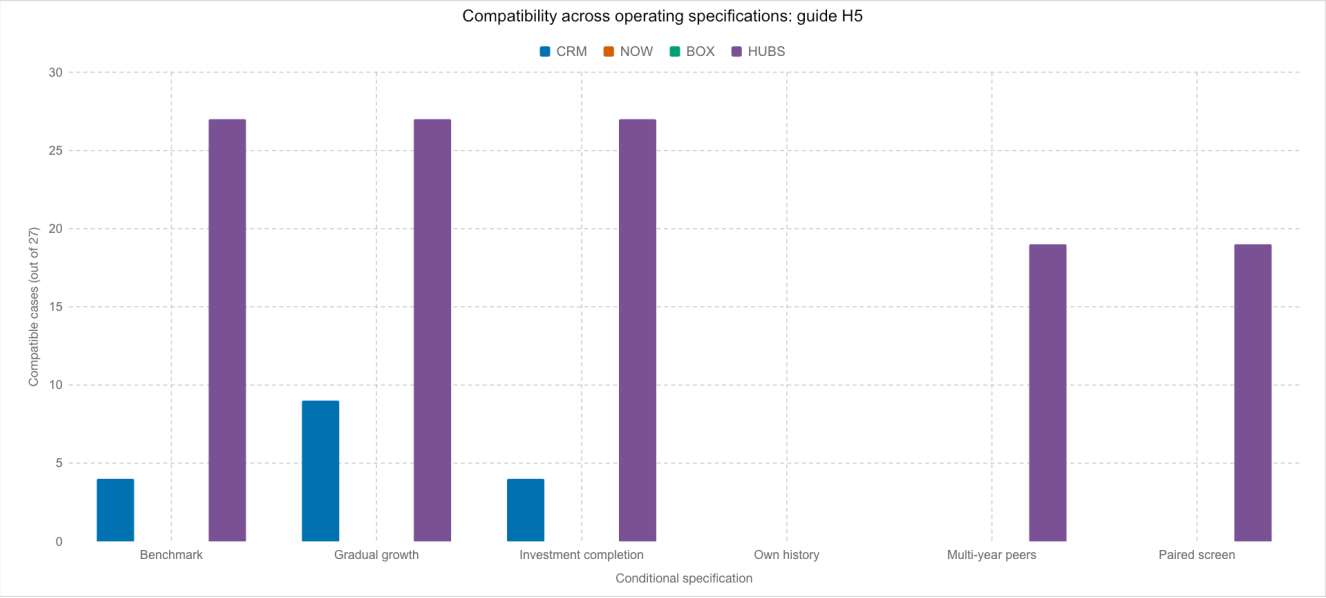
| Operating family | Revenue profile | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|---|
| Sustained own history | Guidance H3 | 0/27 | 0/27 | 0/27 | 0/27 |
| Sustained own history | Guidance H5 | 0/27 | 0/27 | 0/27 | 0/27 |
| Multi-year peers | Guidance H3 | 0/27 | 0/27 | 0/27 | 14/27 |
| Multi-year peers | Guidance H5 | 0/27 | 0/27 | 0/27 | 19/27 |
| Paired endpoint screen | Guidance H3 | 0/27 | 0/27 | 0/27 | 14/27 |
| Paired endpoint screen | Guidance H5 | 0/27 | 0/27 | 0/27 | 19/27 |

*Notes:* Counts are out of 27 for the unchanged investment completion and each stated guidance growth profile. The family labels describe historical scenario rules. No future economic support coverage is established.

Sustained own-history restrictions exclude every firm. This is not evidence that their past margins are permanent ceilings. It shows that matching these proxies requires value outside that particular descriptive support when the other model conditions are maintained.

Under the peer family, Salesforce's remaining three/four investment-completion matches disappear. ServiceNow and Box remain at zero. HubSpot retains 14 guidance H3 cases and 19 guidance H5 cases, rather than 27. The longer revenue convergence remains consequential even under the tighter profitability family.

**Figure S3. Compatibility reversals across the guidance H5 comparisons**




*Preserved Figure S3*

*Notes:* Each bar is a compatible-case count out of the same 27 maintained WACC/g/timing alternatives. The benchmark uses immediate growth and is repeated only as a reference for the five-year comparisons. Missing-height bars represent zero cases, not missing observations. Counts are not probabilities, and adjacent specifications change different declared operating assumptions.

The paired screen adds observed joint endpoint information. It forms a convex hull of mature two-year margin-turnover pairs, supplemented by the target's own paired histories and current state. At each $T$, the actual derived $k_T$ slices the hull and restricts $m_T$; endpoint growth must lie within the declared 0-20% stage range. No independent high-turnover parameter is chosen while holding investment cost fixed.

The hull preserves observed pairings instead of selecting a maximum margin from one donor and maximum turnover from another. Convex interpolation nevertheless introduces unobserved mixtures across firms and acquisition states. The screen applies at the endpoint, not as an empirically validated corridor for every future year. It is an additional conditional diagnostic, not a learned margin-turnover correlation or a production-technology estimate.

Counts are unchanged by this screen. Salesforce's maximum values decline in all 54 guidance H3/H5 cases, while ServiceNow, Box and HubSpot maxima and supported endpoints agree with the peer-only family within recorded numerical precision. A nonbinding joint restriction is not independent validation of the operating path.

**S7.4 Values, boundaries and structural exclusion regimes**

The favorable guidance H5 comparison in Table S13 quantifies the economic scale of the restrictions. Salesforce's peer-family maximum is approximately $165.53bn, below its $199.87bn proxy. ServiceNow remains about $63.83bn short. Box's shortfall increases from approximately $307m under investment completion to $1.09445bn under peer support. HubSpot retains substantial maximum surplus in the favorable case, but not universal compatibility.

**Table S13. Favorable guidance H5 values across complete specifications**

| Firm | EV proxy | Growth/recurrence maximum | Investment maximum bracket | Peer maximum bracket | Peer signed gap bracket |
|---|---|---|---|---|---|
| CRM | 199.868 | 247.375 | [230.349, 230.352] | [165.530, 165.533] | [-34.338, -34.335] |
| NOW | 141.264 | 113.261 | [101.256, 101.258] | [77.430, 77.432] | [-63.834, -63.832] |
| BOX | 5.529 | 5.499 | [5.221, 5.222] | [4.434, 4.435] | [-1.095, -1.094] |
| HUBS | 9.308 | 23.193 | [20.397, 20.399] | [14.328, 14.329] | [5.020, 5.022] |

*Notes:* LOW WACC, g=4% and UNIFORM_ALL timing. Values are USD billions; revised maximum/gap intervals are global numerical brackets rounded outward for display. The relaxed-domain Box witness is excluded from this table. A maximum surplus is not by itself proof of equality, which requires a complete admitted witness and verified support.

At the base case, HubSpot's peer H3 maximum is about $122.4m short of the proxy. Peer H5 remains compatible with an upper rate near 0.022838, compared with 0.121514 under the unchanged investment restrictions. The profile-specific cross-case upper boundaries are 0.057773 and 0.087011. Lower admitted margin levels leave less room for rapid fade on matching paths, conditional on the selected model.

**Table S14. Conditional minimum spread half-lives after the transition endpoint**

| Specification | CRM cross-case union (years) | HUBS cross-case union (years) | HUBS base case (years) | HUBS all-case common (years) |
|---|---|---|---|---|
| Immediate growth | ≈ 57.8 | ≈ 5.7 | ≈ 11.9 | ≈ 35.8 |
| Growth: guidance H3 | ≈ 39.9 | ≈ 4.2 | ≈ 7.7 | ≈ 15.0 |
| Growth: guidance H5 | ≈ 29.5 | ≈ 3.1 | ≈ 5.4 | ≈ 8.7 |
| Investment: guidance H3 | ≈ 51.9 | ≈ 4.5 | ≈ 8.2 | ≈ 17.7 |
| Investment: guidance H5 | ≈ 34.3 | ≈ 3.2 | ≈ 5.7 | ≈ 7.8 |
| Peers: guidance H3 | No admitted match | ≈ 12.0 | No admitted match | No admitted match |
| Peers: guidance H5 | No admitted match | ≈ 8.0 | ≈ 30.4 | No admitted match |

*Notes:* For supported rate interval [0,L], positive-rate spread half-life h = ln(2)/lambda is at least ln(2)/L. Entries show approximate minimum half-lives, rounded to one decimal for readability; the conservative numerical lower value uses the saved outer upper-rate endpoint. Precise reciprocal brackets and source selections are in the duration record and Section S13.6. Every nonempty set shown admits zero decay, so none has a finite upper half-life bound. NOW and BOX have no admitted match in every specification shown. The cross-case union permits any of the maintained cases; the common set is the intersection across all 27, potentially with different operating witnesses. An empty common set does not negate a nonempty union. Base means BASE WACC, g = 3%, UNIFORM_FIRST.

For HubSpot’s base case, the historical comparison implies a minimum spread half-life of about 11.9 years after T; investment with guidance H5 gives about 5.7 years; peer support with guidance H5 gives about 30.4 years. These are lower bounds on the duration of the modeled average-capital ROIC spread, not point estimates, investor beliefs, calendar durations starting October 6, or lifetimes of dollar economic profit. Each component measures time from its own profitability-transition endpoint T, not the revenue-convergence endpoint H_R. Zero decay keeps the spread constant; a zero terminal spread has no nonzero amplitude to halve. Growing capital can keep dollar EP positive even as this spread fades (Section S9.1).

The investment-to-peer comparison changes guidance and, where applicable, the initial floor as well as profitability support (§S7.2). It is a bundled scenario difference, not an isolated cap effect. Empty support means no admitted equality match under that specification, not zero persistence. The original rate table is retained in Section S13.6 for direct comparison.

The revised computations retain two distinct exclusion regimes beyond ordinary below-price envelopes. Eighteen own-history cases have an operating ceiling below the positive-fade asymptotic margin, so all positive lambdas are inadmissible while zero-fade operating paths may remain. Their price-equality sets are still empty. Eighteen paired Salesforce cases leave only negative terminal spreads; values increase toward an unattained infinite-decay supremum already below price. Neither regime is a solver failure or a finite lambda estimate.

The actual supported revised sets consist of 66 case-level numerical intervals across the peer-only and paired catalogues, all with zero inner lower endpoints. They include the same 33 HubSpot case outcomes in two related specifications, not 66 independent financial observations. Global cell coverage rules out unresolved interior gaps at the declared resolution; features inside endpoint brackets remain unasserted. This is stronger than sampled root enumeration and weaker than a formally certified exact topology at every printed digit.

### S7.5 Matched H5 attribution: guidance, upper restrictions and floor admission

The matched comparison was specified before evaluation, beginning with guidance H5 and the already declared base and favorable cases for all four firms. Compatibility changes, maximum-value difference bounds exceeding 1% of the fixed EV target, and an operating-domain inconsistency triggered the sealed expansion rule. All core combinations were then covered across the same 27-case menu. This adds controlled model comparisons; it does not recalibrate profitability rules, prices or investment technology.

H is the historical D8 investment comparison. G replaces its historical R&D guidance bridge with equation (S23) while retaining the old nominal upper caps. The common annual transition floor is min(old floor, coherent guide lower); only ServiceNow changes. O and P replace upper restrictions with the saved own-history or peer rules at the same coherent guidance and floor. RP uses the fixed peer caps with historical guidance as a reverse-order diagnostic. Revenue, capital accumulation, fixed targets, WACC/g/timing, tax, terminal lower bounds, transition shapes/change limits and the spread fade remain matched. Upper ceilings also apply to transition margins; this is an upper-support bundle, not a terminal-only effect.

**Table S15. Compatibility under matched guidance and upper-support restrictions**

| Firm | Historical H | Coherent old caps G | Coherent own caps O | Coherent peer caps P | Old bridge / peer caps RP |
|---|---|---|---|---|---|
| CRM | 4/27 | 4/27 | 0/27 | 0/27 | 0/27 |
| NOW | 0/27 | 0/27 | 0/27 | 0/27 | 0/27 |
| BOX | 0/27 | 0/27 | 0/27 | 0/27 | 0/27 |
| HUBS | 27/27 | 27/27 | 0/27 | 19/27 | 21/27 |

*Notes:* GUIDE_H5; counts out of 27, not probabilities. Every admitted comparison retains complete future-margin admission. H → G changes no case classification for any firm. At coherent guidance, G → O removes four Salesforce and 27 HubSpot matches; G → P removes four Salesforce and eight HubSpot matches. ServiceNow and Box remain excluded. The archived coherent O/P full-grid records are reused after exact restriction/control checks and focused numerical reproduction; the new cells under coherent guidance with original nominal ceilings and the reverse-order cells are evaluated separately.

At the base case, guidance correction changes the maximum by approximately −$17m for Salesforce, −$62m for ServiceNow, +$0.482m for Box and −$119m for HubSpot. Changing peer upper restrictions at that same coherent guidance reduces the maximum by approximately $45.054bn, $16.257bn, $0.548bn and $4.091bn, respectively. These are differences between globally bracketed envelopes over admitted paths, not forecast errors or independent PV corrections. Exact difference bounds for base, favorable and all other cases are retained in Section S15 and the attribution record.

The reverse-order peer diagnostic qualifies a cap-only interpretation. HubSpot goes from 27 to 21 admitted equality cases when peer caps change at the historical bridge, then to 19 when guidance is corrected at those caps. No case gains accompany the losses. The two additional exclusions are BASE WACC/g=2%/YEAR_END and HIGH WACC/g=3%/UNIFORM_ALL. Thus upper restrictions govern the main reversal in this menu, while guidance contributes two further peer-family exclusions. The sequence is conditional on the chosen order and frozen cap rules; it is not an additive causal attribution or a percentage allocation of a binary outcome.

Two failed combinations prevent an artificial complete factorial interpretation. Coherent ServiceNow guidance under the old floor admits no operating path because 15.080% is below the 15.340% annual floor. Lowering only that annual floor restores operating admission while its terminal lower bound remains unchanged. The floor change at historical guidance alters neither maximum values nor compatibility within the numerical brackets, but may enlarge lower feasible paths. HubSpot historical guidance with fixed own-history caps admits no operating domain because its 14.594% guide lower exceeds the 13.808% transition/future upper. Coherent own-history instead admits operating paths but no price match. Neither failure is a zero value or a zero persistence estimate.

Where all four domains are admitted, maximum-envelope interaction is measured as (P−G)−(RP−F), where F is historical guidance/old caps at the common floor. HubSpot's base interaction is approximately −$74.1m, with a numerical bracket [−$74.320m, −$73.953m]. Guidance therefore lowers its maximum more under peer caps than under old caps. The own-history interaction for HubSpot is undefined because the reverse domain is empty. These non-additivities describe the maintained model; they do not validate guidance, hard future supports or capital realization.

### S7.6 Profitability-benchmark pilot: a bounded null result

The benchmark pilot was prespecified before results: guidance H5, all four firms, coherent old nominal caps G and coherent own/peer caps O/P, each at the existing base and favorable cases. Its 24 matched pairs preserve price/risk/timing, capital accumulation, coherent guidance, transition restrictions and the same operating-prefix menu. The alternative changes post-transition NOPAT and its complete admission law, as required by its different estimand.

Expansion required a compatibility change, a bounded maximum-value difference wholly beyond 1% of fixed EV, or a minimum-half-life difference wholly beyond both one modeled year and 20% of the original minimum. None occurred. All 24 classifications remain unchanged; the largest absolute maximum-value difference bound is about 0.7674% of EV. The largest supported minimum-half-life shift is about 1.34 years, less than 4% of its original bound. These are declared stopping criteria, not proof that the dollar changes are economically irrelevant. Salesforce's largest maximum reduction is about $703m against its much larger target.

**Table S16. Pilot classification, unchanged under both profitability benchmarks**

| Firm | G base | G favorable | Own caps: both cases | Peer caps: both cases |
|---|---|---|---|---|
| CRM | No admitted match | Match | No admitted match | No admitted match |
| NOW | No admitted match | No admitted match | No admitted match | No admitted match |
| BOX | No admitted match | No admitted match | No admitted match | No admitted match |
| HUBS | Match | Match | No admitted match | Match |

*Notes:* G means coherent guidance at old nominal caps; O/P mean the saved coherent own/peer upper restrictions. Base is BASE WACC/g=3%/UNIFORM_FIRST; favorable is LOW WACC/g=4%/UNIFORM_ALL. These are selected pilot cases, not 27-case counts. Empty support means no admitted price equality, not zero persistence. No alternative full-menu computation is run because the expansion rule was not met. The original full-menu results remain specific to their original benchmark.

**Table S17. Supported pilot minimum half-lives, after transition T**

| Firm / family | Case | ROIC-spread minimum (years) | EP-rate minimum (years) |
|---|---|---|---|
| CRM / G | Favorable | ≈34.34 | ≈35.68 |
| HUBS / G | Base | ≈5.92 | ≈5.88 |
| HUBS / P | Base | ≈30.35 | ≈31.50 |
| HUBS / G | Favorable | ≈3.47 | ≈3.59 |
| HUBS / P | Favorable | ≈7.97 | ≈7.96 |

*Notes:* Approximate minima from the saved outer upper-rate endpoints; full reciprocal brackets are in the benchmark record. The columns describe different spreads, not estimates of one common economic quantity. Every supported pilot set includes zero decay, so none has a finite upper duration bound. These are not point estimates, investor beliefs, October calendar durations or dollar-EP lifetimes. A component measures time from its own profitability-transition endpoint; fade does not restart at revenue convergence.

The alternative is mathematically supported and gives zero spread an economic-profit break-even meaning. It does not establish an economically preferred forecasting convention or validate investment/cash realization. The pilot provides no evidence that benchmark choice causes the earlier compatibility reversals in these selected cases; it cannot establish insensitivity over untested cases. The new benchmark is retained as a verified comparison, with the original recurrence and benchmark findings preserved.

### S7.7 An assumption-requirement diagnostic for the analyst

The diagnostic asks what endpoint adjusted NOPAT margin would be required to reconcile each fixed valuation under a visibly selected operating story. It fixes H5 revenue, the maintained investment law, coherent first-year guidance midpoint, T=5 LINEAR and the base or already declared favorable risk/timing case. At prespecified

half-lives of 2, 5, 10, 20 and 40 years after T, plus zero decay, it solves the affine price equation for the required endpoint margin. No operating input is calibrated to the price, and the resulting inverse requirement is not a forecast.

Each raw root is tested against transition restrictions and the complete future path for G/O/P. Required endpoint profitability is distinguished from the highest subsequent margin or limiting supremum. The peak has a recorded numerical bracket and omitted-tail certificate; it need not exceed the endpoint. The six samples at each setting do not establish a continuous feasible region. Separately verified complete case sets determine whether the larger admitted transition menu matches; a compatible case can have rejected fixed-T samples.

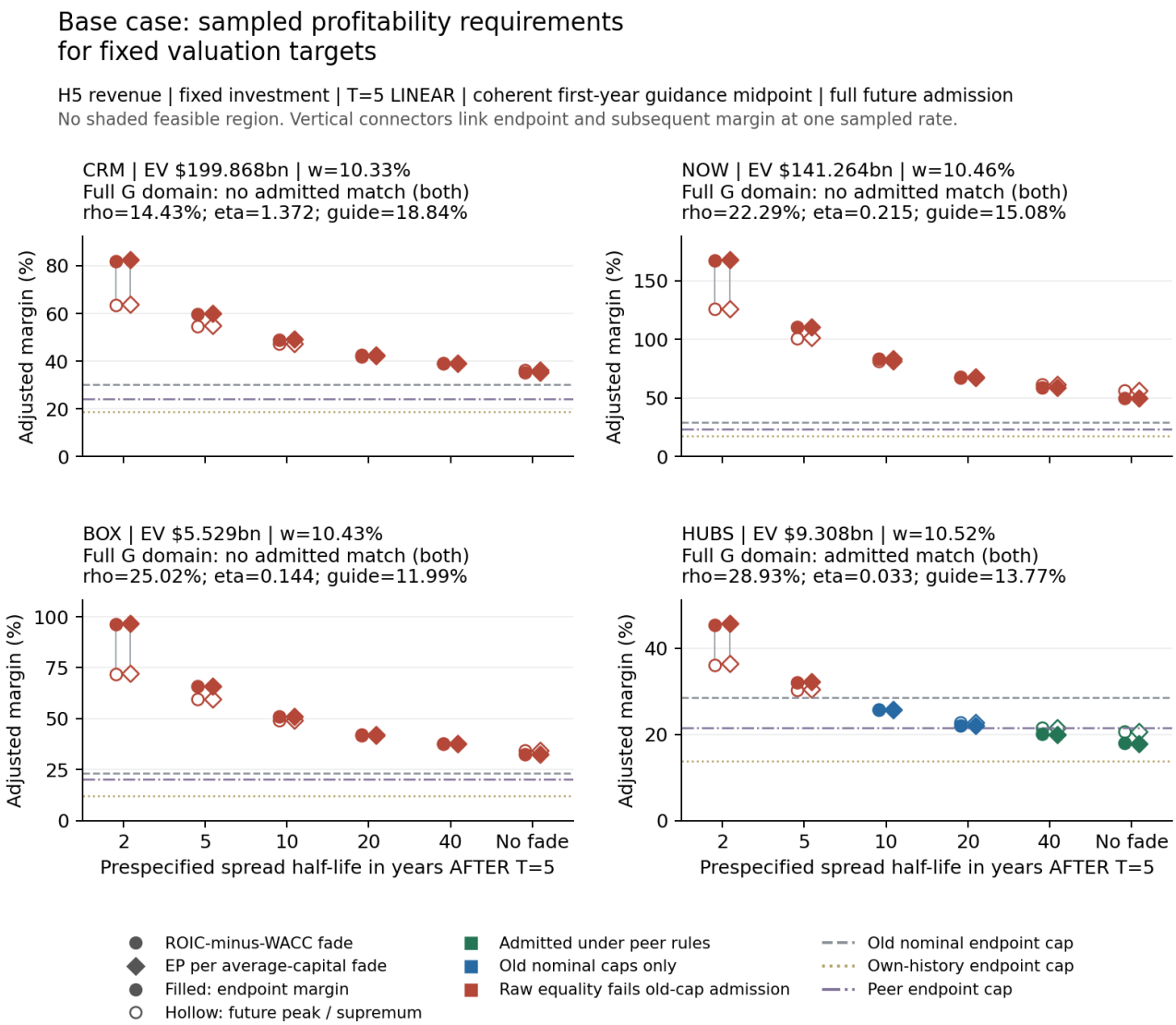
**Table S18. Base-case requirements at a sampled 20-year spread half-life after T=5**

| Firm | ROIC endpoint (%) | EP-rate endpoint (%) | ROIC future peak (%) | EP-rate future peak (%) | Admitted G / P (both) |
|---|---|---|---|---|---|
| CRM | 42.62 | 42.69 | 42.05 | 42.13 | No / No |
| NOW | 67.34 | 67.25 | 67.99 | 67.97 | No / No |
| BOX | 42.28 | 42.27 | 41.90 | 41.92 | No / No |
| HUBS | 22.06 | 21.96 | 22.79 | 22.80 | Yes / No |

*Notes:* Adjusted NOPAT margins in percent. G/P admission is the same under the two benchmarks at these points; O rejects all points shown. The future column concerns years after T, including a limiting supremum where applicable, and is separate from the endpoint. Exact settings, raw equality residuals, failure lists and peak brackets are in ASSUMPTION_DIAGNOSTIC.csv/JSON.

For HubSpot the original benchmark requires about 22.06% at the endpoint and reaches about 22.79% later. That sample passes old caps but fails the peer restrictions. The alternative requires about 21.96% and reaches about 22.80%, with the same admission result. For the other three firms these fixed-setting requirements fail both cap families. These failures expose the assumed operating story; they do not identify actual investor expectations or prove technological impossibility.

**Figure S4. Base-case sampled profitability requirements and full-path admission**




*Preserved Figure S4*

*Notes:* Circles and diamonds distinguish the faded ROIC spread and EP rate. Filled symbols show endpoint requirements; hollow symbols show subsequent peaks/suprema. Green admits peer rules, blue admits old nominal caps only, and red is a raw equality that fails admission under the original nominal ceilings. Vertical connectors compare margins at the same sample; no continuous feasible region is shaded. Horizontal endpoint caps are descriptive scenario comparisons, not validated future hard bounds. Formal requirements can exceed plausible profitability, including 100%, and remain visibly inadmissible. Full-case labels allow other transition/guidance witnesses; they are not inferred from the plotted samples. The prespecified favorable display is in Section S17 and the standalone figures.

The practical lesson is to inspect which profitability definition, growth/investment completion and entire future path make the numerical equation work. An endpoint inside a cap does not establish future admission: the previously saved Box witness in §S6.4 remains a separate concrete counterexample. The diagnostic makes conditional requirements readable without selecting a specification or support because it matches a price.

### S7.8 Two descriptive annual endpoint component comparisons

The completed feasibility audit selected Workday and Docusign at their FY2025 filing origins, March 11 and March 18, 2025, using maturity, earlier donor support and subsequent annual-record availability. Both episodes passed margin dependency verification. **Their FY2026 follow-up periods began February 1, 2025: 38 days for Workday and 45 days for Docusign had already elapsed at origin. This is an annual endpoint comparison, not a wholly future annual-flow forecast.** No elapsed operating flows are reconstructed.

The history-only annual ceiling contains the origin margin and the largest available own adjacent-year mean. The peer annual component additionally uses the unchanged target-excluded, firm-balanced level/headroom statistics, with at least three distinct firms in each donor pool. Both deliberately omit guidance. Section S18 records the formulas, original-vintage accounting and source checks; the full guidance-inclusive D9 restrictions remain those in §S7.1.

**Table S19. FY2026 annual endpoints against guidance-omitted margin components**

| FY2025 origin episode | FY2026 margin (%) | History ceiling (%) | Difference (pp) | Peer component (%) | Difference (pp) |
|---|---|---|---|---|---|
| Workday | 10.415 | 10.457 | −0.042 | 17.133 | −6.718 |
| Docusign | 12.697 | 10.765 | +1.932 | 17.015 | −4.317 |

*Notes:* All levels are adjusted NOPAT/revenue percentages. Differences are outcome minus the stated component in percentage points, computed from unrounded case values and displayed to three decimals. Positive means exceedance. These are two cases sharing donors, not independent sampling observations or an estimated coverage probability. Reported precision, fixed research-stock initialization and unknown residual R&D overlap qualify the comparisons. Turnover and the paired hull were not tested.

Docusign exceeded the history-only ceiling; Workday’s non-exceedance is small. Both endpoints remained below the tested peer component. This does not establish predictive coverage or economic validity, validate a two-year sustained level, test full guidance-inclusive D9, or certify infinite-horizon admission. Historical observations and these component comparisons remain distinct from assumed future restrictions and the conditional valuation results.

## S8. Economic interpretation and limitations

### S8.1 What valuations restrict in this exercise

The results establish a sequence of conditional restrictions, rather than one economically preferred persistence estimate. With a fixed admissible path, market equality usually determines one finite rate. With unknown operating states, it instead restricts a joint set of rates and paths. Changes to the operating completion can widen that set, shrink it or make it empty.

The firm-specific magnitudes carry information that generic sensitivity logic does not. Medium-term growth adds nearly $20bn to ServiceNow’s base maximum under the old capital completion, yet does not eliminate its exclusion. Explicit investment and future admission then reduce the attainable value despite preserving that same growth path. For Box, a relaxed numerical match fails at a future margin inequality, and the fully admitted gap remains economically material. For Salesforce and HubSpot, historically motivated support changes remove previously supported outcomes and substantially contract persistence bounds.

These effects arise through explicit mechanisms. Revenue changes the earnings base and inherited terminal scale. Investment changes both cash deducted from NOPAT and the denominator used to define ROIC. A constant spending share can increase capital per revenue dollar as growth slows. Future margins therefore depend on the completed capital path even after the endpoint margin is fixed. Operating ceilings then alter which margin/persistence combinations are admitted. The maximum of those admitted combinations is a different object from the value of one fixed path.

The raw Box witness demonstrates the importance of complete admission. A price match with a small residual is mathematically successful for its relaxed equation but fails the declared path restrictions. Rejection is justified by the inequality, not by the numerical solver's performance. The ceiling remains an assumption, so the rejection cannot establish that the violating margin is technologically impossible.

Likewise, the historical families reveal dependence without determining which family is economically correct. The original allowance permits profitability beyond own history; the revised rules impose levels and headroom inferred from a small, selected cohort. Their different supports are precisely the issue under investigation. Selecting the family with the desired price result would destroy the independence of the exercise.

**S8.2 Exclusions within the tested menu and failed robustness**

The NOW/BOX exclusions concern only the tested configurations: immediate terminal growth at model year 2 and the H3/H5 convergence profiles, with terminal growth first reached in model year 3 or 5 from the fiscal origin. They do not test longer revenue-convergence horizons or imply exclusion under every operating completion. ServiceNow remains excluded throughout those saved operating and initial investment catalogues and matched gradual-growth comparisons. Its large gaps show that these particular valuation stories require much more value than their admitted paths supply. Box is closer to some boundaries, but still excluded under every fully admitted growth/investment and revised historical-support profile here.

The evidence does not establish economic impossibility. Earlier illustrative terminal-margin relaxations restore Box in one, three or ten cases when the cap is increased by two, five or ten percentage points. Those stresses are not credible forecasts merely because they produce matches. They are legitimate conditional counterexamples to a claim that no operating completion can reconcile Box.

Repeated exclusions also share unresolved inputs. The fixed annual origins, proxy measurement, research conventions, spending assumptions and WACC structure are not independent tests of economic truth. A large gap cannot be described as robust to unknown errors for which no joint bound exists. Nor can a small gap be removed by adding a disclosed stock or annual flow as though it were an equivalent PV adjustment.

Salesforce's benchmark compatible count is specification-dependent: four originally, up to nine under gradual growth, three or four under investment completion, and zero under the two revised profitability rules. Its empty revised sets are not zero-rate estimates. HubSpot's 27-case compatibility survives growth and investment completion for the tested profiles, but not the revised operating supports. Its all-case/common-support interpretation therefore fails outside the original family.

**S8.3 Financial timing and claim measurement**

The price date, share observations, financial balances and annual operating origins do not form one synchronized October state. This maximum surplus is about $0.700bn ($699.853m before display rounding). The saved timing offset is 248/365 years from January 31 to October 6, 2026, about 68% of the first model year. YEAR_END retains the full year-1 flow and adjusts discount time; it does not remove elapsed operations. How that elapsed flow aligns with the July 31 claims snapshot is unresolved. No elapsed-flow correction or synchronized surplus is established. Full-year guidance minus observed first-half revenue describes a quarter-end remainder, not the unknown split between elapsed unreported operations and genuinely future October cash flows. Uniform payment timing does not solve that measurement problem.

The publicly observed interim state is material for ServiceNow. Reported-input NOA rises from $2.858bn at the annual origin to $11.253bn at the reviewed quarter, a $8.395bn increase. Completed acquisitions and their financing already affect the financial snapshot used in the proxy, while the annual operating origin predates them. Updating one side alone is inconsistent. The increase is not an additive correction to the valuation gap, and the sign or magnitude of a coordinated state/flow repair is unresolved.

Salesforce's post-quarter completed acquisitions, Box's hybrid claims and rights, and HubSpot's intervening operating/financing events create other missing bridges. The source reconciliation identifies dated components

and some unpriced rights, but does not supply a complete October EV-error bound. Carrying values, grant-date amounts, repurchase authorizations and registered shares do not resolve fair-value or outstanding-share uncertainty.

These limitations fundamentally prevent a literal current-market-expectations interpretation. They are disclosed here because the measured object is the controlled proxy experiment, not because disclosure makes the original dating economically exact. A synchronized valuation study would require a separately coordinated operating, claim, risk and remaining-flow design.

**S8.4 Cash, tax and capital interpretation**

R&D-adjusted income and capital are coherent constructed objects, but distributable cash equivalence is unestablished. Book capital can change through consumption, working-capital accounting, acquisitions and noncash adjustments. The original recurrence's releases are not evidence of sales or cash recovery. The repaired law makes additions explicit without measuring the ordinary cash-investment/consumption bridge.

**Table S20. Historical constructed FCFF and partial reported-cash comparators**

| Firm / FY | Period end / publication | Constructed FCFF | Reported CFO | Selected cash investment | Partial comparator | Constructed minus comparator |
|---|---|---|---|---|---|---|
| CRM / 2026 | 2026-01-31 / 2026-03-02 | 701.844 | 14,996.000 | 594.000 | 14,402.000 | -13,700.156 |
| NOW / 2025 | 2025-12-31 / 2026-01-29 | -911.047 | 5,444.000 | 868.000 | 4,576.000 | -5,487.047 |
| BOX / 2026 | 2026-01-31 / 2026-03-09 | 9.012 | 356.450 | 41.248 | 315.202 | -306.190 |
| HUBS / 2025 | 2025-12-31 / 2026-02-11 | -133.638 | 760.717 | 183.789 | 576.928 | -710.566 |

*Notes:* USD millions, latest archived annual periods; publication dates are in 2026. Constructed FCFF is adjusted NOPAT minus change in R&D-adjusted book operating capital. The partial comparator is consolidated CFO less PPE cash purchases, also less separately disclosed capitalized-software cash for BOX and HUBS. Selected investment is PPE of 594.000, 868.000, 6.074 and 53.165 respectively, plus software of 35.174 for BOX and 130.624 for HUBS. CRM/NOW software blanks and unmeasured investment categories remain unknown. The comparator also omits acquisition and other investing flows; its selected deduction is not a complete investment measure. These annual flows are not October forecasts or synchronized as-of cash flows. Sources: archived C3 annual cash bridge, C4 component table and annual-filing crosswalk in Section S9.3 and the verification record.

The differences are accounted for, rather than unexplained errors. CFO adds back stock-based compensation and D&A, includes nonoperating income and consolidated tax effects, and reconciles working capital on its reporting scope. Constructed profitability retains an economic compensation cost and uses normalized adjusted-profit tax; book-capital changes include acquisitions, leases, remeasurement and noncash balances. The comparator deducts only the stated cash-investment categories. Accordingly, it is not distributable unlevered FCFF. The decomposition in Section S9.3 closes arithmetically but supplies neither independently additive economic corrections nor a basis for adding these annual differences to enterprise value.

The tax convention creates a related distinction. Relative to an immediate-deduction normalized comparator, constructed FCFF differs by the piecewise tax expression in Section S9. On a joint positive-profit branch that difference is $-\tau \Delta K$. Its algebra does not establish a universally omitted tax shield, usable deductions, a future valuation uplift or an appropriate after-tax return denominator. A capital/tax redefinition would change the estimand and must be coordinated; it is not an independent addback.

SBC presents both operating-cost and claim questions. Retaining it in operating income does not price every outstanding award, and adding CFO's SBC reconciliation back to the constructed model would not automatically correct dilution or financing claims. Research capitalization can also shift the timing of eligible compensation cost. These channels must be reconciled rather than treated as separate favorable adjustments.

The investment parameters remain conditional. The spending share is not a validated future research plan, the NOA ratio is not measured marginal efficiency, and recognized assets do not establish maintenance cash needs. Greater mathematical coherence is useful, but cannot itself certify the economic technology behind the price inversion.

### S8.5 Support coverage, risk and generalizability

Historical maxima and paired means describe what occurred under selected vintages and constructions. Turning them into future ceilings is a maintained scenario restriction whose empirical reliability is unestablished. The peer stage screen, firm-balanced percentiles, headroom intersection and convex hull are disclosed rules. The limited annual component comparisons (§S7.8) do not validate their coverage probabilities. Broad validation stopped because the verified panel lacks origins combining the unchanged three-firm donor gates with two subsequent annual observations. The partly elapsed annual follow-ups cannot supply that missing multi-year evidence or the unavailable forward GAAP guidance bridges (Section S18).

Risk also remains externally specified rather than recovered jointly from one price. Common industry beta, financing categories, constant tax/risk assumptions and discrete WACC stresses do not establish issuer required returns. Broadening WACC into another nuisance parameter would display more trade-offs, not independently identify risk and persistence.

The sample supports case evidence, not statistical population inference. Construction eligibility, business mix, fiscal histories and reporting completeness matter to selection. Scenario counts and reused cases are not independent observations or probabilities. Short-horizon forecast gains in earlier diagnostics are retained, but did not establish a reliable independently fixed multi-year operating path or a general forecasting impossibility.

Finally, the analytical tools and much of the optimization structure have established precedents. The full closest-paper methodological comparison remains unresolved. The paper can therefore claim a transparent case audit and documented failures of robustness, while withholding a first-method, general-theory or causal-moat claim.

## S9. Accounting identities, timing and the cash boundary

### S9.1. Finite valuation identity and the return benchmark

Substitute $F_j = Y_j - C_j + C_{j-1}$ and $EP_j = Y_j - w\,C_{j-1}$, with $q = 1 + w$. The difference between discounted cash and discounted EP is

$$\sum_{j=1}^{H} \frac{F_j - EP_j}{q^j} = \sum_{j=1}^{H} \frac{q\,C_{j-1} - C_j}{q^j} = C_0 - \frac{C_H}{q^H}. \quad \text{(S25)}$$

This proves equation (S8) before imposing transversality. It also identifies the capital term that must remain in an independent finite-stream check. The infinite identity requires $C_H / q^H \to 0$, rather than mere equality of two numerical implementations.

Equation (S9) follows from $C_{j-1} = \acute{C}_j - \Delta C_j / 2$. Under balanced capital growth, beginning-capital zero EP corresponds to average-capital ROIC $2w/(2+g)$, rather than w. With positive lambda under the investment completion, the average-capital spread tends to zero but the limiting beginning-capital EP/revenue is

$$\frac{\lim EP_j}{R_j} = w\,\Gamma_\infty - \frac{w}{1+g}\,c_\infty = \frac{w\,g}{2(1+g)}\,c_\infty > 0. \quad \text{(S26)}$$

This is a conditional consequence of the selected benchmark and growth law, not a claim of actual perpetual economic rents. At lambda zero the return remains the endpoint return and the limit must be evaluated with that return instead.

The origin-date year-end identity is separate from the stylized market-date payment transformation. For elapsed fraction e, a year-end value is multiplied by $q^e$. UNIFORM_FIRST replaces the first year-end contribution by $F_1 \int_0^{1-e} q^{-u}\,du$; UNIFORM_ALL additionally scales later contributions by $w / \log q$. Consistent implementation preserves the constructed comparison, but these transformations do not subtract verified already realized first-year cash or synchronize the capital state to October.

**S9.2. Tax and observed cash comparators**

Because $\Delta K = I^K - A^K$, the construction simplifies to

$$F^* = O - \tau\, max(O + \Delta K, 0) - \Delta N. \text{(S27)}$$

A normalized immediate-deduction comparator is $F^{deduct} = O - \tau\, max(O, 0) - \Delta N$. Their difference is

$$F^* - F^{deduct} = -\tau \{ max(O + \Delta K, 0) - max(O, 0) \}. \text{(S28)}$$

When both pretax measures are positive, this becomes $-\tau\, \Delta K$. Across other branches it is the piecewise expression. This identity does not establish actual tax deductibility, utilization or a universal valuation-bias direction. An after-tax-capital restatement would require coordinated income, capital and spread definitions, not an additional identical shield uplift.

For a reported cash comparator, let Q be pretax nonoperating income, $T^G$ reported tax provision, S reported SBC, Z reported D&A, U the remaining CFO reconciliation and working-capital terms, and $J^{obs}$ the selected cash investment deducted. The identities $NI = O + Q - T^G$ and $CFO = NI + S + Z + U$ imply

$$F^* - (CFO - J^{obs}) = -Q + (T^G - T^*) - S - Z - U - \Delta N + J^{obs}, \text{(S29)}$$

where $T^*$ is normalized tax on adjusted profit. The decomposition organizes measured differences but does not make its components independent causal contributions to a value error. Adding SBC or selected capex alone cannot repair a coordinated cash/claim representation.

More generally, if non-R&D book capital follows $\Delta N = J_N - D_N + X$, with supported cash additions $J_N$, consumption $D_N$ and noncash change X, then consistent income/tax definitions give $F^{cash} = Y - \Delta C + X$. The required bridge is not measured for this exercise. X is never introduced as a plug to fit the proxy.

**S9.3. Archived historical cash decomposition and source crosswalk**

Equation (S29) reconciles Table S20 with the archived measurements. The remaining CFO group U is measured as CFO − NI − SBC − D&A; it aggregates reported reconciling and working-capital items. It is not an invented cash flow or an unexplained-error plug. BOX's negative GAAP tax provision is a tax benefit and is retained with its sign. All amounts below are USD millions; rounded cells need not sum exactly.

**Table S21. Historical cash-gap component decomposition**

| Component: constructed minus partial comparator | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|
| − Nonoperating pretax income | -1,189.000 | -437.000 | -15.540 | -62.084 |
| GAAP provision − normalized research tax | -6.156 | -117.047 | -47.789 | -63.904 |
| − Stock-based compensation | -3,509.000 | -1,955.000 | -233.716 | -528.153 |
| − Depreciation and amortization | -3,631.000 | -738.000 | -32.908 | -136.295 |
| − Other CFO reconciliation / working capital | -399.000 | -1,003.000 | 25.557 | -50.358 |
| − Change in non-R&D book capital | -5,560.000 | -2,105.000 | -43.042 | -53.561 |
| + Selected comparator cash investment | 594.000 | 868.000 | 41.248 | 183.789 |
| Total difference | -13,700.156 | -5,487.047 | -306.190 | -710.566 |

The four independently checked component sums close to less than $10^{-25} using archived decimal inputs. This precision concerns arithmetic, not financial measurement. Compensation must be reconciled with existing awards, future grants, withholding and claims; a CFO addback does not eliminate its economic cost. Tax provisions, cash taxes and normalized operating tax differ in basis and timing. Book investment and selected cash purchases differ in acquisition, software, working-capital and noncash scope. Unknown future tax usability, gross non-R&D investment and capital-realization mechanisms remain unknown.

**Table S22. Annual cash source crosswalk**

| Annual source / publication | Archived accounting and cash crosswalk |
|---|---|
| CRM FY2026 10-K / March 2, 2026 | crm_firm_year.csv, crm_reported_noa.csv, PIT annual origin; CFO 14,996m and PPE cash 594m. |
| NOW FY2025 10-K / January 29, 2026 | now_firm_year.csv, now_reported_noa.csv, PIT annual origin; CFO 5,444m and PPE cash 868m. |
| BOX FY2026 10-K / March 9, 2026 | Transition firm-year/NOA files; archived annual extract lines 2547-2598 for CFO/PPE/software; cash-statement SBC 233.716m and D&A 32.908m. |
| HUBS FY2025 10-K / February 11, 2026 | Firm-year/NOA files; archived annual extract lines 2387-2434 for cash investments; cash-statement SBC 528.153m and D&A 136.295m. |

Primary annual-filing URLs remain in the references; archived C3 §10 identifies the documents and C4 §3 defines the decomposition. The verification record preserves exact copied paths, SHA-256 hashes, numerical inputs and assertions. Cash-statement compensation and D&A rows for BOX/HUBS are explicitly selected rather than substituting a narrower expense or asset-specific amortization field. No new filing extraction, forecast or valuation experiment was used for this integration.

## S10. Baseline tail, inversion and component topology

### S10.1. Alternating capital and the geometric premium

Let $a = 1 + g$, $q = 1 + w$, and $J = T$ under immediate terminal growth. With fixed tail turnover,

$$C_{T+n} = G_T a^n + B_T(-1)^n, G_T = \frac{2\acute{C}_T a}{1+a}, B_T = C_T - G_T, \acute{C}_{T+n} = \acute{C}_T a^n. \quad \text{(S30)}$$

The averaging cancels the alternating component in average capital but not in net investment. Specifically,

$$\Delta C_{T+n} = G_T\, g\, a^{n-1} + 2B_T(-1)^n. \text{(S31)}$$

Its absolute amplitude is undamped. Relative to revenue it declines for g>0, and discounted capital vanishes for $a<q$. A sufficient positive-tail condition is $G_T a > |B_T|$, with finite-prefix positivity checked separately. These statements apply to the maintained positive-growth domain and do not justify extrapolation to negative growth.

With $S_T = s_T \acute{C}_T$, the year-end tail at T is

$$Tail_T = \frac{w\acute{C}_T a - G_T g}{w-g} + \frac{2B_T}{2+w} + S_T \frac{a e^{-\lambda}}{q - a e^{-\lambda}}. \text{(S32)}$$

The alternating PV term can have either sign. Its removal alone therefore does not determine a valuation effect. Transition flows plus the first two tail terms give D, and the last term supplies the signed $E\,h(\lambda)$ premium after timing/discount factors. This establishes equation (S13) while retaining the original capital identity.

**S10.2. Fixed-path cases and the frontier**

For $\delta > 0$, $h(\lambda) > 0$ and $h'(\lambda) < 0$, with image $(0, h(0)]$ over finite nonnegative lambda. When E is nonzero, equality fixes $h = (P - D)/E$, giving equation (S14) only if that value lies in the image. The open lower endpoint excludes equality attained only as lambda tends to infinity. For E=0, value is constant; a dollar tolerance cannot redefine the exact degeneracy.

On the maintained positive-profit transition branch, $Y_j = m_j R_j$ is affine in m and b, and capital is independent of those margins. The general kernel then gives equation (S15), from which equation (S16) is direct rearrangement. A zero denominator requires returning to the unrearranged equality; it can imply exclusion or lack of determination of m. Operating admission remains a separate requirement.

Margin affinity gives polygon-vertex coverage. Benchmark turnover interpolation contributes terms $R_j/[k_0 + (k - k_0)\, j/T]$; their rational derivatives require full-domain sign/curvature or stationary-point coverage. A sampled maximum is insufficient. Transition horizons and shapes are retained as separate components, including mathematically equivalent two-year shape ties.

**S10.3. Sufficient interval theorem**

Consider one compact connected nuisance domain $\Theta$, continuous D and E, and the benchmark kernel h. Assume $max_\Theta D < P$, an admissible witness $\theta_-$ with $V(0, \theta_-) \le P$, and some admissible zero-fade path at or above P. Any E≤0 path remains below P, so a matching path has E>0 and decreases with lambda.

The strict limit inequality supplies $P - D \ge \epsilon > 0$. Compactness bounds E, so matching kernel values are bounded away from zero and their rates uniformly bounded above. Closedness and continuity attain a largest finite matching rate L. Its maximizing-root path is at or above P for every $0 \le \lambda \le L$.

The low witness remains below P throughout: a positive-E witness decreases, while a nonpositive-E witness stays below its D limit. Connectedness implies a continuous valuation image, so the intermediate-value theorem supplies equality at every rate through L. No larger rate matches by construction. The projection is $[0, L]$, including the possible singleton case L=0.

This proves topology under the stated premises, not from connectedness alone. Finite unions of such components share zero and form one case interval. Across the case menu, the union and common intersection are separate set operations; a common rate may use different operating witnesses. A lambda-dependent future domain requires the different reasoning in Section S12.

## S11. Finite growth and split-stock terminal valuation

### S11.1. Carrying the finite prefix

Let $J = max(T, H_R)$. Revenues and capital are valued explicitly through J, and every year after T carries its actual factor $e^{-\lambda(j-T)}$. In the recurrence completion, equation (S30) holds from J with $G_J = 2\acute{C}_J a/(1+a)$, $B_J = C_J - G_J$, and spread at J equal to $s_J = s_T e^{-\lambda(J-T)}$.

The tail is

$$Tail_J = \frac{w\acute{C}_J a - G_J g}{w-g} + \frac{2B_J}{2+w} + s_T e^{-\lambda(J-T)}\acute{C}_J \frac{a e^{-\lambda}}{q - a e^{-\lambda}}. \text{(S33)}$$

The actual $R_J$ and stock are retained. Revenue is not reset to an immediate-growth path, and fade is not restarted at J. Setting the anchor to g recovers the benchmark revenue branch; the finite prefix does not create an infinite high-growth tail.

### S11.2. Investment modes and capital requirements

Under equation (S19), define $U_J = \rho R_J a/(a-v)$, $G_J = \eta R_J + U_J$, $H_J = K_J - U_J$, and $F_J = N_J - \eta R_J$. For $n \geq 1$ ,

$$C_{J+n} = G_J a^n + H_J v^n + F_J. \text{(S34)}$$

This follows by solving the R&D recurrence and using $N_{J+n} = N_J + \eta(R_J a^n - R_J)$. Its characteristic modes contain no negative root. Constructive positivity follows from positive initial R&D capital, nonnegative spending, nonnegative N and the admitted revenue/investment domain; it is not inferred by requiring every modal coefficient to be positive.

For the executed profiles, $G_J > 0$, $H_J < 0$, g>0 and $0 < v < 1$. Tail net investment

$$\Delta C_{J+n} = G_J g a^{n-1} + H_J(v-1)v^{n-1} \text{(S35)}$$

is positive. Finite prefix additions are checked separately. The constant offset contributes no net investment. These conditions establish the modeled stock/flow behavior, not non-R&D gross cash additions or depreciation.

Dividing (S34) by revenue and taking limits gives equation (S21). Discounted stocks converge because $a < q$, $v < q$ and $1 < q$. The resulting cash/EP tail includes the continuing investment required by g.

### S11.3. Closed tail for each mode

Write $C_{J+n} = \sum_\ell c_\ell d_\ell^n$, with $d_\ell \in \{a, v, 1\}$, $z = e^{-\lambda}$ and $s_J = s_T z^{J-T}$. The exact year-end tail is

$$Tail_J = \sum_\ell \frac{c_\ell\{w(1+d_\ell)/2 - (d_\ell - 1)\}}{q - d_\ell} + s_J \sum_\ell \frac{c_\ell(1+d_\ell)z/2}{q - d_\ell z}. \text{(S36)}$$

The formula comes from average capital and net investment of each mode, with geometric sums discounted from J. Its modal coefficients may have different signs while the actual average-capital sequence remains positive. The full physical sequence, rather than the signs of separated algebraic terms, establishes the positive decreasing persistence kernel.

The finite prefix and tail use the same year-end or uniform timing transformation. Independent finite constructed-FCFF and beginning-capital EP streams retain the terminal residual before comparing with the closed value. Agreement verifies this conditional valuation representation.

## S12. Complete restrictions and numerical support coverage

### S12.1. Lambda-dependent future-margin admission

For zero fade, equation (S22) gives

$$m \le \frac{M \Gamma_T}{\sup_{n \ge 1} \Gamma_{T+n}}. \text{(S37)}$$

The supremum can occur at a finite year or at the limiting ratio. A limiting equality is a restriction on all finite future margins, not a finite lambda member. For positive lambda, the same condition can be expressed as

$$s_T \le \inf_{n \ge 1} \left( \frac{M}{\Gamma_{T+n}} - w \right) e^{\lambda n}. \text{(S38)}$$

At a fixed lambda this is an upper terminal-margin half-plane, which intersects the existing margin-guidance polygon. Under strictly positive headroom $M / \Gamma_n - w > 0$, these cap domains expand with lambda. Hence declining value on each positive-spread path does not make the constrained maximum envelope globally decreasing.

After J, the ratio is a constant plus two decaying geometric terms:

$$\Gamma_{J+n} = \frac{G_J (1+a)}{2 a R_J} + \frac{H_J (1+v)}{2 a R_J} (v/a)^{n-1} + \frac{F_J}{a R_J} (1/a)^{n-1}. \text{(S39)}$$

Exact discrete ratio extrema, the finite prefix and the limit cover zero fade. At positive fade, an upper bound on every omitted ratio combines the constant with positive parts of the remaining decaying terms. Once $(M / \Gamma^{upper} - w) e^{\lambda(N+1)}$ exceeds the current candidate spread cap, all later inequalities are weaker. Enumeration then terminates with a tail certificate. A finite record of margins alone would not justify complete admission.

If $w \Gamma_\infty > M$, every positive lambda eventually violates the ceiling, even when a zero-fade constant-return path satisfies it. Zero fade must be evaluated separately. If all admitted terminal spreads are negative, finite values approach their D limit from below; an infinite-decay supremum below P supplies exclusion, but cannot be reported as a finite rate or an attained maximum.

### S12.2. Cellwise feasibility and exclusion

With strictly below-price infinite-limit maxima and positive headroom, let $[a, b]$ be a lambda cell. The domain admitted at b contains domains at smaller rates. A positive-spread path has its largest value at a; a negative-spread path is bounded above by its D limit. Thus a valuation envelope at a over the b-admitted domain, combined with the limit bound, covers every admissible value in the cell from above.

For feasibility throughout the cell, an above-price witness evaluated at b and a below-price witness evaluated at a must belong to the same connected component admitted at a. The above-price witness has positive spread because all D values are below P; it remains above price earlier in the cell. A below-price witness remains below price later, either by declining or by increasing toward a below-price D limit. Both remain admitted as the domain expands, and continuity supplies equality.

The generalized calculation also covers cells below which zero-fade admission is empty and cases without a common low witness at zero. Lower bounds from endpoint minima over enlarged domains can exclude cells whose every attainable value is above P. No sampled absence of a root is interpreted as exclusion.

### S12.3. Precision and scope

The baseline analytical reduction supports a sharp conditional projection under its premises. The later computations supply numerical inner/outer sets and maximum-value brackets. Their lambda boundary resolution is at most $2 \times 10^{-9}$, with ordinary high-precision arithmetic and binary64 envelope guards rather than formally outward-rounded real arithmetic. Interior coverage is retained separately from unresolved boundary brackets.

Maximum brackets have attained finite inner witnesses and upper bounds covering all relevant lambda cells and the remaining limiting region; their target width is no more than $10^{-5}$ of the fixed EV proxy plus the stated numerical guard. In an all-negative-spread case the upper object is an unattained supremum. It is explicitly labeled rather than forced into a finite maximizing parameter.

Numerical roots require residual below $0.001. The $0.02 coverage guard, margin-activity threshold and bracket width are computational devices only. They neither expand exact equality into a price band nor calibrate future operating or market-measurement error. Detailed saved residuals, witnesses and source provenance remain in the companion replication record.

## S13. Extended comparisons and replication record

### S13.1. Earlier illustrative margin relaxations

**Table S23. Earlier illustrative terminal-margin relaxations**

| Illustrative terminal-cap change | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|
| +2 percentage points | 8/27 | 0/27 | 1/27 | 27/27 |
| +5 percentage points | 13/27 | 0/27 | 3/27 | 27/27 |
| +10 percentage points | 23/27 | 0/27 | 10/27 | 27/27 |

*Notes:* Original revenue/capital completion and 27 maintained cases. These expansions are illustrative adjusted NOPAT margin changes, not validated economic supports. Their Box matches are counterexamples to a universal exclusion claim, not reasons to choose the relaxed cap.

An earlier immediate-growth investment comparison with complete future ceilings admits CRM in three cases and HUBS in 26. The gradual-growth investment profiles here retain HUBS in 27. The difference is a revenue-completion difference; the newer result does not silently replace the older immediate-growth finding.

### S13.2. Complete benchmark case sets

**Table S24. Complete historical-benchmark case supports**

| WACC case | Terminal growth case | Timing | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|---|---|
| LOW | LOW | uniform first | Empty | Empty | Empty | [0, ≈0.062963] |
| LOW | LOW | year end | Empty | Empty | Empty | [0, ≈0.073818] |
| LOW | LOW | uniform all | Empty | Empty | Empty | [0, ≈0.077217] |
| LOW | BASE | uniform first | Empty | Empty | Empty | [0, ≈0.082589] |
| LOW | BASE | year end | Empty | Empty | Empty | [0, ≈0.094531] |
| LOW | BASE | uniform all | [0, ≈0.000664] | Empty | Empty | [0, ≈0.098279] |
| LOW | HIGH | uniform first | [0, ≈0.007110] | Empty | Empty | [0, ≈0.103211] |
| LOW | HIGH | year end | [0, ≈0.008572] | Empty | Empty | [0, ≈0.116420] |
| LOW | HIGH | uniform all | [0, ≈0.011984] | Empty | Empty | [0, ≈0.120578] |
| BASE | LOW | uniform first | Empty | Empty | Empty | [0, ≈0.039870] |
| BASE | LOW | year end | Empty | Empty | Empty | [0, ≈0.049194] |
| BASE | LOW | uniform all | Empty | Empty | Empty | [0, ≈0.053478] |
| BASE | BASE | uniform first | Empty | Empty | Empty | [0, ≈0.058189] |

| WACC case | Terminal growth case | Timing | CRM | NOW | BOX | HUBS |
|---|---|---|---|---|---|---|
| BASE | BASE | year end | Empty | Empty | Empty | [0, ≈0.068399] |
| BASE | BASE | uniform all | Empty | Empty | Empty | [0, ≈0.073101] |
| BASE | HIGH | uniform first | Empty | Empty | Empty | [0, ≈0.077266] |
| BASE | HIGH | year end | Empty | Empty | Empty | [0, ≈0.088494] |
| BASE | HIGH | uniform all | Empty | Empty | Empty | [0, ≈0.093678] |
| HIGH | LOW | uniform first | Empty | Empty | Empty | [0, ≈0.019341] |
| HIGH | LOW | year end | Empty | Empty | Empty | [0, ≈0.026647] |
| HIGH | LOW | uniform all | Empty | Empty | Empty | [0, ≈0.031531] |
| HIGH | BASE | uniform first | Empty | Empty | Empty | [0, ≈0.035824] |
| HIGH | BASE | year end | Empty | Empty | Empty | [0, ≈0.044610] |
| HIGH | BASE | uniform all | Empty | Empty | Empty | [0, ≈0.049949] |
| HIGH | HIGH | uniform first | Empty | Empty | Empty | [0, ≈0.053646] |
| HIGH | HIGH | year end | Empty | Empty | Empty | [0, ≈0.063262] |
| HIGH | HIGH | uniform all | Empty | Empty | Empty | [0, ≈0.069120] |

*Notes:* Growth LOW/BASE/HIGH means 2%, 3%, 4%; WACC LOW/BASE/HIGH means firm base minus 100bp, base, plus 100bp. Uniform conventions are those defined in §S4.1. Values are rounded per-year upper endpoints, not point estimates. The benchmark component theorem and complete-domain coverage support each classification.

### S13.3. Alternating book-investment diagnostics

### Table S25. Alternating book-investment diagnostics

| Firm | Revenue completion | Alternating coefficient (USD bn) | Investment amplitude (USD bn) | Minimum net investment/R (%) | Release years / 100 |
|---|---|---|---|---|---|
| CRM | Immediate terminal growth | -3.133 | 6.266 | -8.52 | 18 |
| CRM | Guidance H5 | -0.814 | 1.629 | 1.96 | 0 |
| NOW | Immediate terminal growth | -0.046 | 0.091 | 1.28 | 0 |
| NOW | Guidance H5 | -1.204 | 2.408 | -7.50 | 29 |
| BOX | Immediate terminal growth | -0.020 | 0.039 | -0.63 | 5 |
| BOX | Guidance H5 | -0.024 | 0.048 | -0.58 | 4 |
| HUBS | Immediate terminal growth | -0.069 | 0.138 | -1.84 | 12 |
| HUBS | Guidance H5 | -0.141 | 0.282 | -3.43 | 19 |

*Notes:* These are recorded base-case maximizing witnesses, not minima over every operating path. The release count uses 100 modeled years. Alternating amplitude is $2|B_J|$. Negative capital additions are constructed book releases and are not verified cash distributions.

### S13.4. Principal upper-endpoint numerical brackets

**Table S26. Principal persistence-rate endpoint enclosures**

| Specification | Firm | Cross-case upper endpoint bracket | Saved bracket width |
|---|---|---|---|
| Immediate growth | CRM | [0.011983772, 0.011983773] | 8.67E-19 |
| Immediate growth | HUBS | [0.120577716, 0.120577717] | 8.67E-19 |
| Growth: guidance H3 | CRM | [0.017393364, 0.017393365] | 8.67E-19 |
| Growth: guidance H3 | HUBS | [0.165869913, 0.165869914] | 8.67E-19 |
| Growth: guidance H5 | CRM | [0.023532533, 0.023532534] | 8.67E-19 |
| Growth: guidance H5 | HUBS | [0.221134093, 0.221134094] | 8.67E-19 |
| Investment: guidance H3 | CRM | [0.013351875, 0.013351876] | 5.82E-11 |
| Investment: guidance H3 | HUBS | [0.153938773, 0.153938774] | 5.82E-11 |
| Investment: guidance H5 | CRM | [0.020198324, 0.020198325] | 5.82E-11 |
| Investment: guidance H5 | HUBS | [0.216164974, 0.216164975] | 5.82E-11 |
| Peers: guidance H3 | CRM | Empty | Empty |
| Peers: guidance H3 | HUBS | [0.057773093, 0.057773094] | 1.16E-10 |
| Peers: guidance H5 | CRM | Empty | Empty |
| Peers: guidance H5 | HUBS | [0.087010854, 0.087010855] | 5.82E-11 |

*Notes:* Display limits are rounded outward to nine decimal places from saved inner/outer endpoints. This is presentation bracketing, not formal outward-rounded arithmetic certification. Empty cases contribute no rate. Original mathematical sharpness under stated premises is distinct from numerical approximations and unvalidated economic support.

### S13.5. Data dictionary, source closure and reproducibility

The replication record separates five objects: annual accounting observations and eligibility; dated financial/guidance evidence; the benchmark continuous-case table; the growth and investment case tables; and the historical-support case/evidence tables. Each case has a firm, revenue profile, capital/operating specification, actual WACC/g/timing, fixed target, compatibility status, support/brackets, active constraints and diagnostic states. Blank lambda fields represent empty support, not zero. Unmeasured cash/claim fields remain unavailable rather than becoming numerical zero.

The operating-component study is an additional evidence object, separate from those valuation records. Section S18 links its sealed plan, margin-only source/dependency verification and signed annual endpoint differences. No valuation experiment or full pipeline replay was performed for the present integration.

Original and reconstructed specifications are preserved separately. Reused benchmark rows are not additional independent economic observations. A case is classified compatible only after complete operating admission and value equality within declared numerical accuracy. Raw relaxed-domain witnesses are retained as diagnostics and visibly excluded from supported counts.

Figures use saved records only. Figure S1 reads existing revenue/capital states, supplemented by the observed origin; the matched capital panel preserves the same revenue path. Figure S2 reads the rejected Box witness’s existing margin sequence and the declared ceiling. Figure S3 aggregates compatibility flags. No new frontier sampling, future outcome, price fit or identification experiment is introduced by their production.

The accompanying replication package provides archived input and code dependencies, source hashes, a table-figure crosswalk and portable replay entry points. A standard-library driver replaces only CSV export transport in memory, leaving numerical implementations, inputs, calibrations and tolerances unchanged. The growth, investment and operating-support pipelines completed at two isolated locations with cache reuse disabled for the operating-support calculations. Their 556, 972 and 864 output rows include reused comparator records: the investment pipeline newly evaluates 432 cases, and the support pipeline newly evaluates 648. Case keys,

compatibility, topology labels, compared scalar missingness and persistence endpoints agree with the saved references; the largest compared valuation difference is less than $0.001. These are repeated conditional calculations, not independent financial observations or economic validation. Detailed receipts, comparison criteria and export audits are supplied in the verification report. Original entry points should be invoked through the portable driver to avoid their legacy export dependencies.

Replication should use a new output destination and preserve reference files. A reader must distinguish reproducing calculations from validating economic assumptions. The code and evidence make the exercise challengeable, but successful reproduction does not establish distributable cash equivalence, a synchronized market object or a calibrated future support.

The target-only sensitivity supplement normalizes the existing cross-case upper-envelope shortfalls. It identifies reductions of the fixed target below which exclusion remains, while holding all operating and valuation inputs fixed. Its numerical brackets do not establish sufficient existence of a match at the threshold or represent a coordinated correction to operating states, claims, taxes, cash flows, WACC or future profitability supports. The package has not been publicly deposited; raw third-party documents are not bundled, and redistribution rights remain subject to review.

### S13.6. Duration translation and preserved rate summary

**Table S27. Preserved principal persistence-rate boundaries**

| Specification | CRM cross-case upper | HUBS cross-case upper | HUBS base-case upper | HUBS all-case common |
|---|---|---|---|---|
| Immediate growth | ≈0.011984 | ≈0.120578 | ≈0.058189 | ≈0.019341 |
| Growth: guidance H3 | ≈0.017393 | ≈0.165870 | ≈0.090467 | ≈0.046149 |
| Growth: guidance H5 | ≈0.023533 | ≈0.221134 | ≈0.129280 | ≈0.079419 |
| Investment: guidance H3 | ≈0.013352 | ≈0.153939 | ≈0.084376 | ≈0.039163 |
| Investment: guidance H5 | ≈0.020198 | ≈0.216165 | ≈0.121514 | ≈0.088815 |
| Peers: guidance H3 | Empty | ≈0.057773 | Empty | Empty |
| Peers: guidance H5 | Empty | ≈0.087011 | ≈0.022838 | Empty |

*Notes:* Rates are per year. The cross-case upper summarizes a profile-specific union; the common upper applies only when all 27 case sets are nonempty. NOW and BOX are empty in every specification shown. Approximation marks refer to rounded conditional endpoints or numerical brackets, not economic uncertainty intervals. Outward-rounded technical brackets are in Section S13.

The authoritative source selections are D7 BASELINE_RECURRENCE for ORIGINAL/GUIDE_H3/GUIDE_H5, D8 INVESTMENT_FULL_CEILING for GUIDE_H3/GUIDE_H5, and D9 MULTIYEAR_PEERS for GUIDE_H3/GUIDE_H5. Each firm/profile selection contains exactly 27 cases. Raw diagnostic investment rows and reused D9 comparators are excluded from these selections. Base filters are BASE/BASE/UNIFORM_FIRST. Union endpoints take maxima of case upper bounds; common endpoints take minima only when all 27 sets are nonempty. Saved inner and outer bounds are transformed separately, reversing their order under the reciprocal: [ln(2)/L_outer, ln(2)/L_inner]. No new support topology is inferred from this transformation.

### HubSpot base-case reciprocal brackets (years after T)

**Table S28. HubSpot base-case reciprocal brackets (years after T)**

| Family | Saved upper-rate bracket | Minimum half-life bracket |
|---|---|---|
| Immediate growth | [0.058188582503556685, 0.058188582503556685] | [11.912082246, 11.912082246] |
| Investment: guidance H5 | [0.121514342550653964, 0.121514342608861625] | [5.704241702, 5.704241705] |
| Peers: guidance H5 | [0.022837707423605025, 0.022837707598228008] | [30.350996376, 30.350996609] |

Full serialized decimal endpoints and reciprocal results are retained in DURATION_CASE_BOUNDS.csv and DURATION_SUMMARY.csv; the displayed rate and half-life brackets are approximate decimal presentations. The source calculation uses 60-digit Decimal ln(2) and archived endpoints, not formally outward-rounded real arithmetic. The original numerical certification limits in Section S12.3 continue to apply. For L = 0 there is no finite half-life; for empty support the duration is undefined because there is no admitted match. This task transforms saved results only.

## S14. Literature access

The references retain the published or working-paper versions actually identified. Foundational accounting and implied-cost studies support the broad contextual statements at their recorded source scope, not an independent reproduction of their empirical methods. Inspected archived R&D and forecasting materials inform the measurement conventions and interpretation; broader literature does not supply a missing issuer-specific technology or tax basis.

The 2016 Li-Hall working-paper scalar is distinguished from later publication versions. Author-draft and final-publication evidence are not silently interchanged. Mills is used for the recorded competitive-advantage-period application; Vlčková-Buus supplies the identified topical precedent, not proof of equivalence to this conditional-set construction.

Garcia-Estevez-Puente Vicente remains a metadata/abstract-level comparison. Its exact persistence law, capital charges, investment completion, failed-root treatment, nuisance projection and predictive tests are unverified here. Full-text access failure is not evidence that the paper lacks any of those analyses. No priority over it is claimed.

## S15. Matched-attribution design, restrictions and verification

### S15.1. Cells, admission differences and controls

**Table S29. Matched attribution cell definitions**

| Cell | Guidance | Upper restrictions | Transition floor | Role |
|---|---|---|---|---|
| H | Historical bridge | Old nominal caps | Old | Published diagnostic |
| F | Historical bridge | Old nominal caps | Common | Isolate floor |
| U | Coherent bridge | Old nominal caps | Old | Detect floor conflict |
| G | Coherent bridge | Old nominal caps | Common | Matched comparator |
| O / P | Coherent bridge | Own / peer caps | Common | Support effect |
| RO / RP | Historical bridge | Fixed own / peer caps | Common | Reverse-order diagnostic |

*Notes:* Common floor=min(old annual floor, coherent guidance lower); only NOW changes. Terminal lower bounds stay fixed. O/P and RO/RP use the exact saved cap values, including their existing coherent-guidance hulls. Caps are not recalibrated under historical guidance. The optional paired endpoint screen is off, because it also changes joint margin/turnover and growth admission; its original results in §S7.3 remain a distinct diagnostic. The fixed origin, modeled investment and inferred turnover are not empirically validated by matching these controls.

**Table S30. Matched H5 guidance, floor and upper restrictions**

| Firm | Historical guide (%) | Coherent guide (%) | Old floor → common (%) | Old terminal/future cap (%) | Own terminal/future (%) | Peer terminal/future (%) |
|---|---|---|---|---|---|---|
| CRM | 18.775-18.958 | 18.843-18.843 | 14.272 → 14.272 | 30.308 | 18.843 / 18.843 | 24.076 / 24.076 |
| NOW | 14.898-16.530 | 15.080-15.080 | 15.340 → 15.080 | 29.200 | 17.776 / 17.850 | 23.448 / 23.448 |
| BOX | 11.210-11.872 | 11.990-11.990 | 7.719 → 7.719 | 23.102 | 11.990 / 11.990 | 20.220 / 20.220 |
| HUBS | 14.594-17.231 | 13.723-13.808 | 5.384 → 5.384 | 28.580 | 13.808 / 13.808 | 21.552 / 21.552 |

*Notes:* Adjusted NOPAT margins, percent. The old terminal and transition/future upper caps coincide; own and peer columns show terminal / transition-and-future caps. NOW own-history retains its origin through the 17.850% future/transition ceiling while its terminal cap is 17.776%. Annual floors apply through T, including the fixed origin; no new future lower-margin restriction is introduced. All values are display-rounded; exact dictionaries and source controls are retained in CHANGED_RESTRICTIONS.csv and CALIBRATION_AND_RESTRICTIONS.json.

### S15.2. Maximum-value difference bounds

**Table S31. Maximum-value difference enclosures in the attribution study**

| Firm | Case | Guidance F → G | Own caps G → O | Peer caps G → P |
|---|---|---|---|---|
| CRM | Base | [-0.018805, -0.014840] | [-81.637518, -81.633640] | [-45.055976, -45.052014] |
| CRM | Favorable | [-0.018819, -0.014875] | [-118.043922, -118.040442] | [-64.804522, -64.800572] |
| NOW | Base | [-0.063727, -0.061050] | [-31.714578, -31.711824] | [-16.258622, -16.255861] |
| NOW | Favorable | [-0.056280, -0.053571] | [-46.383222, -46.380501] | [-23.772427, -23.769674] |
| BOX | Base | [0.000428, 0.000536] | [-2.073443, -2.073339] | [-0.547903, -0.547794] |
| BOX | Favorable | [0.000429, 0.000537] | [-3.002983, -3.002874] | [-0.788103, -0.787993] |
| HUBS | Base | [-0.118945, -0.118763] | [-8.486121, -8.485936] | [-4.091051, -4.090867] |
| HUBS | Favorable | [-0.091344, -0.091159] | [-12.408680, -12.408495] | [-5.978096, -5.977911] |

*Notes:* USD billions. Guidance is F → G; cap effects are G → O/P. Difference bounds subtract the opposite original/new inner and outer endpoints and are rounded outward to six decimals for display. They remain numerical brackets rather than statistical or economic uncertainty. Base is BASE/3%/UNIFORM_FIRST; favorable is the prespecified LOW/4%/UNIFORM_ALL corner. CASE_CONTRASTS.csv retains all 27 cases, full decimal bounds, matched controls and failed-domain missingness. INTERACTION_BOUNDS.csv retains the four-cell contrasts, with HUBS own-family interaction undefined.

### S15.3. Prespecification and focused validation

The comparison plan and SHA-256 seal preceded evaluation of newly calculated cells; the H/O/P comparator results were already known. Prespecification did not make those reused results prospective observations. The initial eight representative firm/cases triggered full-menu expansion by the declared criteria: a compatibility change, a bounded maximum change wholly exceeding 1% of the fixed target, or an operating-domain inconsistency requiring menu-wide reporting. All four firms and all core cells were expanded; the criteria were not tuned after outcomes. The 1% criterion is substantive and distinct from numerical tolerances.

The resulting 864 catalogue rows include 324 new nonempty-domain valuation comparisons, 54 structural admission refusals, 24 focused reproductions of saved H/O/P cases, 300 reused saved rows and 162 exact specification aliases. They are not independent financial observations. No complete D7/D8/D9 replay or sample

expansion was performed. Full new case certificates are retained individually; existing coherent O/P records are reused only after exact control/restriction checks and representative reproduction.

All 18 original capital/operating focused tests and seven new matching/adapter tests pass. The 24 focused saved-case reproductions agree on classification/topology, maximum values within $0.001 and rate endpoints within 2e-9. New calculations verify matching revenue/capital paths, stock identities, positive capital/investment, guidance/tax algebra and complete future ceilings. Supported endpoint and interior equality witnesses retain residual below $0.001, and representative paths also pass independent 1500-year constructed-FCFF/EP checks. Signed and zero spreads, zero-only operating fade, limiting values and fail-closed topology assertions retain their existing treatment.

The numerical guard remains $0.02, lambda-cell width 1e-10, endpoint width at most 2e-9 and maximum bracket target 1e-5 of EV plus the existing guard. No tolerance is loosened. A separate one-expression adapter makes transition floors explicit; original code and all 320 isolated source-manifest files remain unchanged. Source/protection hashes are verified before and after execution. Successful checks establish conditional numerical consistency, not distributable cash, current investor beliefs or empirically validated economic support.

## S16. Normalized-economic-profit fade: valuation and admission

### S16.1. Definition, capital growth and matched estimand

Let $\acute{C}_j=(C_{j-1}+C_j)/2$, $r_j=Y_j/\acute{C}_j$, and $EP_j=Y_j-wC_{j-1}$. The alternative defines

$$e_j=\frac{EP_j}{\acute{C}_j}=r_j-w\frac{C_{j-1}}{\acute{C}_j},\ e_T=\frac{mR_T-wC_{T-1}}{\acute{C}_T},\ e_{T+n}=e_Te^{-\lambda n}. \quad \text{(S40)}$$

The terminal margin, transition prefix, coherent guidance and stock accumulation remain matched for the same operating parameters. Future NOPAT becomes

$$Y_{T+n}=wC_{T+n-1}+e_T\acute{C}_{T+n}e^{-\lambda n}. \quad \text{(S41)}$$

The fade clock is n years after T, including any remaining finite revenue prefix. At positive lambda a nonzero normalized-EP spread halves after ln(2)/lambda; zero rate and zero spread retain their separate meanings. On the asymptotic capital growth mode a=1+g, EP break-even average-capital return is $2w/(1+a)$, below w when g>0. For positive e_T, dollar EP grows asymptotically at factor $ae^{-\lambda}$; a declining rate is not a declining-dollar-EP condition.

For the same terminal m, the old/new future NOPAT difference is

$$Y_j^{EP}-Y_j^{ROIC}=-\frac{w}{2}\Delta C_j+\frac{w\Delta C_T}{2\acute{C}_T}\acute{C}_je^{-\lambda(j-T)},\ j>T. \quad \text{(S42)}$$

It can change sign through the path. The new definition is not a discovered error in the old convention, and its parameters measure a different spread.

### S16.2. Finite identity and exact split-stock tail

The beginning-capital identity in Section S9.1 remains $PV(F_{1:H})=C_0+PV(EP_{1:H})-C_H/q^H$ at year end. Terminal capital is retained for finite checks; transversality uses the admitted mode growth below q=1+w, not an arbitrary truncation. It verifies the constructed representation, not observed cash equivalence.

For J=max(T,H_R), let $C_{J+n}=\sum_\ell c_\ell d_\ell^n$, with $d_\ell\in\{1+g,.692,1\}$ and z=exp(−lambda). Average capital is $\sum_\ell c_\ell(1+d_\ell)d_\ell^{n-1}/2$. With $e_J=e_Tz^{J-T}$, the exact year-end FCFF tail valued at J is

$$Tail_J^{EP} = C_J + e_J \sum_{\ell} \frac{c_\ell (1 + d_\ell) z}{2(q - d_\ell z)}. \text{(S43)}$$

Each zero-EP mode has coefficient $[w - (d_\ell - 1)]/(q - d_\ell) = 1$, yielding C_J. The finite prefix is explicit and the inherited fade offset is retained. The unchanged timing transform applies to the whole PV and its actual first-year FCFF; the new capital base and endpoint charge replace the old zero-spread base and average-capital charge together.

Writing $h(\lambda) = PV_{j>T}(\acute{C}_j z^{j-T})/\acute{C}_T > 0$, the raw value is affine:

$$V^{EP} = A_m m + A_b b + A_0 + D^{EP} + (m R_T - w C_{T-1}) h(\lambda), h'(\lambda) < 0. \text{(S44)}$$

Signed monotonicity depends on e_T; zero spread makes value independent of lambda on that fixed path. Equality only at an infinite limit supplies no finite member. The fixed-path inversion checks retain that distinction, including the all-finite-rate zero-spread degeneracy.

**S16.3. Complete future admission and numerical coverage**

Let $\Gamma_n = \acute{C}_{T+n}/R_{T+n}$, including $\Gamma_0 = \acute{C}_T/R_T$, and $B_n = C_{T+n-1}/R_{T+n}$. The alternative margin is $w B_n + e_T \Gamma_n e^{-\lambda n}$. Complete future upper admission requires

$$e_T \le \inf_{n \ge 1} \frac{M - w B_n}{\Gamma_n} e^{\lambda n}, m \le \frac{w C_{T-1}}{R_T} + \Gamma_0 \inf_{n \ge 1} \frac{M - w B_n}{\Gamma_n} e^{\lambda n}. \text{(S45)}$$

This replaces the old wΓ baseline. Endpoint/guidance/transition inequalities and terminal lower bounds remain fixed. After J, B and Γ each have a constant and two decaying geometric modes. At zero fade, the actual margin also has this form, whose discrete extrema occur at the first tail year, the limit and floor/ceiling of its continuous stationary location. Tightening a candidate spread cap at a violating discrete maximum until that candidate passes the global maximum supplies complete zero-fade admission; nontermination would be unresolved.

For positive lambda with positive baseline headroom, omitted-year upper bounds B^u and Γ^u use constant plus positive parts of remaining modal coefficients. Once M−wB^u>0,

$$\frac{M - w B^u}{\Gamma^u} e^{\lambda(N+1)} \text{(S46)}$$

is a lower bound on every omitted spread ceiling. Enumeration stops only when that lower bound exceeds the current cap, with finite inequalities and tail bounds rechecked in Decimal arithmetic. This covers signed spreads; it is not a finite simulated-path certificate alone. If wB_infinity>M, every positive fade eventually fails and zero fade is evaluated separately. Unsupported mixed-headroom/topology cases fail closed rather than receive forced classifications.

The inherited cell-coverage argument is retained only after replacing its signed-spread premise and checking the new headroom. Equality residual remains below $0.001, guard $0.02, cell width 1e-10, endpoint width at most 2e-9 and maximum target width 1e-5 of EV plus the existing guard. These are numerical controls; no formal outward-rounded arithmetic or economic-error interval is claimed.

**S16.4. Pilot validation and stopping rule**

All nine focused benchmark tests pass: independent finite/infinite FCFF and beginning-capital EP streams, terminal-stock residual, exact original/new NOPAT difference, matched prefixes/capital, signed and zero spreads, lambda zero, excluded infinite limits, zero-only admission geometry, and declining rates with growing dollar EP. Three diagnostic tests separately verify sampled equality, full-path failure marking and analytic/numerically

bounded future peaks. All 24 reference pilot reproductions agree with saved classifications/topology, maxima within $0.001 and rates within 2e-9; all 24 new cases check endpoint/interior witnesses and 1500-year streams.

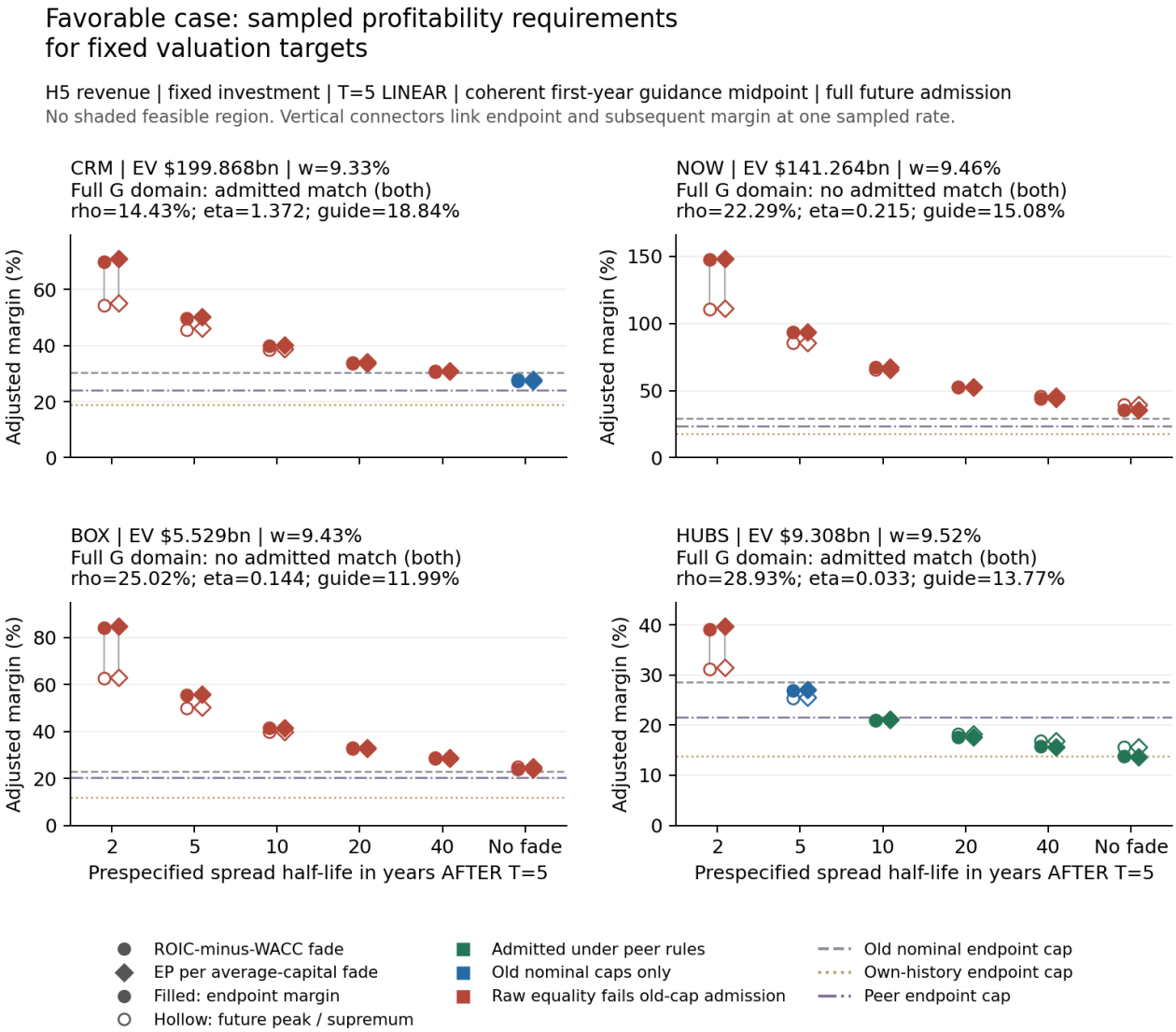
The SHA-256 sealed plan precedes results. None of its three substantive triggers occurs, so the alternative full 27-case menu is not evaluated. Maximum-value changes stay below 1% of fixed EV and minimum-half-life shifts stay below 20%, although two exceed one year. The null result is confined to these selected cases, support rules and coherent-guidance investment completion. Wider repetition is not justified by this rule.

## S17. Sampled requirement diagnostic and accessible records

The inverse requirement at fixed guide b and transition T solves equation (S44) for m. The original benchmark uses its own verified capital base and endpoint charge. All 96 prespecified points retain positive value coefficients and raw equality residual below $0.001. Price fitting is not used to choose a growth, risk, investment or support input.

Every point records endpoint margin, full future maximum/supremum bracket, finite-year versus limiting attainment, and all G/O/P transition/future failures. Zero-fade peaks use exact discrete extrema plus the limit; positive-fade peaks enumerate only until an omitted-tail upper bound falls below the current peak plus 1e-10 in margin units. Limit-only peaks remain labeled suprema. Later margins can be lower than the endpoint. Standalone CSV/JSON and scripts make the display reproducible.

**Figure S5. Prespecified favorable-case sampled requirements**




*Preserved Figure S5*

*Notes:* Same point/marker conventions as Figure S4. The declared favorable corner uses LOW WACC, g=4% and UNIFORM_ALL. It is not a newly selected case. No continuous region is inferred from samples. Full case domains permit other transition/guidance witnesses; red samples in a compatible case remain legitimate failed requirements.

## S18. Operating-support feasibility and annual component evidence

### S18.1. Tested components and their relation to D9

The two targets are fixed FY2025 origins: Workday on March 11 and Docusign on March 18, 2025. Own training windows are FY2020-25 and FY2023-25 respectively. Write m0 for the origin adjusted NOPAT margin. The tested history-only annual ceiling is H=max(m0, maximum available own adjacent-year mean). For each target-excluded donor firm, take the median of max(0, end margin minus start margin) over eligible two-/three-year paths; the interpolated 90th percentile across those firm medians is h. Separately, the interpolated 90th percentile across donor-firm medians of mature two-year margin means is L. The terminal component is C=min(L,m0+h), and the tested peer annual component is Q=max(m0,C). Q corresponds to the study's P notation, without reusing

this paper's valuation-target symbol P. Guidance is deliberately omitted, not set to a numerical zero. D9's actual guidance-inclusive rule remains equation (S24), with its separately defined full future upper restriction.

Mature endpoints require revenue at least USD1bn, two consecutive strictly positive GAAP operating profits, two nonnegative annual growth rates no greater than 20%, and growth-change magnitude no greater than 5pp. Gain paths require every annual record and strictly positive GAAP OP throughout, ending at a mature endpoint; the start need not be mature. Both gain and level pools require at least three distinct target-excluded firms. All available eligible paths are retained; negative gains are clipped to zero before donor medians, not dropped. Workday's gain/level donors are BOX, CRM and ZM; Docusign's gain donors are the same, while its level pool also includes WDAY. No future endpoint maturity or margin selects the two target episodes.

An annual comparison with H or Q documents containment or exceedance of those explicitly annualized components. It is not a test of a subsequent two-year mean or of the complete operating path at a modeled transition T. No second-year outcome is imputed. Supplement Table S19 retains the annual endpoint summary; main-article Table 6 presents its compact counterpart.

**S18.2. Source scope, dependency verification and timing**

The plan was sealed on October 13, 2026 at 21:49:01.265369 UTC before the held-out adjusted-margin calculation (SHA-256 f4a510cd74b984d021b049d3fb275d4421e5a57a867d35a032c9e6b08817fdda). Verification uses original archived official statements, source hashes and actual filing/publication metadata, with current official extracts separately labeled. Only sources available by each fixed origin enter training. Later comparative restatements or corrections are not backdated.

Both episodes passed margin-specific verification. The study resolves 405 repeated margin/maturity dependency links; 384 book-capital/turnover links and 84 other full-row links remain outside these formulas. Margin construction needs revenue, GAAP OP, eligible R&D and its stock/correction chain, period lengths, initialization, and maturity's lagged revenues/profits. It does not need NOA or average capital. No turnover restriction, financing-scope identity or paired-hull outcome assessment is represented as completed.

The retained annual research stock uses .692 retention, source-declared elapsed-period treatment for fiscal stubs, and the existing earliest zero-stock initialization. Adjusted OP retains GAAP compensation and adds eligible R&D less research depreciation; synthetic operating tax is .21 times positive adjusted OP. No quantified overlap correction occurs in the selected chains, which does not establish zero overlap: residual overlap remains unknown. Fixed depreciation/tax and acquired-software/overhead scope remain construction assumptions. All retained training stock/profit identities reconcile with the saved construction. Outcomes advance the verified origin stocks with actual FY2026 R&D rather than importing a later reconstructed stock.

Workday's first original FY2026 10-K is accession 0001327811-26-000014, filed March 6, 2026; its annual release was February 24. Docusign's is accession 0001261333-26-000021, filed March 18, 2026; its release was March 17. The filings' annual statements are reported in millions and thousands respectively and standardized to USD. The fiscal period is February 1, 2025 through January 31, 2026. At the two origins, 38 and 45 whole days had elapsed; the study does not reconstruct their within-year cash or operating allocation. Workday original annual filing; Docusign original annual filing.

A Salesforce FY2021 source reports a March 17, 2021 filing date but a February 24, 2022 acceptance date for accession 0001108524-22-000008. Both are preserved; availability verification uses the later date, still before either study origin, without changing donor membership or input values. Original-vintage limitations, unknown overlap and reported precision remain material qualifications.

Exact source values, locators, receipts, donor statistics, signed differences and calculation checks are retained in outputs/operating_support_research_v1/02_limited_component/: PRESPECIFIED_PLAN.md; SOURCE_LEDGER.csv; DEPENDENCY_VERIFICATION.csv; PUBLICATION_DATE_VERIFICATION.csv; TRAINING_RD_STOCK_CHECKS.csv; DONOR_FIRM_STATISTICS.csv; CASE_RESULTS.csv; OUTCOME_VERIFICATION.json;

FOCUSED_CHECKS.json. The source/dependency tables and plan are copied into this revision's study supplement; prior versions remain intact. These records provide computational and provenance checks, not economic validation.

The saved DONOR_FIRM_STATISTICS.csv lists the chronological pools separately. Workday uses BOX, CRM and ZM for both gain and level. Docusign uses BOX, CRM and ZM for gain, and BOX, CRM, WDAY and ZM for level. Thus BOX/CRM/ZM are shared donors, while WDAY is an additional Docusign level donor. These date-gated pools differ from the October valuation-calibration pools in Section S7.1 and are not additional independent samples.

### S18.3. Feasibility stopping gate and separate earlier evidence

The fixed 15-firm panel contains 72 constructed annual observations and 12 mature origins. Zoom FY2024 is the only mature origin with two subsequent annual records and has no target-excluded gain or level donor at its recorded origin date. No origin in the verified panel combines two subsequent annual records with both unchanged three-firm donor gates. Screening 182 already archived annual-flow/transition records adds no maturity-qualified endpoint outside the constructed panel; those 182 are not a clean PIT sample. Older source availability therefore does not establish an eligible validation sample. Broad collection/validation stopped at that evidence gate.

The one-year Workday/Docusign episodes cannot substitute for mature multi-year coverage. Their original FY2025 releases lack the required forward GAAP guidance reconciliation, so the limited study does not test the full guidance-inclusive D9 rule. No missing guidance, book-capital validation or investment completion is invented. Subsequent annual containment cannot certify every year of infinite-horizon admission.

An earlier, separate history-only check found exceedances in 12 of 16 overlapping origins; none met the maturity diagnostic at origin. It omitted guidance and peers and used selected frozen vintages. Those non-mature component cases are not pooled with the present mature-origin endpoints or treated as mature-peer/full-rule evidence. Both checks support treating historical observations as scenario motivation rather than established permanent economic bounds; neither estimates a population coverage probability. The earlier results are preserved in the feasibility audit's source archive and remain outside the valuation catalogue.

## S19. Completed non-R&D accumulation sensitivity

This section reports the completed design and findings; it does not reproduce instructions to the analyst. The original sealed plan remains unchanged in archival/ETA_ORIGINAL_PRESPECIFIED_PLAN.md. Its identifier is the seal UTC 2026-10-14T01:27:22.141188+00:00, with SHA-256 c0522c9aec60fa88dc58ed51e28 273f02b77f19cef5e7196a06 188a12d4715ca, verified against the retained plan. The seal preceded alternative valuation/admission evaluation, not the already known baseline results.

### S19.1. Existing evidence and fixed cells

The earlier median-three NOA/revenue calibration sensitivity used older revenue/admission conventions; it was related evidence rather than this experiment. The completed matched H5 attribution study supplied exact G/O/P restrictions and saved baseline records. Twenty-four fixed base/favorable A baselines were reproduced before alternative interpretation. Production inputs, opening stocks and saved baseline calibrations remained fixed. G used coherent guidance and original nominal terminal/transition/future ceilings; O/P replaced those ceilings with saved own-history/peer restrictions. The common floor was min(original annual floor, coherent guide lower); terminal lower bounds, annual-change limits, T=2..6, and linear/normalized-exponential shapes were retained. Paired endpoint/turnover screening was off and turnover was derived. Caps and guidance hulls were not recalibrated after changing eta.

Base meant BASE WACC, 3% terminal growth and UNIFORM_FIRST; favorable meant LOW WACC (base minus 100 bp), 4% growth and UNIFORM_ALL. The full menu crossed three WACC cases, 2/3/4% growth and YEAR_END/UNIFORM_FIRST/UNIFORM_ALL timing, retaining fixed EV and the October 6 timing offset.

**S19.2. Scenarios and identities**

A retained the exact serialized latest eta calibration of N0/R0; its serialization residual was recorded rather than replacing it with a rounded or exact-rational ratio. B used eta=(N0-GW0-AI0)/R0 in Decimal arithmetic, with source-verified disjoint opening goodwill and net externally acquired intangibles. Opening N0/K0/C0 retained historical acquisitions. No internal software, synthetic research capital or overlapping balance was subtracted again. Missing category/scope evidence made B unavailable; negative eta was not floored. The maintained domain required N>=0, K>0 and C>0. With positive terminal growth, negative eta implied eventual negative N, documented by finite crossing certificates rather than a truncated admissible tail. No signed-N alternative was introduced. C set eta=0 and held N constant while research accumulation continued; it was an extreme diagnostic.

For each permitted alternative, revenue retained matched H5 controls, N_j=N0+eta(R_j-R0), K_j=.692K_(j-1)+rho R_j, C=N+K, average_C=(C_previous+C)/2, investment=delta_C, FCFF=NOPAT-delta_C and EP=NOPAT-w C_previous. Depreciation .308, latest rho, normalized tax .21, transition margins and guidance translation were unchanged. At T, s_T=m_T R_T/average_C_T-w; subsequent ROIC was w+s_T exp[-lambda(j-T)]. Annual fade began after T; signed/zero spreads retained their original interpretation.

At (J=max(T,5)), the notation of Section S11.2 gave (a=1+g), (v=.692), (U_J= ho R_Ja/(a-v)), (G_J=eta R_J+U_J), (H_J=K_J-U_J), and (F_J=N_J-eta R_J). Ending capital was (G_Ja$^{h+H_Jv}$h+F_J). These modal coefficients were distinct from normalized ratio coefficients and scenario names A/B/C. Valuation tails and (Gamma=ar C/R) geometry, discrete extrema, lambda-dependent ceilings and omitted-tail certificates were recomputed rather than copied from A. Zero-only/asymptotic exclusions and empty polygons were retained. Under g<w, discounted capital vanished; finite FCFF/EP differences equaled the terminal discounted-capital term under the declared timing, and infinite representations agreed. Infinite decay remained an excluded limit.

**S19.3. Pilot, expansion and unavailable cases**

The pilot crossed four firms, two fixed corners and G/O/P: 24 combinations per scenario. Source unavailability, negative-N failures, nonpositive capital, empty admission, unsupported numerical premises and limiting suprema were distinguished; unsupported topology was not classified as economic exclusion. Expansion to all 27 cases for a firm/cell/scenario comparison was triggered by a changed classification, a maximum-difference bracket wholly beyond +/-1% of fixed EV, or changed topology/structural admission. A threshold inside numerical uncertainty triggered conservative expansion. Structural failures received menu-wide analytic records with saved A reused, rather than repeated valuation solves. The 1% threshold was computational triage, not significance. Unavailable sources were not imputed. The design expanded no firms, guidance profiles or other parameters. Section S19.7 reports actual expansions and results.

**S19.4. Checks and attribution**

The retained standards were equality error below USD .001, envelope guard USD .02, lambda-cell width 1e-10, endpoint enclosure width at most 2e-9 and maximum enclosure at most 1e-5 of fixed EV plus guard, with fail-closed topology assertions. Pilot A classifications/topologies/maxima reproduced saved records within USD .001 and rate endpoints within 2e-9. Checks covered N identities, research recurrence, C/average_C, mode reconstruction beyond J, positive stocks and tail investment for admitted paths, Decimal versus float envelopes, full future-margin certificates, equality endpoints/interiors, representative 1500-year FCFF/EP tails, zero/signed spreads, rejected negative lambda and excluded infinite decay. Negative-eta failures used explicit asymptotic N proofs; intermediate total-capital positivity was not established for B (Section S19.8). Protection hashes and calibration/restriction seals were checked. Focused tests were used; the full production pipeline was not replayed.

Saved A maximum witnesses were also reevaluated at fixed T, shape, endpoint/guide margins and lambda under alternative capital laws. Raw values, spread, full-future cap/slack and admission were recorded separately from reoptimized maxima. This diagnostic distinguished valuation and admission channels without claiming additive causal shares. Detailed completed checks and their limits are in Section S19.10.

### S19.5. Interpretation

The completed experiment established conditional sensitivity to the future non-R&D book-accumulation coefficient, not the correct investment policy, organic marginal cash requirements, acquisition-induced mispricing or investor beliefs. C was not a realistic forecast or a bound on inadmissible B cases. Null classifications applied only within tested controls; empty support meant no admitted match. Supported [0,L] sets implied minimum positive-rate half-life ln(2)/L after T, with reciprocal endpoint brackets; zero fade supplied no finite upper duration. Cash/tax, source-vintage, mixed-date claims/risk, unvalidated supports and originality limitations remained unresolved.

### S19.6. Design and reused evidence

The plan was sealed before alternative evaluation (see PLAN_SEAL.json). No equivalent matched A/B/C experiment was found. The earlier B6 latest-versus-median-three study is related but uses older revenue/admission conventions. The completed step2 H5 records supply exact G/O/P restrictions and all baseline27-case records. The24 base/favorable baselines were independently reproduced; no completed full baseline grid was repeated. Existing portable-package/frozen source selections are copied and hash checked.

G means coherent guidance plus old nominal terminal/transition/all-future caps; O means coherent guidance plus saved own-history caps; P means coherent guidance plus saved peer caps. All retain the saved common floor and terminal lower; paired turnover/hull screening is off. Base is BASE WACC/3% g/UNIFORM_FIRST; favorable is LOW WACC/4% g/UNIFORM_ALL. No independently fitted turnover is introduced. These are controlled model cells, not independent empirical observations.

### S19.7. Main A versus C comparison

**Table S32. Eta maximum-value contrasts across the fixed case menu**

| Firm | Cell | A compatible | C compatible | Maximum-value change (% fixed EV), outer range across27 cases |
|---|---|---|---|---|
| CRM | G | 4/27 | 5/27 | -6.04 to 8.96 |
| CRM | O | 0/27 | 0/27 | 9.38 to 14.26 |
| CRM | P | 0/27 | 0/27 | 3.02 to 10.89 |
| NOW | G | 0/27 | 0/27 | 1.72 to 2.76 |
| NOW | O | 0/27 | 0/27 | 1.64 to 2.73 |
| NOW | P | 0/27 | 0/27 | 1.62 to 2.64 |
| BOX | G | 0/27 | 0/27 | 1.16 to 1.94 |
| BOX | O | 0/27 | 0/27 | 1.25 to 2.03 |
| BOX | P | 0/27 | 0/27 | 1.40 to 2.02 |
| HUBS | G | 27/27 | 27/27 | 0.74 to 1.81 |
| HUBS | O | 0/27 | 0/27 | 0.72 to 1.64 |
| HUBS | P | 19/27 | 19/27 | 0.72 to 1.66 |

The range uses the outer numerical brackets for C maximum minus A maximum in each case, divided by that fixed EV; displayed range endpoints are rounded outward to two decimals. It is not a confidence interval, nor the difference of pooled extrema. Every C comparison triggered expansion through the prespecified pilot criterion. Three source-permitted B firms triggered menu-wide analytic domain documentation; Box B was unavailable, without a manufactured result. There were21 expanded firm/cell/scenario comparisons; BOX B remained at its six unavailable pilot rows. The1% rule is triage, not significance.

Salesforce G gains exactly one compatible case: LOW WACC,3% terminal g, YEAR_END. Its baseline maximum is $197.085962 billion and C maximum $200.567623 billion against the unchanged $199.867770 billion target. The new supported set is [0,L] with L inner0.0005613937973976135 and outer0.0005613938556052744 per year. Its spread half-life is at least about1234.690 years after the transition,

conditional on this near-zero-decay model cell; lambda0 supplies no finite upper duration bound. This very long lower bound is a conditional mathematical implication, not an empirical forecast or investor belief. O/P remain excluded in every Salesforce case. C does not uniformly increase its G maximum: changes range from about−6.03% to+8.96% of EV. Across the G case menu, the largest supported rate decreases from [0.0201847652788274,0.02018476533703506] to [0.01339031959651038,0.013390319654718041] per year; this is a union over risk/timing cases, not a single estimated rate. Exact authoritative selections and duration calculations are in DURATION_DIAGNOSTICS.csv.

ServiceNow and Box remain excluded in all three cells, even with zero incremental non-R&D investment. Their maximum values rise, but remain below the fixed targets. HubSpot retains27/27 G,0/27 O and19/27 P compatibility, including the same individual compatible cases. Its supported rate endpoints change: in the BASE/3%/UNIFORM_FIRST P cell, upper inner/outer brackets move from [0.022837707423605025,0.022837707598228008] to [0.02447090484201908,0.024470905016642064] per year. The implied lower half-life moves from about30.351 to28.325 years after T; there is no finite upper half-life bound in either [0,L] set. Other cases can move in the opposite direction; unchanged compatibility counts do not mean unchanged persistence bounds.

**S19.8. Scenario B: source verification and domain failures**

Official archived original filings and SEC publication records verify the same opening fiscal dates/consolidation scopes. Three acquisition-excluded coefficients are negative:

**Table S33. Opening-date acquisition balances and coefficient domains**

| Firm | Fiscal date | N0 ($million) | GW0 ($million) | AI0 net ($million) | eta B | Result |
|---|---|---|---|---|---|---|
| CRM | 2026-01-31 | 56,960 | 57,941 | 6,815 | −0.187742324 | All27 cases in each cell fail complete N>=0 domain |
| NOW | 2025-12-31 | 2,858 | 3,578 | 1,121 | −0.138650399 | All27 cases in each cell fail complete N>=0 domain |
| BOX | 2026-01-31 | 170.047 | 82.290 | Unknown category net balance | Unknown | B unavailable |
| HUBS | 2025-12-31 | 102.724 | 291.452 | 35.225 | −0.071521551 | All27 cases in each cell fail complete N>=0 domain |

With eta<0 and g>0, N_j=N0+eta(R_j-R0) eventually tends to minus infinity. Original GrowthModel prefix admission explicitly requires N>=0, K>0 and C>0; complete-path use cannot stop that requirement at J. No new signed-N completion was used, and no negative coefficient is floored. This analytic failure is separate from an admitted positive-capital path whose value falls short of EV. Total C is still positive at the first N failure in all documented cases: that does not rescue the maintained nonnegative N-stock domain. The rejection is specifically N_j >= 0; the certificates do not demonstrate a total-capital failure at that year. Although the asymptotic C/R ratios computed from the saved parameters are positive for all nine firm/growth combinations, that limit alone does not prove finite-year C or average-capital positivity. The B records stop at the N-domain certificate and contain no complete-path proof of those other inequalities. Intermediate-path positivity therefore remains unestablished here. No exclusive deferred-revenue attribution is supported by a verified decomposition, and C is not a bound on inadmissible negative-eta scenarios. For CRM the first negative-N year is94/64/49 for terminal g=2/3/4%; NOW20/15/12; HUBS3/3/3. These years are indexed from each accounting origin, not from the price date. They are model domain certificates, not projected cash spending.

CRM’s AI balance is explicitly business-combination net intangibles. NOW’s purchased intangible caption is distinct from PP&E computer software. HUBS’s externally acquired caption includes acquired technology and

separately purchased domain/sublease/other rights; it is not exclusively a business-combination technology balance. Internal-use software is separate PP&E. These classifications are disclosed in SOURCE_VERIFICATION.md. No synthetic K or recognized internal software is subtracted a second time. Residual economic overlap in the original R&D stock remains unresolved.

Box's $94.311 million NET intangible caption includes $116.942 million GROSS internally developed software, $3.658 million gross acquired technology and $9.964 million gross on-premises software, less $36.253 million aggregate amortization. A category net acquired-intangible balance is not disclosed in the retained filing; using94.311 or3.658 would violate the prescribed stock scope. The production panel's acquired-intangible label does not resolve this distinction. The correct B result is unavailable, not a guessed negative coefficient. No production accounting record was corrected in this stage.

**S19.9. Valuation and operating-admission channels**

Changing eta leaves opening historical acquisitions in N0/C0, but changes delta_N, average capital, terminal spread, the growing and constant capital-mode coefficients, the zero-spread valuation base, the spread kernel, and Gamma=average_C/R. At fixed J and matched research inputs, the research-residual coefficient H_J is unchanged. At a fixed endpoint margin, the later profit rate is tied to a changed return denominator: lower constructed investment need not monotonically raise the value of the same fading-ROIC path. All 324 saved baseline maximum witnesses remain admitted under C when reevaluated at their fixed T/shape/m/guide/lambda. Their raw values change, sometimes downward, while their recomputed future-cap slack becomes positive. CASE_CONTRASTS.csv records these fixed-witness changes without treating an inadmitted raw root as a result.

Admission also changes substantively: nine HIGH-WACC Salesforce O cases previously admitted only lambda0 because every positive fade violated the asymptotic margin ceiling. Under C, the recomputed positive-fade operating domain is available in all nine, although no valuation equality emerges. Thus the sensitivity operates through both valuation and future-path admission. The Salesforce G count reversal and shifted bounds are joint controlled model effects; no additive causal allocation is identified. B is principally a structural capital-domain failure. Unchanged NOW/BOX classifications and unchanged HUBS counts support only classification robustness within A/C and the fixed controls.

**S19.9.1 Fixed operating coordinates versus reoptimized maxima**

Keep revenue, research stock, opening N0/K0/C0, T, transition shape, endpoint margin m, guide margin, lambda and risk/timing fixed. Scenario labels superscript A/C distinguish paths; they do not denote modal coefficients. For forward $j \geq 1$, the capital identity gives

$$\begin{aligned} C_j^C - C_j^A &= (\eta_C - \eta_A)(R_j - R_0), \\ \acute{C}_j^C - \acute{C}_j^A &= (\eta_C - \eta_A)\left[(R_{j-1} + R_j)/2 - R_0\right], \\ \Delta C_j^C - \Delta C_j^A &= (\eta_C - \eta_A)\Delta R_j. \end{aligned} \tag{S47}$$

At $T$, NOPAT $m R_T$ is unchanged but endpoint return $m R_T / \acute{C}_T$ and spread $s_T$ change. Thus holding these operating coordinates fixed does not hold the ROIC sequence or later NOPAT fixed. For $n \geq 1$, writing $z = e^{-\lambda}$, the original completion implies

$$\begin{aligned} Y_{T+n}^{\eta} &= w(1 - z^n)\acute{C}_{T+n}^{\eta} + m R_T z^n \frac{\acute{C}_{T+n}^{\eta}}{\acute{C}_T^{\eta}}, \\ Y_{T+n}^{C} - Y_{T+n}^{A} &= w(1 - z^n)(\acute{C}_{T+n}^{C} - \acute{C}_{T+n}^{A}) \\ &+ m R_T z^n \left( \frac{\acute{C}_{T+n}^{C}}{\acute{C}_T^{C}} - \frac{\acute{C}_{T+n}^{A}}{\acute{C}_T^{A}} \right). \end{aligned} \tag{S48}$$

Subtract the investment difference in (S47) to obtain the constructed-FCFF difference. Lower eta changes both terms of (S48), its endpoint denominator and the investment deduction; their net discounted effect has no universal positive sign. Complete admission also changes because $m_j = \Gamma_j r_j$. Equation (S36) makes the valuation mechanism explicit: eta changes $G_J$ and $F_J$ ($H_J$ is unchanged at a fixed J and matched research path), the zero-spread tail, the spread kernel and $s_J$. The opening C0 term is unchanged. These algebraic terms are not an empirically estimated or calculated additive allocation of optimized effects.

The saved CRM G base case illustrates the distinction. Its A maximum witness has T=2, LINEAR, endpoint margin 0.28317764672390531256489818350840741153566364468174 and lambda 0.00019931793212890625. It is still admitted under C, but its value changes by -17,156,606,448.87411408862892050913136729323955171390 USD to 150,244,775,360.39704989588217046769604720666452922284 USD. Reoptimization under C instead chooses T=4, NORMALIZED_EXPONENTIAL and zero fade: its maximum inner/outer bracket is [172,875,354,594.38990213057646855607792180397044103395, 172,875,354,594.41006] USD, versus A's [167,401,381,809.27116398451109097682741449990408093674, 167,403,363,531.5989]. These are saved CRM__BASE__BASE__UNIFORM_FIRST__G__A/C.json records and CASE_CONTRASTS.csv selections, not new optimizations. This example does not decompose the optimized difference into independently additive valuation and admission effects.

**S19.9.2 Fading spread and dollar economic profit**

Substitute (S10) into (S9), rather than interpreting the faded rate as dollar EP:

$$EP_{T+n} = s_T \acute{C}_{T+n} e^{-\lambda n} + \frac{w}{2} \Delta C_{T+n}. \quad \text{(S49)}$$

Let $J = max(T, H_R)$. If $a = 1 + g > 1$, $0 < v < 1$, and the leading capital coefficient $G_J$ is positive, then $\acute{C}_{J+h}$ is asymptotic to $G_J(1+a)a^{h-1}/2$ and $\Delta C_{J+h}$ to $G_J g a^{h-1}$. For nonzero $s_T$ the magnitude of the first term of (S49) consequently scales with $(a e^{-\lambda})^n$: it vanishes if that factor is below one, tends to a nonzero magnitude if equal to one, and grows if above one. Its sign remains the sign of $s_T$. This describes only the spread component, not total EP.

With $w > 0$ and $g > 0$ the capital-growth term grows like $a^n$. At positive lambda it eventually dominates the spread component under these leading-mode conditions; at zero lambda the two leading amplitudes can combine or cancel, so total EP need not share either isolated conclusion. A zero spread removes the spread component entirely. Nonpositive/missing leading growth modes, zero growth, signed components and cancellation require their own mode analysis; no universal dollar-profit monotonicity is asserted.

The alternative normalized-EP-rate benchmark in Section S16 instead sets $e_j = EP_j / \acute{C}_j$ and fades $e_T e^{-\lambda n}$. Its dollar EP is $e_T \acute{C}_{T+n} e^{-\lambda n}$, without the additional capital-growth term of (S49). The same growth-factor comparison applies only under the preceding leading-mode/nonzero-amplitude conditions. Its zero EP-rate amplitude gives zero EP, unlike zero original ROIC spread during positive capital growth. This distinction does not expand the separately prespecified 24-pair pilot or its bounded null finding.

**S19.10. Verification and interpretation limits**

All capital paths/tails and admission certificates were recomputed in isolated adapters. Checks independently reconstruct N, research accumulation, C/average_C,100-year mode paths and capital transversality; For admitted A/C paths, positive tail investment follows $G_J > 0$, $H_J < 0$, $1 + g > 1$ and $0 < v < 1$, by equation (S35). The formerly local A/B modal names refer exactly to $\eta R_J + U_J$ and $K_J - U_J$; they have been replaced by $G_J / H_J$, not conflated with normalized ratio coefficients or scenarios. Every new resolved case includes full lambda-cell coverage or an explicit asymptotic certificate, Decimal maximum and equality endpoint/interior checks, and

complete omitted-year future-margin bounds. Pilot witnesses also compare1500-year FCFF and beginning-capital EP with closed tails. Additional independent finite terminal-capital identities and signed/zero/limiting algebra checks are documented separately; their synthetic test margins are not claimed admitted operating paths. Seven capital and eleven operating regression tests pass. There are no unresolved numerical cells and no loosened numerical standards.

Rate endpoints and maximum values are numerical inner/outer enclosures under the inherited topology premises, not certified sharp economic bounds. Empty equality support means no admitted match under these restrictions, not zero persistence. The fade clock starts after transition T and refers to average-capital ROIC minus WACC, not dollar economic-profit lifetime; growing capital can offset a declining rate.

B is an acquisition-excluded historical stock-ratio scenario; it does not estimate organic marginal capital. C is an extreme diagnostic, not a realistic forecast. Changes in N are constructed book investment, not verified cash spending. No result identifies acquisition-caused mispricing, true investor beliefs, economic validity or a correct investment policy. Historical caps remain maintained restrictions with unestablished empirical reliability. Mixed-date valuation/claim inputs, normalized taxes, possible R&D/accounting overlap, cash reconciliation and constant-risk assumptions remain conditional; the two chronological annual endpoints do not validate full D9 or infinite-horizon admission. Methodological priority remains unresolved.

**S19.11. Computation scope and stopping verdict**

The output contains897 model records:348 new resolved optimized solves (24 A reproduction plus324 C),300 reused full-menu baseline records,243 analytic B domain rows and6 unavailable BOX B pilot rows. There were no broad sample expansion, new revenue guidance, fitted price calibration, new manuscript/PDF edits or full production-pipeline replay. Maximum pilot baseline reproduction discrepancy was $0E-38.

The completed findings show that conditional values and some persistence endpoints depend materially on eta, even when compatibility counts mostly survive. Salesforce G is sensitive in one additional case; its O/P exclusions remain robust to the tested zero-eta diagnostic. The acquired-stock subtraction does not produce an admissible investment completion for the three computable firms, and Box cannot be tested as specified with retained net-category evidence. Further investment work would address a genuine economic weakness only if it measures marginal capital/cash requirements or separately justifies a signed-N model; tuning eta to match price would not resolve it. No correct investment policy was identified.

**S19.12. Exact coefficients and source/case dependencies**

The following values are serialized selections from PARAMETER_SCENARIOS.csv; empty values remain unavailable. K0/C0 are unchanged in all scenarios. The full SOURCE_LEDGER.csv includes balances, fiscal/publication dates, scope, accession, source URL, disclosure locator and checksums. HUBS externally acquired intangibles include separately purchased rights; BOX is not assigned the mixed total intangible balance.

**Table S34. Exact saved parameter selections**

| Firm | Scenario | Serialized eta | Domain/source status |
|---|---|---|---|
| CRM | A | 1.371703792895846 | Coefficient admitted |
| CRM | B | -0.18774232390126429861529199277543648404575556893438 | Negative eta; N-domain failure |
| CRM | C | 0 | Coefficient admitted |
| NOW | A | 0.2152432595270372 | Coefficient admitted |
| NOW | B | -0.13865039915649947281217050760656725410453381533363 | Negative eta; N-domain failure |
| NOW | C | 0 | Coefficient admitted |
| BOX | A | 0.14444388759255658 | Coefficient admitted |
| BOX | B | Unavailable | Acquired-net scope unavailable |
| BOX | C | 0 | Coefficient admitted |
| HUBS | A | 0.03280590023332416 | Coefficient admitted |
| HUBS | B | -0.071521550708243886019265051260416713239948314834958 | Negative eta; N-domain failure |
| HUBS | C | 0 | Coefficient admitted |

Machine status keys remain ADMITTED_COEFFICIENT, NEGATIVE_ETA_COMPLETE_N_DOMAIN_FAILURE and UNAVAILABLE_SOURCE_CLASSIFICATION in the complete parameter file. Printed coefficient line breaks do not round or change their serialized digits.

The preserved acquisition-source classification record follows. It verifies original-vintage scope and publication receipts without a later restatement.

Reused archived official filings, checked programmatically against local filing bodies and original SEC submissions records. No later restatement imported. Live web access succeeded for NOW and failed for CRM/BOX/HUBS; failures are not missing balances because archived originals are available. No blocked-source retries.

*Salesforce (CRM)*

GW and business-combination AI are disjoint; net AI=14,139-7,324 million. Internally developed software separate. K is retained unchanged.

Note8, pp80-81; consolidated balance sheet p66; intangible assets acquired through business combinations. Fiscal date 2026-01-31; publication 2026-03-02.

*ServiceNow (NOW)*

Purchased developed technology, customer relationships, patents and other rights are separate from PP&E computer software. Net AI=1,709-588 million. Product software costs are expensed; no internal software subtraction. K retained unchanged.

Note6 pp87-88, Goodwill and Intangible Assets; Note1 capitalization policy and separate PP&E computer software; balance sheet p68. Fiscal date 2025-12-31; publication 2026-01-29.

*Box (BOX)*

UNAVAILABLE: 94,311 thousand NET is not acquired intangibles. GROSS acquired technology=3,658 thousand; internal software=116,942 and on-premises software=9,964; aggregate amortization=36,253. No category net carrying balances. Neither the net aggregate nor a gross acquisition balance supplies the required net acquired-category stock.

Note5 pp88-89; balance sheet p74; intangible net balance combines internal software, acquired technology and on-premises software. Fiscal date 2026-01-31; publication 2026-03-09.

*HubSpot (HUBS)*

Externally acquired intangible caption includes acquired technology, purchased connect.com domain, sublease and other acquired rights; net=72,629-37,404 thousand. Scope includes separately purchased rights, not solely business-combination technology. Internal-use software is separate PP&E. No repeated K/software subtraction.

Note9 pp87-88; Note7 PP&E/internal-use software; balance sheet p66. Fiscal date 2025-12-31; publication 2026-02-11.

The synthetic K research stock may overlap economic technology embodied in acquired companies; this preexisting limitation is not resolved. The coefficient subtraction removes no K or internal-use software and is not an asset revaluation. HUBS includes separately purchased rights. BOX B remains unavailable. Production panel labels do not alone establish acquired-intangible scope.

### S19.13. Reproduction objects and verification

Complete eta reproduction materials (separate replication materials): ALL_RESULTS.csv (separate replication materials), CASE_CONTRASTS.csv (separate replication materials), SOURCE_LEDGER.csv (separate replication materials), PARAMETER_SCENARIOS.csv (separate replication materials), DURATION_DIAGNOSTICS.csv (separate replication materials). The complete897-record catalogue is machine-readable rather than repeated as hundreds of nearly identical printed rows. Case JSON retains transition shape/T, WACC/g/timing, capital modes, full margin certificates, cell coverage and equality witnesses. Negative-N crossing certificates and unavailable BOX B rows are retained explicitly.

The sealed plan checksum is c0522c9aec60fa88dc58ed51e28 273f02b77f19cef5e7196a06 188a12d4715ca. Eighteen focused regression tests and independent opening-NOA, guidance/tax, signed/zero/limiting, FCFF/EP and cell-coverage checks passed in the experiment. Its6,026 protected originals and320 source selections remained unchanged. The current editorial integration checks those saved receipts; it does not rerun them.

## S20. Complete references and primary-source bibliography

### S20.1. Academic literature

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**S20.2. Primary financial and parameter sources**

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Box, Inc. (2026). FY2026 Form 10-K, year ended January 31; Form 10-Q, period ended July 31; August 25 fiscal 2027 results and outlook. Annual filing; quarterly filing; issuer guidance.

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**S20.3. Software and version provenance**

Saved receipts for both the original-location and relocated D7/D8/D9 portable replays record Python 3.9.6. The relocation used a fresh environment without third-party packages. The matched-attribution and alternative-profitability driver records identify Python standard-library scientific calculations, including decimal arithmetic and mathematical functions. Later studies identify the bundled Python executable but do not consistently retain a per-run interpreter version. Its currently verified version must not be assigned retrospectively to those earlier runs. The installed LibreOfficeDev binary reports 26.8.0.0.alpha0; the runtime manifest separately labels its packaged LibreOffice dependency 25.2-headless-codex.1. The table reports the binary actually used for this conversion.

The following versions describe the October 14 preparation workflow, supplemented by versions checked for the October 15 editorial conversion. They are document-preparation provenance, separate from historical scientific execution receipts. They describe document preparation and inspection, not an additional scientific replay:

| Software | Verified version | Role |
|---|---|---|
| Python | 3.12.14 | Document preparation and inspection |
| Pandoc | 3.12.1 | Markdown to editable Word equations and tables |
| LibreOfficeDev | 26.8.0.0.alpha0 | Word rendering and PDF conversion; installed binary version |
| python-docx | 1.2.0 | Word layout and metadata inspection |
| Node.js | 24.19.0 | Local document-rendering workflow |
| Marked | 17.0.5 | Markdown conversion |
| KaTeX | 0.19.0 | Mathematical typesetting |
| Playwright | 1.62.1 | Isolated headless PDF rendering |
| Brave | 154.1.96.61 | Browser engine for local rendering |
| ReportLab | 4.4.9 | Brief and title-page PDF generation |
| pdfplumber | 0.11.9 | PDF text and layout checks |
| pypdf | 6.10.0 | PDF and form handling |
| pypdfium2 | 5.14.0 | Page rendering for visual inspection |
| Pillow | 12.3.0 | Inspection-image preparation |

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